

AWS - ATL089 Procedures Manual

Subcontractor Name: _____



AWS Project Safety Manual

(Site-Specific Programs and Safety Procedures)

August 25th, 2025

**AWS - ATL089
Job #7389-001
11145 Alcovy Road
Covington, GA 30054**

IMPORTANT NOTES TO ALL SUBCONTRACTORS:

- In accordance with Article 9 of the RYAN subcontract, this manual is intended to be used as a general guideline for safety on RYAN jobsites and is not intended to take the place of each subcontractor's responsibility to train their employees in all applicable safety requirements.
- **Article 9.2.3 Additional Safety Personnel.** Contractor reserves the right to require Subcontractor, at Subcontractor's own cost, to add one or more people to its crew who are dedicated to the safety of Subcontractor's employees at the Project if Subcontractor's safety measures are deemed substandard as documented in Contractor's regular safety audits. Safety Representative is defined as one who meets OSHA's definition of a Competent Person and has an OSHA 30 Construction Safety Course, possess a current AED, CPR and First Aid training from a nationally recognized program and have a minimum of five years construction management experience.
- **AWS Global Construction H&S Minimum Requirements (GMR) – v.2.1 Refer to addendum for full list of requirements.**
 - **Section 3.15 Ready to Work/Fitness for Duty**
 - **Section 3.21 Personal Protective Equipment:** Type 1 Class E-rated Safety Helmet that meets (1) ANSI z89.1 and/or EN 397 Certification, (2) Rated Type 1 and (3) Rated Class E, (4) Equipped with Manufacturers-installed 'chinstrap'.
 - **Section 3.5 Hazard Identification, Risk Assessment, Permit Programs & Uncommon Work Areas**
 - **Section 3.24 Simultaneous Operations Management Plans:** Inclusive of Construction Zone (CZ) and Operational Zone (OZ) and any additional requirements while working in those areas.
 - **Section 3.9 Subcontractor Management:** Subcontractor shall have a designated Project Safety Representative on site at all times while work is being performed. Subcontractor shall provide a Project Safety Representative that has an OSHA 30-hour certification, current first aid/CPR/AED training, and 5 years of construction experience. This subcontractor Project Safety Representative may be an employee of the subcontractor, who may also function as a Superintendent, Foreman, or Crew Leader, on the project. The Subcontractor's Project Safety Representatives shall be held accountable for their responsibilities and to ensure that the Subcontractor has a robust Safety Management Program in place on the site that is actively managed and enforced by the Subcontractor's leadership team. Additional Project Safety Managers may be required according to Article 9.2.3 of the Ryan Subcontract.
 - **Section 3.6.1.2 Fall Hazard Control:** Ensure no site personnel work alone while working at heights. Site personnel working at heights shall ensure there are other personnel in the area of operations that can be alerted in the case of emergency.
 - **Section 3.6.1.5 Electrical Safety**
 - **Section 3.6.1.9 Lifting and Rigging:** A Lift Plan for all lifting and rigging activities is required.
 - **Section 3.6.1.10 Ground Penetration, Excavation and Trenching**
 - **PDC load bank operation minimum requirements.**
- **Ryan Energy Control Plan attached as Exhibit 1 superseding all prior Electrical / LOTO safety requirements.**

AWS - ATL089 Procedures Manual

Policy Manual Version Control & Update Log

This form is designed to manage version control and ensure consistent updates in policy manuals. It should be stored in the front section of the manual and updated with every approved change.

Version Number	Effective Date	Section(s) Updated	Description of Changes	Approved By	Next Review Date
1.0	2025-06-16	Fitness for Duty	Added fitness for duty requirements	Jim Person	
1.0	2025-06-16	Lockout Tagout	Added LOTO / Energy Control additions including requirements for Energy Marshall and LDL testing	Matt Gillotti	
2.0	2025-07-28	Personal Protective Equipment	Added Hand Protection Policy	Matt Gillotti	
3.0	2025-08-08	Ladder Use / Work at Heights	Added Work at Heights safety requirements – No A-frame ladders allowed	Bill O'Connell	
3.0	2025-08-08	Lift Use / Work at Heights	Added lift tie-off and safety laser policy	John Gaddini	
3.0	2025-08-08	Minimum Safety Training Requirements	Added OSHA 10 Certification requirement for all trade workers	John Gaddini	
3.0	2025-08-08	Subcontractor Safety Representatives	Added subcontractor safety staffing and qualifications	Bill O'Connell	
4.0	2025-08-18	AWS GMR V2.1	Updated GMR references and addendum to AWS v2.1	Matt Gillotti	
4.1	2025-08-19	Energy Control Plan	Revised 'Electrical Safety' and 'LOTO' with Ryan Energy Control Plan	Matt Gillotti	2025-10-15

Instructions for Use

1. Assign a new version number whenever a change is made to the manual.
2. Clearly state the effective date for the updated version.
3. Specify the exact section(s) that were modified.
4. Provide a concise description of the changes made.
5. Include the name of the person who approved the change.
6. Set a date for the next scheduled review.
7. Archive old versions for compliance and reference purposes.

Use Notes

1. Sections highlighted in **yellow** are **new** to the Project Safety Manual since the previous version was released.
2. Jobsite Meetings are highlighted in **green**.

AWS - ATL089 Procedures Manual



Dear Valued Employee,

As you know, we take safety very seriously at Ryan and we pride ourselves in leading the industry relative to safety performance on the job.

It is our everyday responsibility to work safely on all Ryan jobsites, at projects we own or manage, and at our regional offices and yard facilities. By maintaining safe work environments, we help to secure not only our own safety, but also that of our co-workers and the general public.

We all recognize that construction is a dangerous business and unfortunately, accidents may happen from time to time. However, by instituting a philosophy of safety throughout the organization, we are able to ensure fewer accidents and a much safer work environment. That leads to greater confidence and productivity in the workplace.

We encourage you to actively participate in all safety training opportunities available to you, use tools responsibly, and be alert for any unsafe conditions at all times. If you become aware of potential for a hazardous condition or incident, please take appropriate action and notify your supervisor immediately.

As always, Ryan intends to fulfill the requirements of the Occupational Safety and Health Act. Any information concerning these important standards will be made available at your request.

It is imperative to the leadership at Ryan that we can look you and your coworkers in the eye and say that we are doing everything we can to send you all home safely to your families and friends every night.

Thank you for all you do for Ryan and for making safety a #1 priority.

Sincerely,

A handwritten signature in green ink that reads "Brian C. Murray". The signature is fluid and cursive, with the last name "Murray" being particularly prominent.

Brian C. Murray
Chief Executive Officer

Ryan Companies US, Inc.
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Equal Opportunity Employer

AWS - ATL089 Procedures Manual

Table of Contents:

Minimum Safety Training Requirements	5
Jobsite Orientation Procedures for Subcontractors	5
Subcontractor Safety Representative Requirements	6
Monthly Jobsite Safety Committee Meeting	8
Emergency Management/Traffic Management Plan	8
Incident/Injury/Near-miss Reporting Policy	9
Managed Care Program Information for Ryan Employees	12
Drug Screen Policy	12
Ready to Work/Fitness for Duty	12
Training	12
Jobsite Inspections	13
Jobsite Meetings	13
Disciplinary Program	13
Site Safety Incentive Program	15
Stairways/Access, Egress and Ladders	15
Subcontractor Personal Protective Equipment	17
Use of Handheld Communication Devices while Operating Equipment	18
Mobile Technology	18
Housekeeping	18
Electrical	18
Cranes	18
Use of Crane to Suspend Work Platforms	20
Forklifts	20
Fall Arrest Systems	21
Floor and Wall Openings	21
Material Storage	22
Fire Protection and Fire Prevention	22
Flammable and Combustible Liquids	23
Liquified Petroleum Gas	24
Temporary Heating Devices	24
Welding and Cutting	24
Rigging and Hoisting	25
Controlled Access	25
Operation of Mobile Earth-moving Equipment	26
Trenching/Excavation Operations	27
Masonry	28
Tilt Panel Construction	28
Bad Weather Policy	29
Cold Stress / Heat Stress Policy	29
Jobsite Deliveries	31
Radio Program	31
Steel Erection	31
Environmental Issues	32
Spill Response	32
Do Not Interrupt Adjacent Systems, Utilities, Businesses or Structures	35
Mobile Elevated Work Platforms (MEWPs)	35
Permit Required Confined Space Entry	36
Respirable Crystalline Silica/Silica Exposure Control Plan	37
Use of UAS/Drone(s)	38
Photography Policy	38
Project Specific Tool Tethering Policy	39
Use of ATVs/UTVs	40
Rebar Protection	40
Daily Roof Access Permit Procedure	41
Ground Disturbance Permit Procedure	42
Overhead Power Line Awareness Procedure	45
Safety Organization Structure and Job Description	46
Appendices & Exhibits	48

AWS - ATL089 Procedures Manual

Introduction

Ryan Companies US, Inc. (Company) would like to welcome you to the AWS - ATL089 jobsite. This site-specific Project Safety Manual is to ensure that safety is communicated in a consistent manner, which in turn will ensure the safety of all employees on the jobsite.

Our goal at Ryan is to provide employees with a work environment that is free from recognized hazards and to return them home safe each day. Please review and familiarize yourself with these procedures and expectations to ensure this goal is met.

Minimum Safety Training Requirements

1. All subcontractor and tiered subcontractor personnel performing work on an mission critical project shall have completed an OSHA 10 construction certification training within the last 5 years and provide a copy of the card during registration. Reference OSHA subpart 1926.
2. All subcontractor and/or tiered subcontractor personnel overseeing or managing work (i.e., Lead Person/Foreman/Supervisor/Competent Person/Superintendent/Field Managers, Safety Managers/Representatives and above) shall have completed an OSHA 30 construction certification training within the last 5 years and provide a copy of the card during registration. Reference OSHA subpart 1926.

Jobsite Orientation Procedures for Subcontractors

1. All Subcontractors new to the AWS - ATL089 jobsite must meet with Ryan site supervision to review Ryan safety policies, programs and Article 9 of the Ryan Subcontract - Safety Requirements and Procedures (Exhibit 1). Prior to starting ANY work, a **Preconstruction Subcontractor Safety Meeting (Exhibit 2 – see Procure Forms)** must be completed with Ryan site supervision and the Ryan Jobsite Safety Orientation Video must be viewed by subcontractor's employees and ensure that all their workers and sub-tier(s) subcontractors follow all project safety requirements. Subcontractor's on-site/project leadership (foremen, project management, competent person(s), safety director etc.) must review this Project Safety Manual with Ryan site supervision. Note: Ryan's QAQC Pre-Installation meeting is a separate meeting which does not take the place of the Preconstruction Subcontractor Safety Meeting.
2. All Subcontractors and tiered subcontractors shall submit the following items to Ryan site supervision:
 - Project Specific Safety Manual (hard copy and electronic copy)
 - Project Specific SDS Manual (hard copy and electronic copy)
 - A detailed project-specific Job Hazard Analysis (JHA) (see Exhibit 3 for a sample JHA form) for all scopes of work / site activities must be completed **prior** to commencing any work on site. Additional JHA's or pre-task planning may be required daily or weekly upon request by Ryan Field Management.
 - Certificate(s) of Insurance (as required)
 - **Subcontractor is responsible to provide current training or certification records to Ryan. It is your responsibility to resubmit to Ryan, any changes or updates to certifications for your time on the project.**
 - OSHA, Forklift Operator, Crane Operator, Rigger, and Signal Person certification(s) (if applicable)
 - Annual Crane Inspection/Certification and Assembly/Disassembly plan (if applicable)
3. The individual employee orientation for each Subcontractor and/or any of Subcontractor's tier subcontractor employees must be **acknowledged and completed on the project** by having each individual:

AWS - ATL089 Procedures Manual

- 1) watch the Ryan Jobsite Safety Orientation Video, receive site specific information including emergency evacuation training, and
 - 2) read, complete, sign, and submit the Site-Specific Orientation Form (Exhibit 4) or the Orientation Acknowledgement Form, to site supervision.
4. Safety orientation hard hat stickers and site access badging will be issued to the employee after viewing the Jobsite Safety Orientation Video and upon completing/submitting the Site-Specific Orientation Form (Exhibit 4) or the Orientation Acknowledgement Form to Ryan site supervision. Any employee not displaying these stickers or project badging will be asked to report to Ryan site supervision for the required orientation prior to continuing any work.

Subcontractor Safety Representative Requirements

Purpose

Ensure that all trades working on-site maintain a consistent and effective level of safety supervision aligned with the high standards required for Mission Critical and Data Center construction environments. The policy supports our commitment to zero incidents, continuous hazard awareness, and proactive safety leadership across all work crews.

1. Minimum Safety Supervision Staffing

All subcontractors and trade partners are required to provide **dedicated on-site safety supervision** according to the following crew size criteria:

- **1 Full-Time Safety Representative** for **every 25 trade workers** on-site (measured by the daily average headcount, including all lower tier contracted tradespeople).
- For example:
 - 1–24 crew members → No dedicated Safety Representative required (see Competent Person requirement below).
 - 25–49 crew members → 1 dedicated Safety Representative.
 - 50–74 crew members → 2 dedicated Safety Representative.
 - And so on.

2. Competent Person Designation (for crews under 25)

If a trade partner has **fewer than 25** workers on-site, they are required to:

- Designate a **Competent Person**, as defined by OSHA 1926.32(f), who:
 - Is present on-site during all working hours. Available after hours in the event of a site emergency.
 - Has demonstrable experience relevant to the work being performed.
 - **Provides daily safety oversight** alongside their normal duties.
 - Meets all the 'Dedicated Safety Representative Qualifications' below **except** b. 'Education and Certification'.
 - Is clearly identified during project onboarding and in the daily pre-task plans.

Dedicated Safety Representative Qualifications

Each Safety Representative shall possess relevant certification(s) and education which includes at least one of each in the following categories:

- a. Meet all Competent Person criteria
- a. Education & Certification (one of the following):

AWS - ATL089 Procedures Manual

- a. Certified Safety Professional (CSP)
- b. Associate Safety Professional (ASP)
- c. Construction Health and Safety Technician (CHST)
- d. Occupational Health and Safety Technician (OHST)
- e. Graduate or Chartered Member of the Institute of Occupational Safety and Health (IOSH)
- f. Safety Trained Supervisor – Construction (STS-C)
- g. Bachelor's degree or higher in EHS discipline or EHS related field
- h. Current OSHA 500 training card
- b. Training (all of the following):
 - a. 1926 OSHA 30-hour Construction Industry Outreach Training Course within the last 5 years. Training shall be conducted by an OSHA authorized education center or authorized trainer.
 - b. First Aid / CPR / AED First responder Trained from a Nationally recognized organization.
 - c. For Electrically-Qualified (*see Energy Control Program*) related scope – NFPA70E training within the last 3 years.
- c. Experience (all of the following):
 - a. Minimum of three years' experience with similar project size and construction type.
 - b. Minimum five years' safety management experience in commercial construction.
- d. Proficiency in conducting Job Hazard Analyses (JHAs), safety inspections, and incident investigations. Entry into digital management platform required.
- e. Capable of interacting with the GC's and Owner's safety teams and participating in coordination meetings.

The Sub/Contractor Safety Representatives to be approved by the GC. Resume(s) submitted to GC Partner two weeks prior to the start of work. Provisional exceptions to these requirements may be made and documented by the General Contractor (GC), but may be revoked at any time by the GC.

Responsibilities

Dedicated Safety Representatives and Competent Persons must:

- Conduct **daily safety inspections** of their trade's work areas. Complete a minimum of 5 safety inspections per week with any deficiency observations closed within 48 hours.
- Lead or participate in **daily safety huddles**, weekly Safety Committee meetings, and weekly toolbox talks.
- Participate in all on-site Safety Walks (following weekly Committee meetings).
- Enforce and ensure compliance with site-specific safety procedures.
- Coordinate with the GC's safety team regarding **incident reporting, corrective actions, and compliance audits**.
- Submit all required safety planning documentation, including high-risk permits, JHA/PTPs, crew certifications, toolbox talk records.
- Lead or facilitate training for crew requisite to their specific scope of work.
- Ensure all new employees are paired with experienced workers through a minimum 90-day onboarding period to learn their tasks and safety protocols.
- Ensure all employees are fit for duty on a regular basis.
- Track, share, and act on jobsite safety trends and lessons learned.
- Maintain accurate documentation of safety observations, corrections, and training.

Compliance and Enforcement

Failure to comply with this policy may result in:

- **Corrective Action Notices (CANs)** issued to the trade partner.
- **Work stoppage** until compliance is achieved.

AWS - ATL089 Procedures Manual

- Possible **removal from site** for repeat violations.
- **Hold payment** until accurate staffing ratios and qualifications are met.

All safety personnel and Competent Persons are subject to regular audits by the General Contractor (GC) and Owner's Risk/Safety teams.

Monthly Jobsite Safety Committee Meeting

1. The AWS - ATL089 Jobsite Safety Committee will consist of the following:
 - Ryan Safety Representative
 - Ryan Field Manager/site supervision
 - Subcontractor Safety Representative and/or site supervision at a ratio of 2:1 (workforce to management) will be required.

The AWS - ATL089 **Monthly Jobsite Safety Committee Meeting** will be held on the **First Wednesday** of each month at **10:00 am** at the Ryan jobsite office.

2. Jobsite Safety Committee members will inspect the AWS - ATL089 jobsite prior to the monthly Jobsite Safety Committee Meeting.
3. Safety-related items including issues, violations, injuries, near-misses, and corrective actions will be discussed at the monthly meeting. See Jobsite Safety Committee Agenda – Exhibit 16/Procore Forms.
4. Documentation of meeting minutes and attendees will be distributed and kept in Procore Meetings.
5. Ryan Best Practice would be to hold the Jobsite Safety Committee Meeting a week before or just prior to scheduling the All-Job Meeting so any issues or findings can be discussed at the All-Job Meeting.

Emergency Management (Reference GMR Section 3.18 for additional AWS requirements)

1. Ryan site supervision shall complete and distribute the Emergency Response Plan for Construction Sites (Exhibit 5). All contractors on site must comply with OSHA first aid training requirements and provide documentation on demand (please provide a list of current first aid trained personnel to Ryan site supervision). Post the Emergency Response Plan on the jobsite in several locations.
2. Updates/revisions to Exhibit 5 should be completed as jobsite conditions change.
3. A Jobsite Emergency Contact List (Exhibit 6) should be updated/maintained throughout the project.
4. Ryan site supervision shall complete and post a site map for the project. The following is a minimum list of items to identify on the map.
 - Field offices
 - Muster Points (Indoor and Outdoor)
 - Medical trailer (if applicable)
 - Shelter locations
 - AED location(s)
 - First Aid Kits
 - Narcan / Naloxone

AWS - ATL089 Procedures Manual

- Eyewash station(s)
- Rescue equipment
- Mobile Technology Zones
- Designated smoking areas
- Security checkpoint
- Gas shut-off location(s)

Traffic Management Plan

Project – specific traffic management plan (TMP) is required to be completed by Ryan site supervision. The TMP shall be kept current and identified on a map of the project. The TMP must include the following;

1. Physical security arrangements
2. Speed limit
3. Speed limit controls
4. Pedestrian safety measures
5. Employee parking
6. Contractor parking
7. Visitor parking
8. Overflow parking
9. Shelter / storage
10. Courier and / or delivery pick – up and drop off points
11. Safety equipment arrangements
12. Large vehicle deliveries
13. Emergency response arrangements
14. Temporary change design
15. Hydrant / standpipe systems
16. Gate access location(s)
17. Safe Walkways for access and egress
18. Traffic devices

Incident Reporting Policy (Reference GMR Section 3.17 for additional AWS requirements)

Introduction

This procedure establishes a uniform method for **reporting** any/all (Ryan and non-company/subcontractor) incidents or injuries, equipment/property damage, near-miss incidents and/or vehicle accidents.

Reporting Requirements and Procedures

A Jobsite Emergency Contact List is provided for your use (Exhibit 6). In the event of a serious injury or fatality, including after-hours, contact:

- 911
- Ryan Jobsite Manager and Field Coordinator
- Safety Director and Jobsite Safety Representative

Any/all (Ryan and non-company/subcontractor) incidents, injuries, equipment/property damage, near-miss incidents, and/or vehicle accidents **must be reported immediately following any occurrence**. The following procedures apply:

1. Injury to a Ryan Employee

- a. Call 911 for any life-threatening injury.

AWS - ATL089 Procedures Manual

- b. For a non-serious injury to a Ryan employee, contact WorkPartners at 1-800-359-5020 for recommended course of action or treatment.
- c. If a clinic visit is required, Ryan Jobsite Manager shall notify the closest clinic network provider and inform them of injured Ryan employee.
- d. In the event of a non-serious injury, a Ryan representative shall transport the injured Ryan employee to the designated clinic or medical provider and shall inform them of the work-related injury. Injured employees should not drive themselves to a clinic.
- e. Ryan Jobsite Manager shall be responsible for ensuring the injured Ryan employee (and medical provider) receives "Managed Care Program Information for Ryan Employees" (see page 7) prior to the injured Ryan employee's clinic (or urgent care/hospital) visit.
- f. The Ryan Jobsite Manager shall complete the "Incident Report" (Exhibit 7, see Procore Incident) in Procore. If a hard copy incident form is completed, it must be scanned and attached to an Incident in Procore. **Complete all sections of this report that pertain and submit within 24 hours.**
- g. The Ryan Jobsite Manager shall utilize notification through the Incident Reporting. Incident Notification should go to Ryan VP of Field Operations and Safety, Ryan Safety Director, Regional/Sector Field Coordinator, and the Ryan Claims Administrator, at a minimum.
- h. The Ryan Jobsite Manager shall email a PDF version Ryan Incident Report (Exhibit 7, see Procore Incidents) to the Ryan Claims Administrator (claims.administrator@ryancompanies.com).
- i. The Claims Administrator shall complete the "First Report of Injury" and shall notify the insurance company. The Claims Administrator will also distribute reports to the following personnel:
 - Ryan Vice President of Field Operations and Safety
 - Ryan Safety Director/Regional Safety Director(s)
 - Ryan Field Coordinator
 - Ryan Risk Management
- j. The Ryan Jobsite Manager shall ensure that corrective action is taken to prevent similar incidents or injuries.

2. Subcontractor Employee Incidents and/or Injury

- a. The Ryan "Incident Report" (see Procore Incident or Exhibit 7) shall be completed in the event a non-employee is injured, or a non-company incident occurs. If a hard copy incident form is completed, it must be scanned and attached to an Incident Report in Procore. **Complete all sections of this report that pertain and submit before 24 hours.**
- b. The Ryan Jobsite Manager shall utilize notification through the Incident Reporting. Incident Information Notification should go to Ryan VP of Field Operations and Safety, Ryan Safety Director, Regional/Sector Field Coordinator, and the Ryan Claims Administrator, at a minimum.
- c. The Ryan Jobsite Manager shall attach the subcontractor's incident/injury report, as well as any follow-up information pertaining to the non-employee to the Procore Incident of record created. Subcontractors are required contractually to provide an incident/injury report, with corrective action and root cause.
- d. The Ryan Jobsite Manager must ensure that a corrective action plan has been provided from the subcontractor and that the subcontractor's jobsite manager/competent person has implemented the corrective action on site to prevent similar incidents or injuries.

3. Lost Time Injury

For employees with injuries resulting in lost work time, the Safety Director shall be contacted immediately with all pertinent information. A "return to work authorization" from the treating physician must be provided prior to an employee's return to work following a lost time injury.

4. Tracking of Incidents

- a. Ryan's Safety Department shall be responsible for maintaining the OSHA 300 log for all injuries relating to Ryan employees.

AWS - ATL089 Procedures Manual

- b. Every employer other than Ryan is responsible for maintaining and posting their OSHA 300 log.
- c. The Ryan Safety Department shall be responsible for tracking incidents and safety statistics company-wide.

5. Equipment and Property Damage

- a. The Ryan Jobsite Manager shall utilize notification through the Procore Incident Reporting. Incident Information Notification should go to Ryan VP of Field Operations and Safety, Ryan Safety Director, Regional/Sector Field Coordinator, and the Ryan Claims Administrator, at a minimum for all Ryan and non-company accidents causing damage to equipment and/or property.
- b. The Ryan Jobsite Manager shall ensure that corrective action is taken to prevent similar incidents or injuries.
- c. The Ryan Claims Administrator shall forward information to the insurance company and distribute reports to the following:
 - Ryan Safety Director
 - Ryan Field Coordinator
 - Ryan Risk Management
 - Ryan Vice President of Field Operations and Safety

6. Near-miss Incident

- a. A near-miss is a potential hazard or incident in which no property was damaged, and no personal injury was sustained, but where, given a slight shift in time or position, damage or injury easily could have occurred. Near misses also may be referred to as close calls, near accidents, or injury-free events.
- b. The Ryan Jobsite Manager shall complete the "Incident Report" (see Procore Incidents) for any near-miss incidents that occur.
- c. The Ryan Jobsite Manager shall utilize notification through the Procore Incident Reporting. Incident Information Notification should go to Ryan VP of Field Operations and Safety, Ryan Safety Director, Regional/Sector Field Coordinator, and the Ryan Claims Administrator, at a minimum.
- d. The Ryan Jobsite Manager shall ensure that corrective action is taken to prevent similar incidents or injuries.

7. Anti-Harassment Policy

- a. Violations of Ryan's Anti-Harassment Policy must be reported in the Procore Incident Report and submitted to the Ryan Claims Administrator, Field Coordinator and Safety Director.

8. Motor Vehicle Accident

- a. Any accident involving a Ryan-owned vehicle or privately-owned vehicle used for Company business shall be reported promptly.
- b. Personnel involved in a motor vehicle accident shall complete the "Motor Vehicle Accident Report" and forward it to the Ryan Claims Administrator.
- c. In the event of injury to an employee or non-employee involved in a motor vehicle accident, the "Incident Report" (see Procore Incidents) shall be completed and forwarded to the Ryan Claims Administrator. The Motor Vehicle Accident Report form can be found in Tools, or the safety kit located in all Ryan vehicles.

9. Theft

- a. The Ryan Jobsite Manager shall utilize notification through the Procore Incident Reporting for all incidents involving theft of Ryan property. Please attach any/all police reports to the Procore Incident Report. Incident Information Notification should go to Ryan VP of Field Operations and Safety, Ryan Safety Director, Regional/Sector Field Coordinator, and the Ryan Claims Administrator, at a minimum.

10. Incident Reporting Timeline Requirements

AWS - ATL089 Procedures Manual

- a. All Incidents that occur on the project require a report to AWS in accordance with the 2-hour, 24-hour, 2-week timeline structure and is to be submitted by Ryan Companies.

Managed Care Program Information for Ryan Employees

Ryan Companies US, Inc. has a Managed Care Program through its workers' compensation insurance company. When seeking medical attention for a worker's compensation injury, provide the clinic with the following information:

Insurer:

Insurance Carrier: Zurich American Insurance Company
Policy Number: See Gidget Wegener (Ryan Claims Administrator)

EMAIL OR FAX ALL MEDICAL DOCUMENTATION TO:

Claims.Administrator@RyanCompanies.com
Fax: (612) 492-3501
Phone: (612) 492-4411

All Regions:

If you have any questions pertaining to the following, please contact Jim Person at (612) 492-4361 (office) or (612) 369-3476 (mobile), or Gidget Wegener at (612) 492-4411 (office).

- Worker's compensation wages
- Worker's compensation medical care
- Medical bills
- Return-to-work/light duty

Drug Screen Policy

Each Contractor, its Subcontractors and employees of every tier, are prohibited from reporting for duty under the influence of any intoxicating substance which will in any way adversely affect their working ability, alertness, coordination or response, or adversely affect the safety of any person at the jobsite. It is the responsibility of each Contractor and its Subcontractors and employees of every tier to ensure that employees reporting for duty at the jobsite are free from the influence of any intoxicating substance. Employees who report to work while under the influence will be dismissed from the jobsite.

Ryan employees are subject to the Ryan Substance Abuse Policy.

Ready to Work/Fitness for Duty (Reference GMR Section 3.15 for additional AWS requirements)

Fitness for Duty requires our project personnel to be in a physical, mental, and emotional state, that enable the individual(s) to perform their assigned duties and must also ensure that all individual(s) are physically able to carry out the tasks that are assigned to them. **Subcontractors must develop a process which gives you reasonable assurance that the individuals working on the project who are employees of the subcontractor or who are tasked by your scope of work to perform duties on the project are physically fit to safely perform their assigned duties without risk or harm to themselves or others.**

Training

AWS - ATL089 Procedures Manual

1. Ryan Companies US, Inc. will offer OSHA Outreach training to Subcontractor employees. Invitations will be sent notifying Subcontractors as to when classes will be held.
2. Training on this Project Safety Manual will be provided at the time of the Preconstruction Subcontractor Safety Meeting.
3. Training will be offered at the monthly All-Job Safety Meeting. Please note that this training does not relieve any Subcontractor of the obligation to train its employees.
4. The content/topic and attendees for each training session will be documented.

Jobsite Inspections (Reference GMR Section 3.25 for additional AWS requirements)

1. Ryan's Loss Control Representatives may include the Safety Director, Field Coordinator, Regional Safety Consultant(s), Field Manager(s) or a Ryan employee from the jobsite who, through training and/or years of experience, is able to recognize jobsite hazards.
2. Ryan's Loss Control Representative(s) will conduct on-site inspections periodically (See Procore Safety Inspection). Reports will be generated by the Loss Control Representative and kept in the jobsite safety file/Procore.
3. Each Subcontractor is responsible for appointing personnel (Competent Person, Safety Representatives and Managers) to conduct daily safety inspection of their work area(s) and crew actions, completing a minimum of (5) inspections weekly, and providing bi-weekly audits of JHA/PTPs. Submission of inspections and audits shall be hard copy or digital, as directed by Ryan Supervision and reviewed daily with site safety leadership.

Jobsite Meetings (Reference GMR Section 3.10 and 3.20 for additional AWS requirements)

1. All Subcontractors are responsible for holding a **weekly safety meeting** for their employees. Safety meetings shall be documented for content and attendance and be submitted to Ryan site supervision.
2. All Ryan employees and Subcontractor employees of every tier are required to attend a monthly Ryan total project ("**All-Job**") **Safety Meeting**, which will be held every **Second Wednesday** of the month at **7:00 am** at the AWS - ATL089 jobsite.
3. All Subcontractor foremen are required to attend the **Weekly Supervisory ("Sub-Foremen") Meeting** held every **Wednesday** at **11:00 am** at the Ryan construction office, and attend the **Monthly Safety Committee Meeting** (2:1 workforce to management).
4. Plan of Day (POD) meetings are required and will be held at a location determined by Ryan Companies field management staff.

Disciplinary Program

Safety requires constant attention from everyone on the project. Each employee must follow their company's policies and procedures, the safety rules and regulations of Ryan Companies US, Inc., Federal and State agencies, as well as the owner for whom the work is being performed. To ensure a safe workplace for each employee, Ryan Companies US, Inc. has developed a Disciplinary Program to enforce these safety rules and regulations. Please also refer to the subcontract agreement, Article 9.

AWS - ATL089 Procedures Manual

In the event of a **Safety Violation**, the employee will be subject to one of the following disciplinary actions:

NON-SERIOUS VIOLATION - A violation that has a direct relationship to project policies, safety or health but is not serious in nature.

- 1st violation: Verbal Warning
- 2nd violation: Verbal and Written Warning to Your Company
- 3rd violation: Verbal and Written Warning to Your Company

Your company must follow up with a written statement detailing what it intends to do in order to correct the offending action and ensure that a repeat safety violation does not occur. Your company may opt to remove you from the project or terminate your employment.

- 4th violation: Removed from the project by your Company and/or Ryan.

SERIOUS VIOLATION - A serious violation exists when the conduct constitutes a serious violation of any policy, regulation or requirement which could result in serious injury to any person or serious damage to any property.

- 1st violation: Verbal and Written Warning to your Company

Your company must follow up with a written statement of what it intends to do in order to correct the violation, so it does not happen again.

- A. Termination of employment
- B. Send you to a different job
- C. Keep you on-site after review of the problem

If the decision is for you to stay and there is a 2nd violation: Verbal and Written Warning to your Company.

Your company will receive both verbal and written notices of the nature of the problem, and you will be removed from the project by your company and/or Ryan.

In each case the employee will be provided a written Safety Violation Notice (Exhibit 9/Procore Forms). A copy of the notice will be issued to the employee's supervisor and the Ryan Safety Director.

Nothing in this policy prohibits the immediate dismissal or removal from the jobsite of any employee whose conduct constitutes a serious violation of the safety requirements, or which could cause serious danger to themselves, co-workers, property, equipment, or the employees of others.

HARASSMENT POLICY

It is the policy of Ryan Companies US, Inc. to maintain a working environment free from offensive or degrading remarks or conduct based on gender, age, race, national origin, religion, disability, marital or public assistance status, sexual orientation, genetics, or any other status protected by applicable state or federal law. Offensive behavior prohibited by this policy includes, as examples, requests to engage in illegal immoral or unethical conduct. Offensive and harassing behavior are prohibited and will not be tolerated for the protection of jobsite employees, public by-standers and local businesses.

If you are subjected to offensive behavior of any kind, you are strongly encouraged to object to the behavior and should also immediately report it to your supervisor or other appropriate management personnel. Should

AWS - ATL089 Procedures Manual

there arise any instances of unacceptable conduct, Ryan Companies US, Inc. will call for immediate removal of offending persons from the jobsite.

ENFORCEMENT

Implementation of the Disciplinary Program is to be carried out by the Safety Director, Field Coordinator, project superintendents and foremen. It is the responsibility of each person mentioned to see that all safety requirements are followed by all personnel and that warnings are issued as required under this policy and procedure. There are no exceptions.

ISSUANCE OF SAFETY VIOLATION NOTICE

When issuing a Safety Violation Notice, meet with the employee(s) to discuss the safety infraction. Inform the individual(s) of the rule or procedure that was violated and the corrective action to be taken. Complete the Safety Violation Notice (Exhibit 9) in its entirety and issue the copies as directed on the form.

Questions regarding the Disciplinary Program policies and procedures are to be directed to the Safety Director.

SUPERVISOR RESPONSIBILITY

The on-site supervisory personnel will be judged on the same merits as field employees and due to the safety stewardship, will bear responsibility for overall site conditions.

Field Coordinators, the Safety Director, and the Safety Consultant will make physical inspections of the jobsites. If an inspection of activities and areas under your control indicates violations showing overall lack of commitment to Ryan Companies US, Inc. safety requirements, the project superintendent and/or foreman will be issued a written warning. Two written warnings will be considered cause for disciplinary lay-off or demotion. Three written warnings will be considered cause for dismissal. Failure to institute corrective procedures after top management has ordered them shall be considered insubordination and shall be cause for dismissal. It is required that the suspension and termination process be completed under the direction of the Safety Director.

ADDITIONAL SAFETY PERSONNEL

Per Article 9.2.3, Ryan reserves the right to require Subcontractor, at Subcontractor's own cost to add one or more people to its crew who are dedicated to the safety of Subcontractor's employees at the Project if Subcontractor's safety measures are deemed substandard as documented in Ryan's regular safety audits.

Site Safety Incentive Program

1. All personnel are eligible to receive a token of recognition and appreciation for the best safety or corrective action suggestion presented to Ryan site supervision. This award may be given each month. Safety suggestions may be submitted on a Ryan Safety Suggestion Form (see Ryan Safety Manual) and be turned in to the supervisor or the Site Safety Suggestion Box as may be located in the jobsite office. The consensus of jobsite personnel will determine the best suggestion(s)/winners.
2. Important safety milestones will be determined by the Ryan site supervision.

Stairways/Access, Egress and Ladders

1. **Stairways, Access/Egress and Stair Towers.** A stairway, ramp, or approved access system must be provided for all personnel access points with a vertical break in elevation of 19 inches or more.

AWS - ATL089 Procedures Manual

If permanent stairs are not available as the building is erected, a temporary scaffold/stair tower must be provided. Each building/structure level requires that **two (2) means** of unobstructed access/egress are provided at all times to/from any uppermost floor that has been planked or decked; one means of access/egress may be the primary stairs, the second means of access/egress may be a ladder (not to exceed a height of 24 feet) or a temporary scaffold/stair tower. Note: more than one temporary scaffold/stair tower may be required per building design; (i.e. for buildings greater than 200,000 square feet, more than one stair tower is required).

- **Access to steel stairways** will be restricted until stair treads/pans are filled.
- **Stairways** with four or more risers or rising 30 inches or more shall be equipped with at least one handrail.

2. General Ladder Safety, Use, Inspection and Maintenance.

Important: *The use of A-frame (step) ladders is prohibited on this project.* Only **platform ladders** meeting the specifications below are permitted when ladders are deemed the safest and most appropriate method for access or task completion.

Ladder Alternatives – Preferred Methods for Working at Heights

Before using a ladder, the following engineered alternatives **must be considered and prioritized**:

- Aerial lifts (boom or scissor)
- Stair towers
- System or frame scaffolding
- Baker (mobile) scaffolds

Use of ladders is discouraged and must be task-justified by the supervisor and planned for in the Jobsite Hazard Analysis (**JHA**), and Pre-Task Plans (**PTP**). When ladders are used, the following requirements apply:

i. Ladder Type, Rating, and Use Requirements

- Only **Type 1A fiberglass platform ladders** are permitted on this project.
- Aluminum and other metal ladders are strictly prohibited.

ii. Inspection and Maintenance

- Ladders must be inspected by a competent person prior to each use.
- Damaged ladders shall be tagged "Do Not Use" and removed from service immediately.
- All manufacturer's use and warning labels must be present and legible.
- Rungs and platforms must be kept clean and free of slippery substances.
- Side rails and/or platform guardrails must extend at least 36 inches (3 feet) above the landing surface and be secured to prevent movement.

iii. General Ladder Use Practices

- Place ladders on a stable base with clear access at top and bottom.
- Maintain a pitch of 1 foot out from the support structure for every 4 feet of ladder height.
- Always face the ladder and maintain three points of contact when ascending or descending.
- A hand line must be available to hoist tools and materials — do not carry loads by hand while ascending/descending.
- Do not use ladders in passageways, doorways, or traffic areas unless protected by barricades, dedicated spotters, or other means to prevent displacement.

AWS - ATL089 Procedures Manual

- When working near interior/exterior edges or openings, **personal fall protection** is required if the distance to the fall hazard is less than or equal to 1.5 times the height of the ladder and at all times when working above 6 feet from the ground level.
- Ladder access points must be protected with guardrails and either self-closing gates or offset entries to prevent accidental falls into ladder openings.

iv. Additional Restrictions

- A-frame step ladders, extension ladders, and pail-shelf ladders are **not permitted** for working at heights. Ladders may continue to be an alternate egress means for below grade trench work when appropriately planned.
- No person shall straddle, sit on, or stand on the top platform guardrail or cap of any ladder.
- Don't overreach, lean to one side or try to move the ladder while you are on it.
- Do not exceed the ladder's rated load capacity.
- Ensure ladder is completely open with spreaders locked and any platform guards in-place before use.
- No more than one worker may occupy a ladder at any time.
- Always secure ladders from movement and ensure a second person is present to stabilize, if required.

Job-Built Ladder Requirements

Any job-built ladder must meet the **ANSI A14.4-2009** standard for wood ladders and be constructed under the supervision of a competent person. Use of job-built ladders requires project-specific approval.

Ryan Employee PPE (Reference GMR Section 3.21 for additional AWS requirements)

To easily identify Field supervision and Safety representatives on Ryan jobsites, white hard hats are to be worn by Field Managers who are Assistant Superintendents and up, as well as all Safety Directors and Safety Coordinators.

- Attire for non-Field employees while on a Ryan jobsite includes long pants, work boots with ankle support, ANSI-rated safety glasses, an ANSI Class 2 Safety vest/garment and a **yellow** hard hat.
- Required attire and personal protective equipment for Field Operations supervision employees (Assistant Superintendent and above) on the jobsite includes long pants, work boots with ankle support, ANSI-rated safety glasses, an ANSI Class 2 Safety vest/garment (or ANSI Class 3 if working near traffic), work gloves (minimum ANSI Cut Level 4) and a **white** hard hat.
- Craftworkers/self-perform employees are required to wear personal protective equipment while on the jobsite, including long pants, work boots with ankle support, ANSI-rated safety glasses, an ANSI Class 2 Safety vest/garment (or ANSI Class 3 if working near traffic), work gloves (minimum ANSI Cut Level 4), and a **yellow** hard hat.

Subcontractor Personal Protective Equipment (Reference GMR Section 3.21 for additional AWS requirements)

Subcontractor's employees shall report to the Project site with proper apparel and safety equipment. All PPE shall be inspected by the subcontractor's Competent Person prior to use. Subcontractor's employees shall:

- Wear a hardhat at all times while on the Project site, until otherwise notified by Contractor's Superintendent.
- Wear ANSI approved safety eyewear at all times while on the Project site. This shall include the use of additional eye protection/face shield as recommended by the equipment or tool manufacturer.

AWS - ATL089 Procedures Manual

- Wear adequate work shoes (above ankle height) at all times on the Project site. Tennis shoes or casual shoes will not be permitted.
- Wear proper clothing (not less than long pants and a short-sleeved shirt) and wear additional protective clothing as required by the work being performed by the employee.
- Wear ANSI Class 2 compliant high visibility apparel at all times on the project. All high visibility apparel must meet ANSI Class 2 or ANSI Class 3 requirements.
- Wear gloves, minimum ANSI Cut Level 4, at all times while on the project. Additional levels of hand protection may be required as documented in an approved JHA and approved by a competent person to address chemical, welding, impact, electrical, and other high-risk activities. Subcontractors are required to follow all 29 CFR 1926.28, 1910.137, ASTM D120 or any other regulatory requirement(s).

Use of Hand-held Communication Devices while Operating Equipment (Reference GMR Section 3.26 for additional AWS requirements)

Distracted use or operation of equipment is a recognized safety hazard. If your company will be operating any equipment that may require the use of hand-held communication devices, such as: two-way radios, CB radios, cell phones, walkie-talkies, etc., you are responsible to assess and manage that use to prevent distraction hazards, as well as ensure that all employees are trained on distraction hazards, know how to run their equipment safely, and that they clearly understand this policy.

Mobile Technology

The use of mobile technology such as cellphones and tablets can distract the user from hazards around them. Restricting the use of mobile devices on the project will increase the level of concentration of project personnel. Using mobile technology except as necessary to protect safety (e.g., in the event of an emergency) or for the performance of work-related tasks in a safe and secure manner is always required.

The following are the minimum requirements for mobile technology use on the project.

1. No walking while talking / texting using a mobile device.
2. Mobile technology is not to be used in equipment / vehicles / ATV / UTV's when in motion.
3. Mobile devices are not permitted while attending safety meetings or while receiving safety related information.
4. The use of mobile devices is not permitted around explosive or flammable atmospheres, while decanting or working with flammable or combustible material(s).
5. Utilize designated Mobile Technology Zones for all non-emergency communications

Housekeeping

During the course of the AWS - ATL089 construction, all Contractors will institute a housekeeping program. This may be daily or as requested by Ryan site supervision. Good housekeeping practices will be followed in order to keep the work areas, passageways, and stairways in and around the building or other structures free from debris of all types. No dry sweeping or compressed air shall be used as per the OSHA standard on silica. Subcontractors generating any silica dust must control, collect and properly dispose of all silica material. Please refer to Article 9 of the Ryan Subcontract.

Electrical (Reference GMR Sections 3.24 and 3.6.1.5 for additional AWS requirements)

Follow all energy control requirements outlined in the **Energy Control Plan** section of this document.

AWS - ATL089 Procedures Manual

Cranes (Reference GMR Sections 3.6.1.9 for additional AWS requirements)

1. All crane operators must comply with OSHA 29 CFR 1926.1427 Operator Qualifications and Certifications (i.e. NCCCO certified or equivalent) after November 2017.
2. Effective February 7, 2019: In addition to training and certification, all crane operators will need to be evaluated by their employer for the equipment they will be operating. For an example, please see the Ryan [Crane Operator Evaluation Form](#), Exhibit 11/ Procore Forms.
3. The Contractor or its Subcontractors of every tier shall comply with the manufacturer's specifications on limitations applicable to the operation of any and all cranes. Where manufacturer's specifications are not available, the limitations assigned to the equipment shall be based on the determinations of a qualified engineer competent in the field. Such determinations will be appropriately documented, recorded and available at the jobsite.
4. Rated load capacities, operating instructions (owner's manual) and specialized hazard warnings shall be conspicuously posted on all equipment. They shall be visible to the operator while at the control station.
5. An illustration of the hand signals to crane operators shall be posted in a designated location visible to all employees.
6. The Contractor or Subcontractor shall designate a **qualified and competent person(s)** who shall inspect all machinery and equipment prior to and during use to make sure it is in safe operating condition. Deficiencies will be corrected before continued use.
7. A working radius (crane and load) must be maintained at least 20 feet from any overhead power lines, or in accordance with options listed in [29 CFR 1926.1408](#) and applicable Tables. See Ryan Field Manager for [Overhead Power Line Awareness Permit \(Exhibit 17/Procore Forms\)](#). The swing radius of the crane shall be barricaded with signage.
8. The Contractor or its Subcontractors of every tier shall maintain and provide copies of both the annual, monthly, and daily inspection(s) of the hoisting equipment during the Preconstruction Subcontractor Safety Meeting.
9. Qualified riggers and signal persons are required whenever workers are within the fall zone and are hooking/unhooking or guiding a load or doing the initial connection of a load to a component or structure. Qualification documentation must be provided during the Preconstruction Subcontractor Safety Meeting.
10. The Contractor or Subcontractor shall ensure that each signal person meets the qualification requirements of 29 CFR 1926.1428(c), prior to giving any signals. Qualification documentation must be provided during the Preconstruction Subcontractor Safety Meeting.
11. No person shall be allowed to ride on the material hoist, hook, or load.
12. Contractor shall coordinate and comply with height restrictions relating to cranes, concrete pumps, or any other equipment used on the jobsite that could potentially interfere with other operations.
13. Contractor shall supply a [Critical/Engineered Lift Plan](#) (see Ryan Safety Manual/Tools and Procore Forms) if a lift has one or more of the following conditions:
 - Hoisting personnel.
 - Tandem picks/lifts.
 - The net weight of the load exceeds 25 tons.
 - The value of the load exceeds \$50,000.
 - The gross weight of the load exceeds 75% of the crane's rated capacity.
 - The replacement time for a damaged load exceeds two months.
14. A fire extinguisher must be kept in the crane at all times.
15. **Ryan crane operators have the authority and responsibility to refuse lifting any load they feel does not comply with safety requirements.**
16. All tower cranes used on Ryan jobsites will have load-indicating devices (digital scales).
17. A tag line shall be used on all loads.
18. Weather conditions should be constantly monitored, and crane operations shut down when conditions may create safety concerns. (Ex: wind, rain, snow, etc.)

AWS - ATL089 Procedures Manual

19. If operations will require vertical lifts involving a **helicopter crane**, follow Ryan Helicopter Crane Procedures safety inspection checklist in Procore Forms (see also Tools) **and provide Ryan Legal and Ryan Corporate Safety Director with adequate notice in advance of the lift.**

Use of Crane to Suspend Work Platforms

The use of a crane to lift, lower and/or suspend work platforms **shall be permitted only when other means of reaching the work area are not feasible** and only when the operation can be performed in compliance with applicable state or federal regulations (OSHA). If the Contractor deems it necessary to use a crane-supported work platform, the following Submittal Procedure must be followed:

Submittal Procedure

1. In compliance with OSHA Subpart CC 1926.1431(a), a plan (Personnel Basket Pre-lift Test Procedure) for the use of crane-suspended work platforms shall be submitted to Ryan Companies US, Inc., Jobsite Manager, for review with the following (see Ryan Safety Manual/Tools):
 - a. Statement of why conditions, methods, or operations require the use of a crane-suspended work platform.
 - b. Description of the crane and work platform to be used and the manufacturer's instructions and requirements in the use of the crane to lift, lower and/or suspend personnel on the work platform.
 - c. Documented emergency plan in the event of crane failure.
2. Copies of the latest annual inspection report as well as the latest monthly inspection report of the crane to be used shall be submitted to Ryan Companies US, Inc., Jobsite Manager prior to the use of the work platform. Contractor shall ensure that daily inspections are made, and submit to Ryan Companies US, Inc., Jobsite Manager, a monthly crane inspection report so long as the crane and suspended work platform are in use.

Forklifts (Reference GMR Sections 3.22 and 3.23 for additional AWS requirements)

1. All forklift operators shall be certified and evaluated to comply with OSHA 1910.178 forklift regulations prior to operation of forklifts on the jobsite. **Subcontractor must have hard copies of certifications available on site for review.**
2. Refresher evaluation shall be provided when:
 - a. Operator has been observed operating forklift in an unsafe manner.
 - b. Operator has been involved in an accident or near miss.
 - c. Operator is assigned to drive a different type of forklift.
 - d. Condition of workplace has changed.
 - e. Re-certification is required every three years.
3. When a forklift is left unattended, load-engaging means shall be fully lowered, controls shall be neutralized, power shall be shut off, brakes set, and LP operated forklifts shall have fuel shut off.
4. All traffic regulations shall be observed, including authorized speed limits.
5. Only loads within the rated capacity of the truck shall be handled.
6. Forklift inspections shall be conducted prior to the work shift or its use.
7. Elevated work platforms used with forklifts will be required to meet the requirements of that forklift manufacturer and must follow ASME- B56.6a. The responsibility to know and follow these rules lies with the employer of the crew using the elevated work platform.

AWS - ATL089 Procedures Manual

8. Work will be immediately stopped if any unsafe acts are observed, such as: free-rigging, seat belts are unfastened, passengers are riding on the forklift, distracted driving/operation, unsafe speeds, etc.
9. Forklifts operated on roadways must obey traffic regulations, include a strobe-light and slow-moving vehicle sign.
10. All attachments shall be approved by the manufacturer and rated with an identification for the use. Operators may be subject to additional training requirements when specific attachments are used in a manner that would reclassify the equipment as a crane. Refer to 29 CFR Subpart CC 1926.1400 (c)(8).

Fall Arrest Systems (Reference GMR Sections 3.6 and 3.21 for additional AWS requirements)

1. **Any** worker exposed to a fall distance of six (6) feet* or greater shall be provided with adequate means of fall protection. Fall protection may consist of, but is not limited to, personal fall arrest equipment (body harness, shock-absorbing lanyards and self-retracting lifelines) or physical barrier protection such as guardrail systems, catch platforms, parapet walls meeting OSHA height and strength requirements, or a combination (100% fall protection). ***Note:** Some OSHA state plans may have more stringent fall protection requirements, i.e. **Washington State's requirement is four (4) feet**. Check with your local OSHA area office for specific requirements.
2. Contractors performing structural erection activities (such as pre-cast concrete and steel erection) shall submit a task-specific Fall Protection Plan, which complies with the six (6) foot fall protection requirement. Note: Washington State requires fall protection at four (4) feet.
3. All fall arrest equipment shall be inspected prior to each use for wear, damage and other deterioration, with special attention given to the retractable device and engineered anchorage points. **Defective equipment shall be removed from service.** Note: Perimeter cables are to be used for guardrail purposes only.
4. Lifelines, body harnesses, and shock-absorbing lanyards shall be used only for worker safeguarding, not for positioning.
5. Shock absorbing devices shall not be connected together in fall arrest systems (lanyard to lanyard or lanyard to retractable).
6. Any lifeline, body harness, shock absorbing lanyard, or anchorage point subjected to forces of arresting a fall shall be **removed from service, tagged "danger, do not use" and removed from the jobsite immediately.**
7. Anchorages used for attachment of personal fall arrest equipment shall be independent of any anchorage being used to support or suspend platforms and capable of supporting at least 5,000 pounds (22.2 kN) per worker attached.
8. All personal fall arrest equipment and anchorage points shall meet or exceed OSHA/ANSI standards.
9. **All contractors are responsible for the proper training of their workers in personal fall arrest systems, recognition of potential fall hazards, and understanding the need to properly inspect and maintain all fall protection equipment/components. A competent person for fall protection must be on site while fall protection is being used.**
10. All contractors shall have a Rescue Plan providing for the prompt rescue of workers in the event of a fall. This Rescue Plan will be available upon request.

Floor and Wall Openings (Reference GMR Section 3.6 for additional AWS requirements)

1. **All floor holes / openings greater than 2 inches** in any direction will either be covered with a secure cover marked "Danger Floor Opening/Hole/Cover – Do Not Remove" or will be protected by standard guardrails (42 inches), mid-rails (21 inches) and toe boards. All holes/openings in decks, around columns and dock pits shall be protected by guardrails or covers. Note: Warning lines are to be used for leading edge and roof work only. Warning lines must meet OSHA Subpart M 1926.502(f). Plastic red tape does not meet the requirements of the standard for warning lines.

AWS - ATL089 Procedures Manual

2. Standard guardrails and toe boards will protect **wall openings** and entrances to shaft-ways where applicable. Ryan requires adding extra guardrails 21 inches on center above the 42-inch top-rail. (This protects workers on ladders and/or stilts adjacent to shaft-ways.)
3. **Wall openings** with framing greater than 19 inches on center, shall have a top and mid-rail installed.
4. Stairwells will be protected on any open side(s) with standard guardrails and will be equipped with adequate handrails.
5. A controlled access zone will be designated when any hole covers are removed.
6. Elevator shafts and mechanical shafts shall be controlled access zones. Decking over these shafts must be considered at the beginning of the project as to frequency and strength of the temporary deck/cover.
7. There are numerous site-specific safety requirements that apply to **wood framed construction** which shall be addressed in the Subcontractor's Job Hazard Analysis.
8. **Warning lines are to be used for leading edge and low-slope roof work only.** Warning lines for iron workers and roofers shall be set not less than **fifteen (15) feet** from the leading edge and **fifteen (15) feet** from roof edge(s). All other trades shall have warning lines set at **fifteen (15) feet** from the roof edge(s). If roofers are using motorized equipment, warning lines must be no less than **fifteen (15) feet** from the roof edge(s). Rope grabs are not permissible for fall protection use outside of the warning line. Anyone working outside of the roof warning line system shall use a leading edge self-retracting PFAS. **Roof/Safety monitors are not permissible in lieu of fall protection.**
9. Perimeter safety cable systems shall be installed by qualified workers with turnbuckles every 50', and corners must have the appropriate length stiffeners, so that the system can be tightened to specification without deflection of the vertical corner post. Perimeter cables and other fall protection guardrail systems shall be frequently inspected. Edge protection or similar must be installed to prevent any chance of fall from heights, after a concrete pour.

Material Storage

1. All materials shall be stored in such a manner that they are secured from sliding, falling, or collapsing to a lower level, or blowing off a building. **Subcontractors shall designate personnel to inspect material(s) daily to ensure proper storage.**
2. Aisleways/walkways/exits/egress access shall be kept clear to provide free movement of material handling equipment and employees.
3. Maximum safe load limits of floors where material storage or equipment is stored in the buildings and other structures shall not be exceeded.
4. Materials shall not be stored on scaffolds or runways except for the materials needed for immediate operations.
5. **Material stored on metal decking shall be stored on top of structural support to prevent failure of the metal deck. Material shall not be stored until metal decking is secured and inspected by the subcontractor's competent person.**
6. Material stored inside buildings under construction shall not be placed within 6 feet of any hoist way or inside floor openings, nor within 10 feet of an exterior wall which does not extend above the top of the material stored.

Fire Protection and Fire Prevention (Reference GMR Section 3.6 for additional AWS requirements)

1. All Contractors and their Subcontractors of all tiers shall be responsible to provide for the development of a fire protection program in their area of work. This program will supplement and complement the fire protection program for the overall jobsite (i.e. Hot Work Permit program).

AWS - ATL089 Procedures Manual

- Subcontractor is responsible for providing their own fire extinguishing equipment for hot work procedures/tasks. Fueling of generators, equipment and tools will also require fire extinguishing equipment to be provided.
- 2. Good fire prevention practices are to be observed throughout the jobsite. This shall include:
 - The observance of designated smoking and non-smoking areas.
 - **Smoking / vaping will be prohibited in enclosed buildings.**
 - The storage of combustible and flammable materials not to exceed OSHA or State/local requirements.
 - Good housekeeping practices.
 - Electrical wiring and equipment in compliance with the National Electrical Code and other applicable codes.
 - Clearance around portable heating units (refer to the manufacturer's recommendations).
 - Carbon monoxide monitoring shall be performed while using portable heating units.
 - Observance of minimum travel distances to portable fire extinguishing equipment.
- 3. OSHA requires that a **fire extinguisher** rated not less than 2A shall be provided for each 3,000 square feet of building area and placed by each stairwell level.
- 4. All welding or torch use shall have a dedicated fire extinguisher stationed in the work area.
- 5. Contractor shall provide a fire watch when performing welding, torching, grinding, and any other fire hazard operations. The fire watch durations will be specified on the project specific hot work permit. Required on all AWS Projects.
- 6. Ryan will set up an inspection by the local Fire Department to familiarize the Fire Department with the jobsite.
- 7. No material shall be stored to prevent the Fire Department access to standpipe system(s) on the jobsite.
- 8. A standpipe system(s) shall be constructed and located within one floor of newly constructed floors.

Flammable and Combustible Liquids (Reference GMR Section 3.6 for additional AWS requirements)

1. Flammable and combustible liquids shall be stored and dispensed in compliance with regulations and rules established by the governing regulatory bodies.
2. Any leakage or spillage of flammable or combustible liquids shall be cleaned up immediately and disposed of properly and safely.
3. No smoking, matches, or open flames will be permitted within 50 feet of any area where flammable liquids are used, transferred, or stored; be aware of conditions which may warrant an even greater clearance/distance.
4. Employees who conduct fueling must be authorized and properly trained.
5. The fueling area must be kept clean and included in stormwater inspections; SWPPP requirements must be followed. The fuel tank storage area must be shown on the site plan.
6. **Fuel tanks must be:**
 - Compliant with all state, local and OSHA requirements.
 - Clearly labeled for content with no smoking/open flames signage.
 - Equipped with a sign providing emergency contact information posted near the tank.
 - Double walled; secondary containment (110% must be contained) is required (either a dike or containment box). Proper disposal of secondary containment liquid is required (draining of liquid onto the ground is not allowed).
 - Grounded and protected from contact by vehicles on all sides.
 - Protected at the end of each day with restricted access and tank/dispenser secured.
 - Accessible for safe re-fueling and/or maintaining/obtaining fuel level measurements.
7. A spill kit is required next to any fuel tank. The subcontractor will be responsible for any/all clean-up related to a spill or leak from any fuel tank.

AWS - ATL089 Procedures Manual

8. A written spill plan is required. NO Fuel Tanks in ATVs/UTVs.
9. The tank must be put in a safe and sensible location (interior of site, not near property lines, away from high traffic, etc.) and in an area that has low runoff potential and/or could easily contain a spill.
10. Small quantities of flammable liquids which may be used at various locations on the jobsite shall be handled in approved, labeled safety cans. Only approved safety cans are allowed on-site. Approved safety cans must be UL approved and equipped with flame arrestors and self-locking latches. Fuel storage is prohibited inside any building.
11. An approved portable fire extinguisher shall be made available not less than 25 feet, nor more than 75 feet from any flammable liquid storage area(s).
12. Any fuel/chemical spills must be reported to the Ryan Environmental Department immediately. The subcontractor responsible for the spill will be responsible for all required reporting to Federal, State, or Local agencies.
13. Ryan site supervision and/or the Ryan Safety Department should be contacted for any additional safety requirements – additional requirements may be imposed as deemed necessary.

Liquefied Petroleum Gas

1. No storage of liquefied petroleum gas will be permitted within the building unless it is for immediate use, such as that required for a heater that is in use.
2. Storage of liquefied petroleum gas outside the building shall comply with all rules and regulations of the governing regulatory bodies.
3. All liquefied petroleum gas containers shall be protected from external damage and secured in an upright position.

Temporary Heating Devices

1. Ventilation shall be such so as to assure a fresh air supply in sufficient quantities to maintain the health and safety of the workers. Each Subcontractor shall be responsible for monitoring air quality as it affects their work and have a mitigation plan in place prior to the start of work.
2. All temporary heating devices shall be inspected by a qualified person and any faulty conditions shall be corrected prior to the device being used.
3. Adequate clearance must be maintained around the temporary heating device.
4. A fire extinguisher shall be positioned adjacent to each heating device.
5. All temporary heating devices, including weed burners, shall be equipped with an automatic shut-off device. (thermo couple)
6. All gas valves for temporary heaters and risers/branch lines that feed heaters shall be marked with high visibility material and shut-off valves must be labeled with emergency shut-off signage.
7. Kerosene/salamander heaters are prohibited from use on Ryan jobsites due to flammability of the fuel source. Pod/mushroom heaters are also prohibited. Both direct-fire forced-air heaters and indirect-fire heaters may be used (LP/natural gas).

Welding and Cutting (Reference GMR Section 3.6 for additional AWS requirements)

1. When transporting compressed gas cylinders, valve protection caps shall be in place.
2. Valve protection caps shall not be used for lifting cylinders from one level to another.
3. Compressed gas cylinders shall be stored and transported and used with the cylinder secured in a vertical position.
4. Cylinders containing oxygen or acetylene, or other fuels shall not be taken into confined spaces.
5. Storage cylinders shall be separated, capped, and secured as required by Federal, State and local agencies.

AWS - ATL089 Procedures Manual

6. All hose used for carrying acetylene, oxygen or other fuel gas shall be inspected at the beginning of each working shift. Defective hose shall be removed from service.
7. Regulators and gauges shall be kept in proper working order.
8. All arc welding and cutting cables shall be of the completely insulated flexible type capable of handling the maximum current requirements of the work in progress. Cables in need of repair shall not be used.
9. All ground connections shall be inspected to ensure that they are mechanically strong and electrically adequate for the required current.
10. Whenever practical, all arc welding and cutting operations shall be shielded **by non-combustible or flameproof screens which will protect employees and other persons working in the vicinity from the direct rays of the arc.**
11. Suitable fire extinguishing equipment shall be immediately available at all welding worksites and each Contractor shall develop a Fire Watch Program.
12. Adequate exhaust ventilation shall be maintained at all welding and cutting worksites.
13. Torch gauges shall be removed, and safety caps put in place at the end of the work-shift.

Rigging and Hoisting (Reference GMR Section 3.6 for additional AWS requirements)

1. Ryan jobsite supervision shall develop a job specific hoisting plan that designates ground personnel for crane operations.
2. Subcontractors shall designate personnel to perform hoisting operations or shall use Ryan's hoisting program.
3. All rigging shall comply with the 1926.251 Subpart-H OSHA standard.
4. A jobsite rigging storage area shall be designated.
5. All rigging shall be inspected daily to the manufacturer's standards and an inspection log maintained (Exhibit 8).
6. All damaged rigging shall be tagged and taken out of service. **(Do not allow employees to take damaged equipment home.)**
7. Rigging when not in use shall be removed from the work area and properly stored in rigging storage areas.
8. All material hoisting equipment (**picking beams, trash boxes, material cages, bolt boxes, torch cages, and material platforms, etc.**) shall be engineered and labeled with safe loading capacity and placement over beams.
9. No hoisting shall be allowed over the **public** or **traffic** without prior approval from Ryan site supervision.
- 10. A tag line shall be used on all loads.**
11. Crane operators will have **FINAL SAY** regarding making a lift.

Controlled Access (Reference GMR Section 3.5 and 3.6 for additional AWS requirements)

1. A controlled access area shall be established on floors that are not completed. A controlled access floor is determined by:
 - Guardrail or perimeter cable incomplete.
 - Decking installation on floor incomplete.
 - Decking installation on floor above is incomplete.
2. Signage shall be posted on floors by Gang ladders, at stair towers and in the personnel lift designating these floors.
3. Controlled access floors are limited to iron erection crews. If work activities warrant working on these floors, please contact Ryan site supervision for permission.
4. The controlled access designation to a floor will be removed when Ryan site supervision and iron erection supervision have reviewed floors.

AWS - ATL089 Procedures Manual

Operation of Mobile Earth-Moving Equipment

This section identifies minimum safety requirements for the safe operation of mobile earth-moving equipment used for earth moving, building, road construction or demolition, including, but not limited to, bulldozers, motor graders, scrapers, loaders, skid-steer loaders, compaction equipment, backhoes, end dumps, side-dumps, and dump trucks. This section pertains to operators of the equipment and exposed employees, including, but not limited to, grade checkers, grade persons, rod person, stake hops, stake jumpers, and blue toppers working in the area.

Training Requirements

Mobile earth-moving equipment operators and all other employees working on the ground exposed to mobile earth-moving equipment shall be trained in the safe work procedures pertaining to mobile earth-moving equipment and in the recognition of unsafe or hazardous conditions.

Training programs shall be developed and taught by competent individuals who have knowledge, training, experience and the demonstrated ability to identify existing and predictable hazards related to the subject matter.

Training programs must include the following elements:

1. Safe working procedures on how to approach mobile earth-moving equipment, whether in use or idling, including:
 - a) Visual, voice, or signal communication that shall be made with the operator prior to approaching earth-moving equipment.
 - b) Maintaining one's visibility to the operator while approaching the equipment; and
 - c) Operator responsibilities, such as placing the transmission in neutral, setting the parking brake, parking equipment side by side, and indicating that it is safe to approach the equipment.
2. Identification of the operator's blind spots in operating various types of earth-moving equipment.
3. Instruction for mobile earth-moving equipment operators regarding conducting daily equipment inspections according to the manufacturer's recommendations and checking the area around the equipment for a clear path prior to beginning operation. Daily equipment inspections must be provided to Ryan site supervision.
4. Safe operating procedures of equipment, including traveling, backing, parking, loading for transport, maintenance, and operation.
5. Safe work procedures when working around or adjacent to overhead or underground utilities as described in Code of Federal Regulations, title 29, parts 1926.600 (a) (6) and 1926.651 (b).
6. Additional hazards that could be created by changing conditions.

Employees shall be trained initially before beginning work that exposes them to mobile earth-moving equipment. Employee training records shall be retrained by the employer for the duration of the project.

ANSI Class 2 Compliant High Visibility Personal Protective Equipment (PPE)

A high visibility garment meeting ANSI Class 2 standards will be worn at all times by workers when exposed to moving equipment. This is a requirement on all Ryan projects in all States.

AWS - ATL089 Procedures Manual

Each employee working on the ground who is exposed to mobile earth-moving equipment shall be provided with and required to wear an ANSI Class 2 compliant high visibility warning vest or other ANSI Class 2 rated high visibility garment.

For work during hours of darkness or low light conditions of less than one foot-candle, this protective equipment must include ANSI Class 2 compliant retro-reflective material.

In Minnesota, high visibility apparel shall comply with the specifications in part 5207.0100, which is ANSI Class II. Employers must know the rules of the State they are working in.

All required PPE will be provided by the employer for the employee.

Equipment Requirements

All mobile earth-moving equipment shall comply with Code of Federal Regulations, title 29, part 1926.602 (a) (9) (ii) for back-up alarms or signal persons if applicable.

When mobile earth-moving equipment is operated during times of darkness or low light conditions, the equipment, if designed to function equally in both forward and reverse directions, such as compaction equipment, bulldozers, motor graders, loaders, and skid-steer loaders, shall be equipped with at least two headlights for forward travel and adequate rear lights for reverse travel unless other adequate lighting is provided.

Contractor Responsibility

If the mobile earth-moving equipment contractor exposes other contractor's employees to the hazard of mobile earth-moving equipment, the controlling employer, such as general contractor or construction manager for the project shall coordinate a joint contractor-employee safety awareness meeting between contractors and employees on site.

The employee safety awareness meeting shall be documented, identifying when the meeting was held and who attended, including a brief summary of what was reviewed. Documentation shall be retained for the duration of the project.

Trenching/Excavation Operations (Reference GMR Section 3.6 for additional AWS requirements)

General

1. Every employer that is trenching or excavating must have a properly trained competent person on site at all times.
2. All trenches shall be opened wide enough to achieve stable bank conditions. That is, slopes at least 1-1/2:1 in type C soil, 1:1 in type B soil, and 3/4:1 in type A soil, or,
3. If it is not possible to cut back to the angle of repose, all trenches 5 feet or more in depth shall use shoring or trench shields. **For the State of Washington, all trenches 4 feet or more in depth shall use shoring or trench shields.**
4. In trenches 4 feet or more in depth, ladders or steps shall be provided so that no employee will have to move more than 25 feet laterally to exit.
5. Ladders should extend at least 36 inches above ground and be secured.
6. Excavated and loose material shall be back a minimum of 2 feet from the edge of the cut.
7. Excavation contractor is responsible to call for utility locates before commencing any excavation work.

AWS - ATL089 Procedures Manual

8. Excavation around utility lines shall be done by hand or hydro-vac.
9. Blasting is frequently involved in trenching operations and shall be conducted in accordance with applicable regulations and codes.
10. Excavating equipment shall rest on a firm base away from the edge of the cut.
11. Spotters shall be used to direct truck movements in and around the area.
12. Stop logs or barricades shall be used where there is a danger of vehicles approaching or equipment being placed too near the edge of a trench or elevation change.
13. Where possible, trench should be opened in lengths in which work can be completed and the trench closed at the end of the day.
14. When trenches are left open overnight, they shall be barricaded or covered with suitable materials and lighted during non-working hours. Means to eliminate accumulated groundwater or rainwater shall be available in order to maintain a safe trench or excavation.
15. Care should be taken to prevent employees from working too closely together to minimize the danger of being struck by tools or materials. 12 foot spacing is recommended.
16. Tests are to be conducted for hazardous atmospheres and adequate precautions shall be taken to prevent employee exposure to atmospheres containing less than 19.5% oxygen and other hazardous atmospheres.
17. Emergency rescue equipment, such as breathing apparatus, a safety harness and line, or a basket stretcher, shall be readily available where hazardous atmospheric conditions exist or may reasonably be expected to develop during work in an excavation. This equipment shall be attended when in use.
18. Any excavation that is over 20 feet in depth, the cave-in protection system shall be designed by a licensed professional engineer. The design documents shall be kept on-site during the erection or construction of the system.
19. Frequent inspection shall be made to make certain that shoring, bracing, and sheeting are in good condition. Daily inspection of soil conditions should also be made due to weather changes, by the competent person.

Masonry

The following are general safety requirements for masonry work:

1. Limited Access Zone: Before work is started on a masonry wall, a limited access zone is required to be established. The zone shall run the length of the wall and shall extend out from the base of the proposed wall the height of the wall plus 4 feet. Therefore, a proposed wall 50 feet long by 20 feet high would have a limited access zone 50 feet long by 24 feet wide.
2. This access zone shall be located on the side of the wall that does not have scaffolding.
3. This access zone shall only be occupied by workers who are actively engaged in constructing the masonry wall. No other work can be performed while the wall is being built within the zone limits.
4. When the masonry wall passes 8 feet in height, bracing shall be installed to prevent overturning or collapse, unless there exists a means of supporting the wall without bracing (such as an adjoining wall). Bracing installed shall remain in place until other permanent supporting elements such as adjoining walls or roof structures are in place. See article 5.19 of the Ryan Subcontract.
5. When the masonry wall is properly braced, the limited access zone may be withdrawn.

While there are no specific references to training in this section, safety training and discussion in the referenced regulations is necessary before working on concrete or masonry construction.

Tilt Panel Construction (Reference GMR Sections 3.5 and 3.6 for additional AWS requirements)

Panel Layout and Placement

AWS - ATL089 Procedures Manual

A site usage and layout plan should be done for each project.

Panel embeds such as brace points and pick points shall be attached to the steel reinforcement in such a manner as not to come loose or collapse under pressure from individuals standing on them at the time of concrete placement.

Tilting and Hoisting

Panic plates (bolt on plates for lifting) **WILL NOT** be permitted for safety reasons; therefore, proper supervision from the Concrete Subcontractor shall apply. If conditions require the use of a panic plate(s) it will need approval of the Ryan Field Coordinator, the Structural Engineer for the project, and the manufacturer or engineer of the lifting and bracing hardware supply company.

Prior to the start of the first lift, a safety orientation meeting including the concrete subcontractor(s), the erection company, and the crane company/operator is required. If new personnel are added at any time to the crew, the meeting will need to be repeated. In addition, a daily panel erection meeting is required to provide organization and awareness prior to the start of each day's lifting/erection of tilt-up concrete panels.

Observation of safety violations by the lifting crew will immediately require a safety meeting by the Concrete Subcontractor with Ryan personnel present.

ANSI Class 2 compliant (high visibility) safety vests will be required for the lifting crew. Any personnel not wearing a safety vest are restricted from the lift and crane zones and the immediately surrounding areas.

Crane Safety During Tilt Operations

Crane records are required pre-lifting and daily reports throughout the duration that the crane is on-site. Faulty crane equipment will automatically require that a crane stop work for repair and/or replacement of faulty equipment. All other regulations as prescribed in the CFR part 1926 will apply to crane operations (see [Daily-Monthly Crane Safety Checklist](#)- Ryan Safety Manual/Procure Forms).

Bad Weather Policy

1. All work activities will stop on floors not enclosed when wind speeds are determined by Ryan site supervision to be at unsafe levels.
2. The Ryan Lightning policy requires a work shutdown whenever lightning is detected within 15 miles of the project. Work may only resume at least 30 minutes after the last recorded lightning strike on radar. Competent person from subcontractor(s) working on the project are responsible for monitoring hazardous weather conditions, including the potential for lightning, and notifying their crews accordingly.
3. Site supervision should download a free electronic weather application on the company smartphone (such as NOAA Radar Pro, which is free for both iOS and Android operating systems) to be alerted to any impending weather conditions which may impact the project/jobsite location.
4. Subcontractors will designate personnel to secure materials and equipment on floors not enclosed.
5. Crane activity will stop until hazardous weather has ceased.
6. If air quality conditions are in question, check with your local Environmental Protection Agency (EPA).

Cold Stress Policy

AWS - ATL089 Procedures Manual

When the outdoor weather conditions and/or temperature is more than -10 degrees below zero (Fahrenheit) and/or the windchill factor makes the “feels like temperature” -30 degrees below zero or colder, work shut down will be at -10 degrees below zero Fahrenheit and/or -30 below zero or colder wind chill. Reference the Wind Chill Chart provided below to determine wind chill conditions. Conditions to the right of the red line represent work shut down conditions.

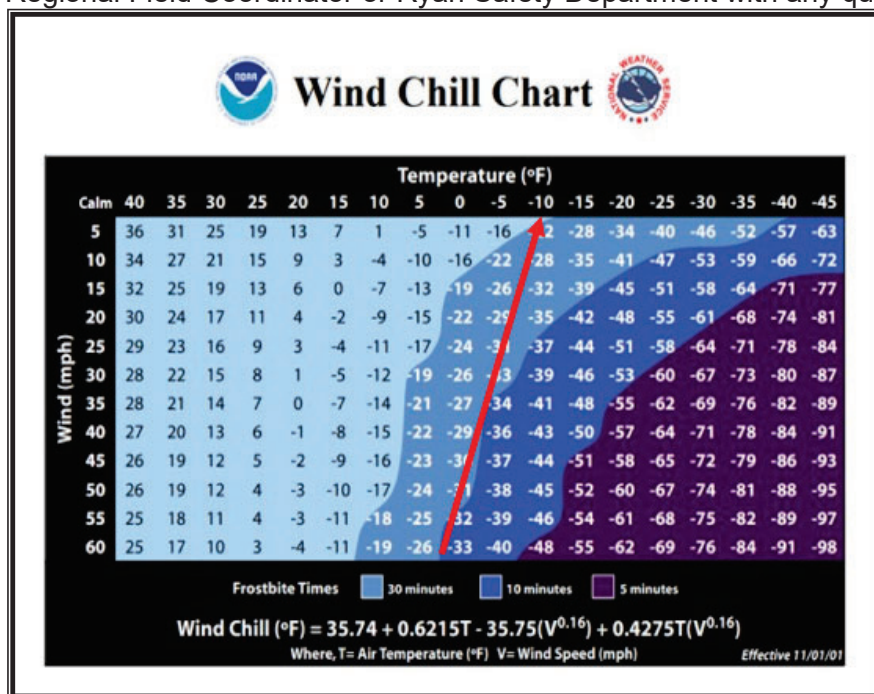
Hydraulic/heavy equipment (cranes, lifts, forklifts, skid steers, etc.) should not be started at -5 degrees below zero (°F) or colder; see the manufacturer’s recommendations for cold weather operation.

During any extreme cold weather events, steps must be taken to protect outdoor workers from the conditions such as:

- Staggering start times, monitoring personnel and scheduling periodic warm-up breaks
- Know the signs of frostbite or hypothermia
- Dress for the conditions; minimize skin exposure
- Consider special JHA or pre-task analysis for contributing cold factor(s)
- Hydraulic/heavy equipment may be started if stored prior in warm conditions/enclosures and operated outdoors per manufacturer’s recommendations. See chart below.

Equipment Type	Ambient Temp	Operations Recommendation
Tower Crane	-13 below or colder	Do not operate unless Mfg recommendations specify otherwise
Manlift	-13 below or colder	Do not operate unless Mfg recommendations specify otherwise
Other Hydraulic Equipment	-5 below or colder	Do not operate unless Mfg recommendations specify otherwise

Please contact your Regional Field Coordinator or Ryan Safety Department with any questions.



Heat Stress Policy

AWS - ATL089 Procedures Manual

This program is designed to reduce the risk of work-related heat illnesses and applies to all work performed in any location/region. **Precautions must be taken at all jobsites to prevent heat related illnesses, making adequate provisions for water, rest, and shade for all workers.**

Washington, Minnesota, and California have specific laws governing occupational heat exposure. Federal OSHA **does not** have a heat stress standard, however the General Duty Clause (Section 5[a][1] of the Occupational Safety and Health Act of 1970) requires employers to provide a place of employment that is “free from recognized hazards that are causing or are likely to cause death or serious physical harm to employees.”

- **California.** California’s Heat Illness Prevention Standard requires employers to provide training, water, shade, and planning. A temperature of 80°F triggers the requirements. See [CalOSHA's website](#). See the full text of the [California heat standard](#). Work performed outdoors in California must comply with the Heat Illness Prevention Standard T8 CCR 3395.
- **Minnesota.** The standard applies to **indoor** places of employment with respect to air flow and temperature. See the [full text of the regulation](#).
- **Washington.** See Washington State’s [Outdoor Heat Exposure Rule](#). See the [full text of the regulation](#).

See the Ryan Safety Program for the full Heat Stress Policy.

Add the OSHA-NIOSH Heat Safety Tool - **Heat Index** app to your mobile phone:

<https://apps.apple.com/us/app/osha-niosh-heat-safety-tool/id1239425102>

Jobsite Deliveries (Reference GMR Section 3.28 for additional AWS requirements)

1. All deliveries shall be coordinated through Ryan site supervision.
2. All delivery personnel shall wear proper attire – long pants, shirts and proper work boots.
3. All delivery personnel shall wear a hardhat and safety glasses.
4. All deliveries should be coordinated to prevent blocking of traffic.
5. All personnel helping to direct delivery truck outside of the job-fence shall wear an ANSI Class 2 compliant high visibility vest.

Radio Program

Ryan site supervision shall develop a radio communications program. The following is a list of assigned channels:

1. Main channel	4.	7.
2. Layout channel	5.	8.
3.	6.	9.

Steel Erection (Reference GMR Sections 3.5, 3.6 & 3.28 for additional AWS requirements)

This section sets forth the steel erection requirements of Ryan Companies US, Inc., as the General Contractor. Steel Erection contractors are required to comply with the 1926.750 Steel Erection Standard and any site-specific Ryan Companies US, Inc. regulations.

Duties of Controlling Contractor

1. A steel erection contractor shall not erect steel unless the steel erection contractor has received written notification that the concrete in the footings, piers and walls, or the mortar in the masonry piers and walls has attained, on the basis of an appropriate ASTM standard test method of field-cured samples, either

AWS - ATL089 Procedures Manual

75% of the intended minimum compressive design strength or sufficient strength to support the loads imposed during steel erection. Ryan Companies US, Inc. shall be responsible to send this written notification (Exhibit 10).

2. Ryan Companies US, Inc. shall be responsible to provide and maintain adequate access roads into and through the jobsite for the safe delivery and movement of derricks, cranes, trucks, and other necessary equipment and material to be erected, as well as means and methods for pedestrian and vehicular control.
3. Ryan Companies US, Inc. shall be responsible to provide a firm, properly graded, drained area readily accessible to the work with adequate space for the safe storage of materials and the safe operation of the erector's equipment.
4. Anchor bolts shall not be repaired, replaced or field-modified without the approval of the project's structural engineer of record. Also, prior to the erection of a column, Ryan Companies US, Inc. shall provide written notification to the steel erector if there has been any repair, replacement or modification of the anchor bolts of that column (see Exhibit 10).
5. Ryan Companies US, Inc. will coordinate overhead protection during steel erection through use of the steel erector's site-specific erection plan.
6. The fall protection perimeter safety cable provided by the steel erector shall remain in the area where the steel erection activity has been completed, only after Ryan Companies US, Inc. has inspected and accepted control and responsibility of this cable prior to authorizing persons other than steel erectors to work in the area (see Exhibit 10).
7. On multi-story construction, columns shall be stood on the ground, hoisted vertically, and set in place during steel erection operations.

Environmental Issues

1. Ryan site supervision and project management shall communicate directly with the Ryan Environmental Specialist and assess the project for any known or potential environmental issues prior to the start of the project. Any and all environmental issues relative to this project should involve the Ryan Environmental Specialist, Jonathon Blaha at 612-492-4430.
2. In the event environmental concerns have been identified, please see #5 below, and/or any attached addendum to this AWS - ATL089 manual for further information.
3. Please be aware that although environmental issues may not be known at present, the potential to encounter environmental contamination exists at all project sites. In this case, contact the Ryan Environmental Specialist as listed above.
4. Environmental addendums or exhibits may be added to this site-specific Project Safety Manual as needed.
5. Additional project-specific environmental notes: Low level impacted soil and groundwater may be present at the site along the western site boundary adjacent to the Georgia Plating facility.
6. The owner retained WSP as its environmental consultant for the project. WSP has been involved with the environmental investigation activities at the site and is overseeing all environmental issues. WSP's investigation documented contamination in the soils and groundwater. WSP prepared a Phase I Environmental Site Assessment (Phase I ESA) dated July 31, 2023 which identified the following Recognized Environmental Condition (REC) for the site – the Georgia Plating facility adjacent to the property on American Way which has operated as a plating facility with current operations including zinc electroplating and E coating. An irritant odor was apparent while walking the southern boundary of that property during the Phase I ESA. WSP prepared an Limited Phase II Environmental Site Assessment (Phase II ESA) dated September 22, 2023 which identified low level soil impacts within soils adjacent to the plating facility, and impacts in site groundwater in the vicinity of the plating facility. At the time of the Phase II ESA, none of the soil or groundwater impacts exceeded regulatory levels. Soil impacts consisted of volatile organic compounds, and per and polyfluoroalkyl substances (PFAS) compounds, and groundwater impacts consisted of metals and PFAS compounds. A copy of the environmental

AWS - ATL089 Procedures Manual

reports will be kept at the job site and/or made available by contacting Ryan's Job-Site Manager or Ryan's Environmental Specialist named in item 1 above.

7. In the event that unanticipated impacts are noted during construction, STOP work immediately in the area and notify the Ryan Job-Site Manager.
8. Workers are advised to be observant and on the lookout for obvious signs that impacted materials have been encountered or unearthed. Those signs may include:
 - Previously unidentified USTs;
 - Strong or unusual chemical odors;
 - Obvious physical signs of industrial or other wastes, including ash, cinders, tars, sludges, powders, resins, liquids or containers (i.e. drums, pails, etc.);
 - Unlabeled drums or containers;
 - Buried metal objects such as tanks and ground water production wells;
 - Above ground metal objects such as vent pipes;
 - Buried building materials that may contain asbestos containing materials (ACM); and
 - Co-workers who suddenly become ill on the Property.
9. Please notify Ryan's Job-Site Manager prior to starting any work on the project so that construction and environmental concerns can be properly addressed.

Spill Response

The following are procedures to follow if a spill/release is discovered at a Ryan owned or managed Facility:

1. Stay calm.
2. **Call 911** if there is a threat to life or property.
3. **Report the spill immediately to Jon Blaha or Steve Banovetz of Ryan Environmental or a Terracon or Stantec representative listed below.** You must make contact with one of these individuals. They will handle notification to regulators and clean up contractors, if needed.

Contact	Office	Cell	Home
Jon Blaha (Ryan)	612-492-4430	612-735-7610	763-545-8714
Steve Banovetz (Ryan)	612-492-4873	651-356-1627	651-773-3185
Mark Miller (Terracon)	651-600-3309	612-209-3852	651-688-0735
Mike Willey (Terracon)	763-489-3126	612-269-3439	612-269-3439
Aaron Benker (Stantec)	763-479-5132	612-719-3282	612-719-3282

In the event you cannot reach Ryan Environmental, Terracon, or Stantec, Go to STEP 12 BELOW

4. Define the problem:
 - leaking valve
 - ruptured hose
 - hole in tank
5. Stop the source of the release - Only if you can do so safely:
 - shut off pump
 - close valves
6. Eliminate sources of ignition:

AWS - ATL089 Procedures Manual

- shut off motors and engines
 - no smoking
7. Quickly analyze and assess the situation:
 - Where will the spill go?
 - What problems might result?
 8. Site Control: Keep people away from spill and affected areas
 9. Attempt to contain the spilled material - Only if you can do so safely:
 - Place a drip pan or bucket under a leak in a valve
 - Use sorbent material on small spill (absorb socks, floor dry, kitty litter, pads, sand)
 - Block or dike any nearby drains or other pathways
 10. Clean up the spill/release using on-site equipment if spill is small and manageable.
 11. If the spill/release is beyond your ability to manage properly, a qualified Clean-up Contractor must be retained. Clean-up Contractors provide 24-hr. services including: contain and recover spills/releases on land and to surface waters; mitigate petroleum vapor and product release to sewer systems; able to handle cleanup of unknown material/product; access to containment booms, sorbents, vacuum truck, and pumps for spill/release clean-up. The following is a partial listing of Clean-up Contractors available in your region.

Contractor	Phone Number	Coverage Area
Clean Harbors	1-800-645-8265	Nationwide
BELFOR Environmental	1-800-930-0011	Nationwide
EnviroServe	1-800-488-0910	Great Lakes, Minnesota, Iowa, Denver, Kansas City, Tampa, Phoenix, Seattle
HEPACO	1-800-888-7689	Eastern US
ERI	1-480-967-2802	Arizona
SWS Environmental	1-877-742-4215	Florida, Texas
AMPOL	1-800-482-6765	Gulf Coast
Bodine	1-800-637-2379	Illinois, Iowa
Seneca	1-800-369-5000	Iowa
Bay West, Inc.	1-800-279-0456	Minnesota
Ocean Blue	1-800-990-9930	Southern California
National Response Corp. (NRC)	1-800-899-4672	Washington

12. **As identified in #3 on Page 1, report the spill immediately to Ryan Environmental, Terracon, or Stantec** who can handle notification to regulators, if needed. **In the event that you cannot reach Ryan Environmental, Terracon, or Stantec, then notify your state emergency agency listed below** (note this list covers states with active Ryan projects).

State	Phone Number	When to Report
Arizona	1-602-771-2330	Immediately
California	1-800-852-7550	Immediately
Colorado	1-877-518-5608	Within 24 hours
Florida	1-800-320-0519	Immediately
Georgia	1-800-241-4113	Immediately or 24 hours for petroleum

AWS - ATL089 Procedures Manual

Iowa	1-515-281-8694	Within 6 hours
Illinois	1-217-782-3637	Immediately
Indiana	1-888-233-7745	Within 2 hours
Kansas	1-785-291-3333	Immediately
Louisiana	1-225-925-6595	Within 1 hour
Minnesota	1-651-649-5451	Within 24 hours
Missouri	1-573-634-2436	Earliest practical moment
New Mexico	1-505-827-9329	Immediately
North Carolina	1-800-858-0368 or 919-814-0375	Immediately
Ohio	1-614-224-0946	Within 30 minutes
Oregon	1-800-452-0311	Immediately
South Carolina	1-888-481-0125	Within 24 hours
Tennessee	1-800-262-3300	Immediately
Texas	1-800-832-8224	Immediately
Washington	1-800-258-5990	Immediately
Wisconsin	1-800-943-0003	Immediately

You will need the following information:

- Name, address, and telephone number of the person reporting and the responsible party;
 - Specific location of the incident;
 - Date and time the incident occurred or was discovered;
 - Name of the chemical/material released;
 - Source and cause of the release;
 - Total quantity discharged;
 - Medium into which the substance was discharged;
 - Amount spilled into water;
 - Weather conditions;
 - Name of the carrier or vessel, the railcar/truck number, or other identifying information;
 - Number and type of injuries or fatalities;
 - Whether an evacuation has occurred;
 - Estimation of the dollar amount of property damage;
 - Description of current and future cleanup actions; and
 - Other agencies notified or about to be notified.
13. If the spill consists of oil or an oil product and it contacts a "Water of the United States", the spill must be reported to the National Response Center in addition to the state agencies noted in #12 above.

US EPA National Response Center	1-800-424-8802
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14. If necessary and depending upon the magnitude of the spill, you may need to call the following Ryan personnel to report the spill:

John Gaddini – Corporate Safety Director	(612) 919-8639
Gidget Wegener – Sr. Director, Risk Management	(612) 492-4411
Laura Graf – Chief Legal Officer (if necessary)	(612) 492-4440
Colleen Kucera – SVP, National Marketing Leader (if necessary)	(602) 322-6154

AWS - ATL089 Procedures Manual

Do Not Interrupt Adjacent Systems, Utilities, Businesses, or Structures (Reference GMR Section 3.6 for additional AWS requirements)

Ryan site supervision and Subcontractor management should evaluate jobsite operations that could affect surrounding businesses and owners, i.e. powerlines, fiber optics, gas lines, etc.

Job Safety/Hazard Analysis (Exhibit 3) should be completed prior to the start of work activities.

- Will work activity involve existing service or utility that could potentially affect an up or downstream user or any other system or operation in the area?
- Does work activity involve shut-off, alteration, removal or blocking, or disconnection of an alarm, smoke, fire, heat detector; sprinkler*; FM200, or protection system?
- Will work activity involve hot work such as welding, cutting, grinding, air arcing, brazing, soldering, or touch-work? Is the Hot Work Permit in place?
- Will work activity include core drilling that could affect service?
- Are contact personnel identified and/or designated in the event of a shut down?
- If sprinkler systems are “live” while working, please be advised that a “shut-gun” or similar device should be on hand in order to close any open sprinkler heads and/or stop a leak.

Mobile Elevating Work Platforms (MEWPs) (Reference GMR Sections 3.6, 3.28 and 3.29 for additional AWS requirements)

***Note: Plan ahead. Utilize the proper lift in the appropriate configuration in order to allow the worker to reach the point of work.**

1. All operators and occupants of Mobile Elevating Work Platforms (MEWPs) must wear a full-body harness equipped with an ANSI Class 2 Self-Retracting Lifeline (SRL). The SRL must be secured to the manufacturer-approved anchorage point(s) on the boom or platform, immediately upon entering the lift.
 - a. ANSI/ASSP Z359.14 Class 2 Self-Retracting Lifelines (SRLs) are required for fall arrest in MEWPs. Lanyards or shock absorbing lanyards are not permitted.
2. MEWPs must be equipped with a self-closing mechanism entry gate (meeting ANSI A90.20 standard requirements)
3. Employees must keep both feet on the platform at all times while working within the lift.
4. All vertical mast lifts and Scissor lifts shall be equipped with a Perimeter Safety Zone Warning Light System that projects a visible colored light pattern on floor, which acts as a warning to personnel working in close vicinity of an active scissor lift. The use of this system does not eliminate the need for ground spotters.
5. All Mobile Elevating Work Platforms must have proof of a current annual inspection posted on the machine.
6. Anyone who operates a Mobile Elevating Work Platforms must be trained by that person’s employer on how to run the machine, how to recognize an unsafe machine and how to remove it from use.
7. All Mobile Elevating Work Platforms must have an owner/operator manual on the machine.
8. Aerial lifts must be inspected by the qualified operator prior to each use with the inspection checklist on-hand at all times while the lift is in use.
9. Changes may not be made to MEWPs equipment without prior written approval received from the manufacturer.

Example: Perimeter Safety Zone Warning Light System

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Permit Required Confined Space Entry (Reference GMR Sections 3.6 and 3.28 for additional AWS requirements)

Because confined spaces are potentially dangerous, OSHA requires that employers must evaluate all confined spaces in which their employees work to determine whether hazards exist or whether the work to be done in the space can create hazards.

If a confined space contains an actual or potential hazard or could otherwise interfere with a worker's ability to leave the space in an emergency, it is a "permit-required confined space", or "permit space".

Employers must take certain precautions whenever workers enter a permit space. These include 1) specifying the precautions to be taken to protect the workers in the space; 2) training the workers who are covered by the standard to give them the knowledge to protect themselves and others; and 3) planning how to rescue injured workers promptly and safely. OSHA requires workers to have a permit to enter these spaces.

Each permit space is unique and a JHA evaluation must be performed and submitted to Ryan site supervision prior to entry.

1) Existing permit space:

Ryan site supervision will inform subcontractors of known permit-required spaces and corresponding entry requirements by posting "danger permit required confined space" signage and will identify such permit spaces during pre-construction and/or jobsite orientation(s), including any known hazards and any precautions or procedures to be followed when in or near permit spaces.

Ryan site supervision will coordinate entry operations with affected subcontractors, as well as when permit space entry occurs during other activities at the jobsite that might create a hazard in the space.

Subcontractor's competent person (including any tiered subcontractors) must inform Ryan site supervision of the subcontractor's confined space program followed and any hazards encountered in permit spaces (see #9 of Exhibit 2). **All employers must ensure their workers are properly trained and informed of the confined space requirements as outlined in 29 CFR 1926.1200, Subpart AA.**

2) Permit Space to be Constructed:

Subcontractors (including any tiered subcontractors) must notify Ryan site supervision when any permit space (temporary or permanent) is created in the course of construction (see #9 of Exhibit 2).

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Subcontractors' (and/or tiered subcontractors') competent person must identify any/all permit-required confined spaces and notify Ryan Site Supervision of such location(s) prior to any work activity which may be required to take place therein as defined by 29 CFR 1926.1200 Subpart AA.

Subcontractors who do not have confined space workers must make sure their workers stay out of any permit spaces present on the jobsite unless the workers are authorized for entry. No person shall enter a permit-required confined space without authorization by their employer and/or Ryan site supervision.

Respirable Crystalline Silica

Exposure to respirable crystalline silica can cause silicosis, lung cancer, other respiratory diseases, and kidney disease. Exposure can occur during common construction tasks such as using masonry saws, grinders, drills, jackhammers, and handheld powered chipping tools; operating vehicle-mounted drilling rigs; milling; operating crushing machines; and using heavy equipment for demolition or certain other tasks.

Employers can either use a control method laid out in Table 1 of the construction standard or they can measure workers' exposure to silica. Note: if Table 1 is not being used, objective data from air sample testing for each task must be available at the jobsite and provided upon request. Regardless of which exposure control method is used, all construction employers covered by the standard are required to establish and implement a written exposure control plan for each task that involves exposure, including methods used to protect workers and procedures to restrict access to work areas where high exposures may occur.

Prior to commencing any work, silica exposure control plan(s) must be turned in to Ryan field supervision for each task generating respirable silica dust.

Designate a competent person to implement and enforce the written exposure control plan.

Restrict housekeeping practices that expose workers to silica where feasible alternatives are available. No dry sweeping or compressed air as per the OSHA standard on silica. Subcontractors generating any silica dust must control, collect, and properly dispose of all silica material.

Train all of your workers in the recognition, avoidance, and control of crystalline silica as per the OSHA standard 29 CFR 1926.1153. See Exhibit 15, Silica Exposure Plan.

Use of UAS / Drone(s) - Not allowed on Mission Critical Projects

If Subcontractor utilizes drones in connection with its work, Subcontractor shall comply, or cause its subcontractor to comply, with the requirements of Exhibit 1 - Article 9 - Section 9.4.11.

Photograph Policy

The Contractor is not allowed to take any photos within the OZ. If photos are required, the Contractor will submit a written request to the Construction Manager describing the request and justification. The Construction Manager will then request approvals through the AWS approval process. This process may take up to 2 weeks and approval is not guaranteed.

Photos may be taken within the Construction Zone (CZ) prior to PFHO. The following requirements must be followed prior to, during, and after the photos are taken:

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1. Photos/videos/thermal images are only allowed to be taken within the CZ.
2. Photos/videos/thermal images are only allowed to be taken by the following Amazon employees: Construction Managers, Critical Projects Engineers (CPI), Commissioning Engineers, Controls, and Design Engineering. Additionally, the following vendors are allowed to take photos/videos/thermal images: General Contractor Superintendent, Amazon Commissioning Agent, Controls vendors, CPI vendors, owner-requested testing agencies, long lead equipment vendors, and Design Engineering consultants.
3. Photos/videos/thermal images are only allowed to be taken PRIOR to the PFHO milestone. No pictures will be taken down the network aisles where the network racks are placed at NEA and no pictures will show a rack (network rack, patch rack, or any service rack).
4. Photos/videos/thermal images are only allowed to be taken to document construction related issues requiring resolution, such as engineering discrepancies, RFI's, potential design changes, equipment issues, punch-list items, field conditions, and for providing photographic/video validation/proof of installations/testing/acceptance of equipment and/or building scope completion.
5. Photos/videos/thermal images can only be taken with cameras WITH NO geospatial and wireless capability that only use media/storage capability, and no features for photo sharing.
6. Phones cannot be used to take photos.
7. Photos/videos/thermal images are to be limited with no visibility of the OZ, portions of the live buildings, or anything that would give an indication of location.
8. No photos/videos/thermal images will be taken that show any of the following in its entirety: AMCOP panels, BACOP panels, BOB (BACOP Output Box) or close-up images of their HMIs.
9. Photos/videos/thermal images will focus only on the item needing a photo, no wide-angle shots are allowed.
10. Photos/videos/thermal images are only to be shared on an as-needed basis and by only the AWS Construction Manager, blue badge Commissioning engineer, or blue badge Controls Manager.
11. MVP builds, including but not limited to APA, LCK, ALE, NCL, and DCA require prior photo approval by the AWS Cluster Security Manager for any photos regardless if they are in the CZ or before PFHO.
12. All photos that will be used in reports, meetings, or for project coordination purposes must have the stamp "Amazon.com Confidential" placed on them.

Camera Details:

- The CM will provide a camera to authorized users upon request. The user will return the camera to the CM at the completion of the project.
- Must be a standalone, memory card type, and with NO geospatial and wireless capability.

Picture/Video Disposition:

- All Photos/videos/thermal images are to be downloaded from the memory card daily and will only be stored on WorkDocs with the defined folder structure and access controls that are already established for sharing construction/commissioning related documents/files with the contractor and associated partners.
- GC and CM to work together to support Design Team/Consultants when on site with uploading of photos to WorkDocs after site inspections/visits.

Camera Sanitization:

- Cameras will be sanitized within 24 hours of photography/videography event and between users.

Thermal Imaging Requirements:

- Amazon Commissioning Engineers (ACx) and Commissioning Authority (CxA) are allowed to take thermal images of the equipment under load during Commissioning to comply with Commissioning Specification and Busduct Plan as a proof of installation/testing/acceptance of equipment.
- CxAs are allowed to use their own IR cameras after being reviewed and approved by the Amazon

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Commissioning Engineer (CxE). Wireless, geospatial and photo sharing capabilities of IR Camera must be disabled on-site if IR camera is equipped with these features.

- All thermal images and photos captured during commissioning are to be downloaded from the IR camera memory card and will only be stored on WorkDocs/Procore with the defined folder structure and access controls that are already established for sharing commissioning/ construction related documents/files with the contractor and associated partners. The camera will be sanitized (all photos deleted) prior to being taken off site.
- IR Thermal Images that will be used in reports must have the stamp "Amazon.com Confidential" placed on them.
- All personnel supporting Thermal Imaging in the Construction space shall comply with rest of the requirements of CZ photo policy.

Project Specific Tool Tethering Policy (Dropped Object Projection)

Accident prevention is essential to the successful completion of any project. Ryan Companies' implementation of tool tethering requirements will be determined by field management on a project-by-project basis.

In general, to help ensure that workers, pedestrians and passing vehicles are safe from harm; if they cannot be separated by a safe distance horizontally, all workers are expected to cooperate and comply with the following safety protocols:

1. Hard hat, hand tools and power tools must be tethered when:
 - a. Working from any height and within 15' of the perimeter of the building
 - b. Working within 15' of any interior shaft-way(s)
 - c. Working from an aerial lift
2. Make sure that the correct tethers are being used for the tool you are tethering. Each tether has a max load designation on the tag, which should correspond with the weight of your tool.
3. Do not tether any tools heavier than 5 lbs. to yourself. Hand tools (such as hammers, Klein's (pliers), measuring tape, etc.) can be tethered to your belt or harness; power tools should be tethered to the perimeter cable, guardrails, or a fixed member of the structure.
4. All hand tools in your tool pouch must be tethered, even when not in use. If possible, remove any tools from your tool pouch that are not necessary for your current task.
5. If you are working over a high traffic area (even inside the building), barricade or demarcate the areas below your work with appropriate signage and tether your hard hat, hand tools and power tools.
6. If possible, materials should also be secured/tethered. When installing/removing material at or near the building perimeter or any shaft-way, tether or clamp the material to a fixed structure prior to removal and/or installation. Make sure the area below your work area is protected/barricaded/demarcated accordingly. All employees are expected to follow these protocols at this jobsite. Please bring any safety concerns or issues to your supervisor or jobsite supervision.

Use of ATVs / UTVs (Reference GMR Section 3.28 for additional AWS requirements)

Any person who operates an ATV (single passenger) or UTV (side-by-side/multiple passengers) on Ryan jobsites must be trained to safely operate their specific machine, including conducting regular inspections and maintenance as may be required prior to use. **Maximum 10 MPH speed limit enforced on all jobsites.**

Acknowledgements of training, inspection, and maintenance requirements are included on Exhibits 2 and 4 of this Project Safety Manual. See Exhibit 12 for ATV / UTV Use Checklist.

AWS - ATL089 Procedures Manual

Rebar Protection and Other Protrusion Hazards

Any protrusion which presents an impalement hazard must be protected, including rebar, form stakes, conduit, all-thread on roof drains, wall anchors, posts, and column anchors, etc.

There are two common types of accidents associated with exposed rebar and other protrusions, they are impalements and abrasions.

Exposed vertical or horizontal rebar must be protected upon installation and subcontractor shall not leave the work area unprotected. The work area must be barricaded/marked with signage, and access-restricted to protect from impalement hazards.

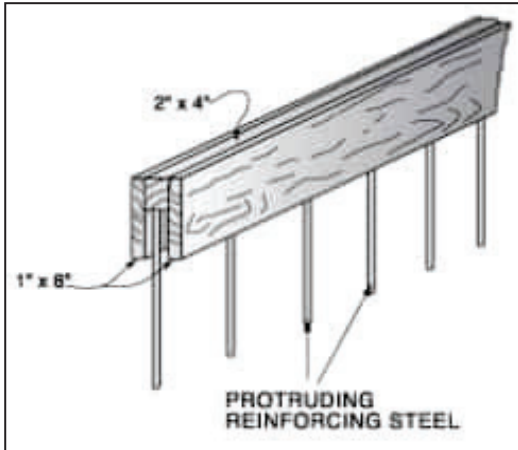
Protection of horizontal and vertical protruding rebar can be provided in many ways, using rebar caps, wooden troughs, Carnie Caps, or bending the rebar.

- Bend rebar so exposed ends are no longer upright.
 - “Candy canes” are an acceptable means of protection when engineering allows for this type of protection/guarding.
- When employees are working at the height above exposed rebar, provide fall prevention (ie guardrails).
- Guard all protruding ends of steel rebar with:
 - **Steel Plate Reinforced Rebar Caps** (required for impalement protection)
 - **Note:** For **California** refer to 1712 [Requirements for Impalement Protection](#).
 - Plastic mushroom caps **are prohibited** from use.
 - A single wooden 2"x4" is not acceptable as a means of impalement hazard protection on vertical rebar.
 - **Wooden Troughs** (see diagram below) Long 2 x 4 wood caps or other manufactured troughs can be used to effectively protect exposed rebar.
 - **Carnie Caps*** (see diagram below) Carnie Cap is another effective way to cap exposed rebar and reduce the danger of impalement. In this arrangement, two caps are placed every eight feet and fixed wooden plank spans between the two caps to protect against impalement. It can be used in horizontal, vertical, and inclined applications. (*Do not use Carnie Caps in California)
- Strictly limit the access to the work area where formwork, protruding rebar, and rebar cages are present.

Wooden Caps / Troughs

Carnie Caps

AWS - ATL089 Procedures Manual



Daily Roof Access Permit – Procedure (Reference GMR Sections 3.6 and 3.12 for additional AWS requirements)

PURPOSE: To establish a “ROOF ACCESS PERMIT” procedure and clearly define “SAFE ROOF WORK PRACTICES” for project sites. (See Exhibit 13 included in this Manual)

SCOPE: These procedures apply to any work to be performed on the roof.

GUIDELINES: At the beginning of each workday, subcontractors intending to perform roof work activities requiring roof access shall have their competent person notify Ryan’s Site Supervision or designated Safety Coordinator of all intended activities and the location(s) of such activities and complete a Roof Access Permit (see hard copy or Procore Forms).

Each subcontractor’s competent person will be responsible for securing the proper permit(s) from Ryan’s Site Supervision prior to beginning any roof work activities.

Ryan’s Site Supervision and/or Ryan’s designated safety coordinator is responsible for administering and ensuring compliance with the submitted roof access permit.

OBTAINING PERMITS: The following procedures shall be followed for obtaining and closing out daily roof work activities/permits:

- Each subcontractor’s Competent Person shall notify Ryan’s Site Supervision prior to the beginning of each work shift.
- Ryan’s Site Supervision shall issue a permit once the appropriate information has been obtained and documented within the “Roof Access Permit”.
- Ryan’s Site Supervision will maintain this permit in the jobsite office until the activity is completed, but no longer than the end of the issuing work shift/day.
- These permits will be used by Ryan’s Site Supervision and/or other appropriate parties to assess the roof work area and assure compliance with the submitted permit as submitted by subcontractor’s competent person.
- Subcontractor’s Competent Person shall close out and initial the permit once the roof work activities are completed, not to exceed the end of the work shift.
- Ryan’s Site Supervision will be responsible for ensuring that all roof access permits issued are closed out prior to ending the work shift.

AWS - ATL089 Procedures Manual

- Subcontractors and/or its employees found in violation of this procedure or any portion thereof will be subject to immediate disciplinary action(s) up to and including immediate removal from the project.

MAINTAINING THE PERMIT: Pre-Work Inspection and JHA considerations:

- An inspection of the roof work area shall be conducted by each subcontractor's competent person who has requested a permit.
 - Competent person shall ensure that load limits are not exceeded with equipment, material and/or personnel.
 - Openings created must be protected and/or covered with appropriate **MATERIALS SUPPLIED BY THE SUBCONTRACTOR.**
1. Fire extinguishers are available, fully charged, and operable; and personnel are assigned, equipped and trained in the use of the equipment. A separate permit shall be required for Hot Work.
 2. If the planned work cannot be performed safely, that work shall not be done until the conditions meet the requirements to safely perform that work.
 3. Any equipment intended to be used for roof work must be removed from service if the competent person observes any defects. Such equipment must be replaced or repaired.
- Violations noted during inspections may result in the revocation of the issued Roof Access Permit.
 - A Safety Violation Notice will be issued to the violating subcontractor.

CLOSING THE PERMIT: Each subcontractor's Competent Person will be responsible for closing out their permit with Ryan's Site Supervision. Failure to closeout this permit may delay issuance of future permits. Copies of the permits must be maintained on site until the end of the project; there is no archiving requirement.

Contact your Regional Safety Director with any questions.

Ground Disturbance Procedure (Reference GMR Section 3.6 and 3.12 for additional AWS requirements)

The purpose of this Ground Disturbance Procedure is to promote Safety First by pre-planning, reviewing any plans with site supervision, and avoiding impact to public or adjacent properties/community.

1. "Ground Disturbance" includes the following activities that move dirt, pavement, sod or rock – any activity that penetrates the ground at any depth by any means.
 - a. Digging / Excavating / Trenching
 - b. Drilling / Boring
 - c. Grading
 - d. Driving stakes / posts or form pins
 - e. Hand digging / Vacuum truck
2. The ground disturbance crew and work must be supervised by a competent person identified by the subcontractor with appropriate training to supervise the activity to be performed.
3. A Site Utility Map, Site Logistics Plan, and BIM models are to be posted in the jobsite office marked with existing utilities and is to be updated periodically by Ryan site supervision or their designee as new lines are installed, or site conditions change.
 - a. Lines are to be marked using the APWA Utility Color Codes (see color code diagram below)

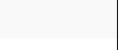
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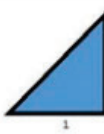
- b. Site Utility Map or Site Logistics Plan is to be reviewed with the subcontractor and any tiered subs by the Ryan Field Manager as part of the Daily Ground Disturbance Permit review process (see Exhibit 14 of this Project Safety Manual or Procore Forms).
 - c. Newly installed facilities (non-energized) utility lines must also be updated or located on the Site Logistics Map/plan.
- 4. Ryan site supervision orientates subcontractors to the project site and known utility locations, however any subcontractor performing ground disturbance must have their own utility locate completed prior to digging (Call B4 U Dig/811). Subcontractor must provide the locate ticket number on the Daily Ground Disturbance Permit.
 - a. Existence of any overhead power lines must also be identified, and an Overhead Power Line Permit must be completed/submitted. See Exhibit 17 of this Project Safety Manual or Procore Forms.)
 - b. Include utility emergency contact information on the Daily Ground Disturbance Permit**
- 5. Live underground utilities must be exposed safely by use of a hydro-vac or potholing for 100% positive identification. No work shall occur within 30' (9 m) of a known utility prior to potholing and confirming the existing utility location. All excavations within 10' (3 m) of a located live utility shall be completed by a vacuum truck method only. Newly installed facilities and/or (non-energized) utility lines must also be exposed safely.
- 6. Daily Ground Disturbance Permits (Construction Zone and Operational Zone) are available in Procore Forms and must be completed and signed by the competent person on behalf of the subcontractor performing the work **and reviewed by Ryan site supervision.**
 - a. Permit shall be reviewed with crew and operators before work starts.
 - b. Permit shall be readily available on the project premises throughout the Ground Disturbance activity.
 - c. Daily Ground Disturbance Permit is valid for the workday with a renewed permit required (each day) for that week.
 - d. A Job Hazard Analysis completed and available for review upon request/by the competent person must also be submitted with the Daily Ground Disturbance Permit.
 - e. Competent person must return/close out the permit at the end of the workday by signing and submitting the hard copy to Ryan Site Supervision or by signing/closing in Procore.
- 7. All soils are to be treated as Type C unless adequate reason and evidence can be shown to treat it as Type A or B through documented (visual and manual) testing.
 - a. The soil can NOT be classified as Type A if it is subject to vibration from heavy traffic, pile driving, or similar effects; or if the soil has been previously disturbed.
 - b. Benching is not permitted in Type C soils.**
- 8. In the event the potential for unknown environmental or archeological conditions are uncovered or discovered (ie asbestos contaminated soils), stop work and notify Ryan Field Management.
- 9. When an excavation reaches five feet or deeper, protective measures must be put in place – trench box, sloping, shoring, benching, etc. If the potential exists for oxygen deficiency or toxic atmospheres, air monitoring must be performed regardless of depth.
- 10. The permit must be created in or printed from Procore, and reviewable (hard copy or i-pad) at the site of activity and/or made available for review upon request.
- 11. Access/egress must be maintained within 25 feet of any worker in a trench/excavation of 4 feet deep or greater.
- 12. Fall protection must be provided for all excavations six feet or more in depth.
- 13. Protection is required for barricading/marketing excavations and/or trenches at the end of each shift or when the excavation/trench will remain “open for any length of time without supervision/unattended”.

AWS - ATL089 Procedures Manual

Flagging/spoil piles are also acceptable means of protection. Keep heavy vehicles/equipment and/or the public away from trenching/excavation areas.

14. Excavations that are 20' or deeper must have professional engineer approval (for the State the work is being performed in) for the protective method to be used (sloping, shoring, trench box, combination, etc.).
15. When excavation conditions or methods change (i.e. after a rainstorm, or an unknown utility is uncovered), the Daily Ground Disturbance Permit is to be reviewed, re-evaluated and re-authorized by the competent person before work continues.
16. During backfill, utility marking tape is to be placed two feet above the line. Tracer wire or RFID tags are to be used to mark the lines, appropriate to requirements for the project.

Utility Location Color Code: Flags or paint in the following colors denote designated facilities	
	Red: Electric Power Lines, Cables, Conduit and Lighting Cables
	Yellow: Gas, Oil, Steam, Petroleum or Gaseous Materials
	Orange: Communication, Cable TV, Alarm or Signal Lines, Cables or Conduit
	Blue: Water, Irrigation and Slurry Lines
	Green: Sewers and Drain Lines
	Pink: Temporary Survey Markings
	White: Proposed Excavation

Type A	Type B	Type C
3/4 : 1	1 : 1	1 1/2 : 1
53 degrees	45 degrees	34 degrees
		
For every foot of depth, dig out 9"	For every foot of depth, dig out 12"	For every foot of depth, dig out 18"
<i>Sloping in A Soil</i>	<i>Sloping in B Soil</i>	<i>Sloping in C Soil</i>
		
<i>Benching in A Soil</i>	<i>Benching in B Soil</i>	NO BENCHING IN TYPE C SOIL
		
The deepest rise is no more than 5', the rest are no more than 4'	The max for each rise and run is 4 feet.	
TRENCHES DEEPER THAN 20 FEET REQUIRE ENGINEERING FOR PROPER PROTECTION		

Overhead Power Line Awareness Procedures and Permit (Reference GMR section 3.6 - Risk Mitigation for additional AWS requirements)

Safe Working Clearance and Procedures

PURPOSE: To establish a Safe Working Clearance and Procedures for cranes and other high-reaching equipment operating adjacent to Overhead Power Lines at the Project, per OSHA 29 CFR 1926.1408-1412, 1926.600 standards and ANSI/ASME B30.5-2007.

SCOPE: These procedures apply to all cranes, high-reaching equipment, material handling equipment, hauling equipment, and earthmoving equipment being operated on the jobsite.

GUIDELINES: Any contractor operating cranes, high-reaching equipment, material handling equipment, hauling equipment, and earthmoving equipment shall notify Ryan's Jobsite Manager at the beginning of each workday of all intended activities in relationship to the overhead power lines. Each subcontractor's competent person/foreman and/or superintendent will be responsible for filling out an [Overhead Power Line Awareness Permit](#) (click link, see Exhibit 17, or refer to the forms section of the Safety Manual) with Ryan's Jobsite

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Manager and/or designated person prior to operating equipment each day, change in location, and throughout the workday as weather conditions change (temperature, wind, humidity, etc.). Ryan's Jobsite Manager and/or designated person will be responsible for administration and ensuring compliance with this program throughout the day.

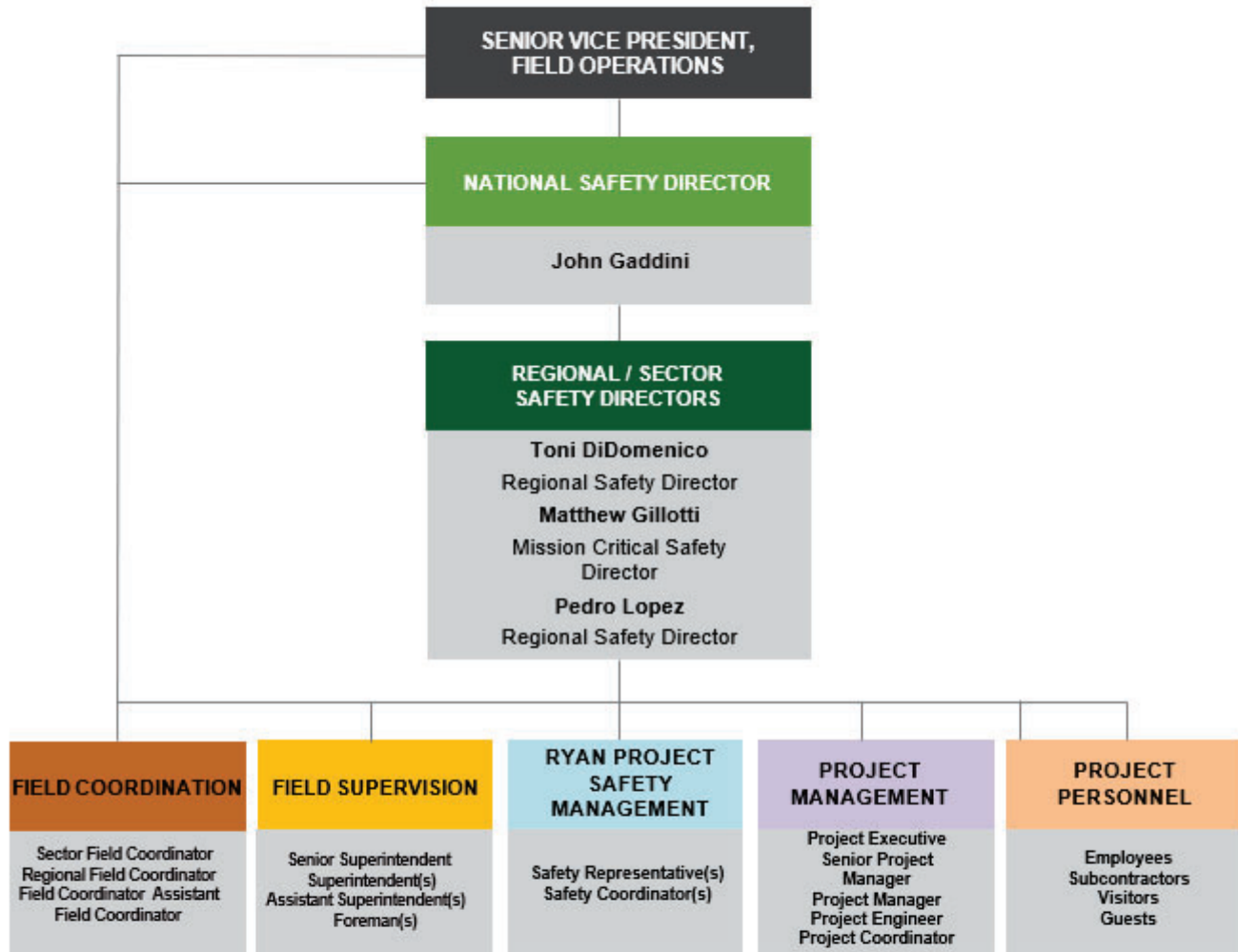
REQUIREMENTS AND PROCEDURES:

- A working clearance of 20 feet between the electrical wires and any cranes, high-reaching equipment, material handling equipment, hauling equipment, and earthmoving equipment used in or near the power line easement, and a clearance of 20 feet to the physical proximity of workers must be maintained at all times. **Note:** Work zone is defined as the area 360 degrees around the equipment, up to the equipment's maximum working radius. Refer to OSHA 29 CFR 1926.1408 Table A-Minimum Clearance Distances.
- Competent person must verify the voltage of the overhead powerlines by contacting the Energy Provider for written verification of the powerline(s) voltage rating.
- Refer to OSHA 29 CFR 1926.1411 Table T-Minimum Clearance Distances While Traveling With No Load. A JHA must be filled out for traveling with a crane.
- Competent person shall instruct each of their employees on the exact locations of the power lines in relationship to the building and the Energy Provider's power line drawings.
- Each company must have a certified person in First Aid and CPR onsite while equipment is being operated.
- All materials being stored/staged adjacent to overhead power lines must be grounded by a grounding rod and cable system.
- There shall be no flammable or combustible liquids stored onsite or refueling of vehicles and equipment within 50 feet of the power line easement.
- Employers should evaluate the need for specialized PPE for rigging personnel.
- Each company must have a designated Spotter (competent person) assigned to monitor the distance between the high-reaching equipment and the overhead power lines. This individual must be able to accurately judge the distance between the energized power lines and the high-reaching equipment.
- Danger signs stating "High Voltage Power Lines Overhead" must be posted near the site entrance and along the perimeter of the overhead power lines running east, west, north and south of the jobsite. Warning lines with flags must be erected to visibility show where the power lines buffer zones are located.
- Note: High voltage proximity may affect pacemakers and defibrillator devices.
- Refer to the Ryan Corporate Safety Manual for additional requirements and guidelines for Overhead Power Line Awareness Procedures and Permit.

AWS - ATL089 Procedures Manual



Safety at Ryan
Safety Team



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NATIONAL SAFETY DIRECTOR It is the responsibility of the Safety Director or Management level individual responsible for safety to provide all levels of management and supervisors with the necessary service relative to safety activities and required advice for the promotion and use of an effective safety program. The Safety Director is accountable to the Construction Vice President.

SECTOR AND REGIONAL SAFETY DIRECTORS It is the responsibility of the Sector and Regional Safety Director or Management level individual responsible for safety to provide all levels of management and supervisors with the necessary service relative to safety activities and required advice for the promotion and use of an effective safety program. Sector and Regional Directors are accountable to the National Safety Director.

FIELD COORDINATOR The Field Coordinator is responsible for the administration of the safety program on his/her assigned projects and is accountable to management and the safety committee.

PROJECT SUPERINTENDENT Responsibility for the safety performance and record keeping of an assigned project is vested in the Superintendent, who is accountable to the Field Coordinator.

PROJECT SAFETY MANAGER / REPRESENTATIVE / COORDINATOR Responsibility for the safety performance and record keeping of an assigned project is vested in the Project Safety Manager, who is accountable to the Project Superintendent.

FOREMAN The responsibility for the crew's safety, the safe condition of the assigned work area and the equipment's safe operation is vested in the Foreman, who is accountable to the Superintendent.

PROJECT MANAGER Responsible for the initial pre-bid safety considerations and anticipated safety costs.

EMPLOYEE / SUBCONTRACTOR(S) / VISITOR(S) / GUEST(S) In order to provide a safe working environment, it takes the cooperation of all employees, subcontractors, visitors and guests. Any safety program will fail if there is no support from all levels of the company. We must work together and help each other if we are going to be successful. Safety is everyone's responsibility.

AWS - ATL089 Procedures Manual

Appendix:

Exhibit 1 – Energy Control Program.....	50
Exhibit 2 - Article 9 – Safety Requirements and Procedures.....	78
Exhibit 3 - Preconstruction Subcontractor Safety Meeting	83
Exhibit 4 - Job Hazard Analysis.....	88
Exhibit 5 - Site Specific Orientation – Special Safety Regulations	89
Exhibit 6 - Emergency Response Plan for Construction Sites.....	91
Exhibit 7 - Jobsite Emergency Contact List.....	92
Exhibit 8 - Incident Report	93
Exhibit 9 - Rigging Inspection Log.....	97
Exhibit 10 - Safety Violation Notice	98
Exhibit 11 - Written Verification for the Steel Erection Contractor	99
Exhibit 12 – Crane Operator Evaluation Form.....	100
Exhibit 13 – ATV/UTV Use Checklist / Form.....	101
Exhibit 14 – Daily Roof Access Permit.....	102
Exhibit 15(a) – AWS CZ Daily Ground Disturbance Safety Permit.....	105
Exhibit 15(b) – AWS OZ Daily Ground Disturbance Safety Permit.....	107
Exhibit 16 – Silica Exposure Control Plan.....	109
Exhibit 17 – Safety Committee Meeting Agenda.....	111
Exhibit 18 – Overhead Power Line Awareness Permit.....	112
Exhibit 19 – Cattle Gate Procedures.....	114
Exhibit 20 – Temporary Guardrail Systems Removal Permit.....	116
Exhibit 21 – Ryan Companies Safe Lift Permit Plan.....	118
Exhibit 22 – Ryan Companies Photograph Policy.....	120
Exhibit 23 – Working at Heights Permit.....	121
Exhibit 24 – Amazon MOP Example.....	124
Exhibit 25 – AWS Global Construction H&S Minimum Requirements.....	130

AWS - ATL089 Procedures Manual

Exhibit 1 - Energy Control Program

ECP EXECUTIVE SUMMARY

PURPOSE

The Energy Control Program (ECP) exists to support our project sites in the pursuit of zero accidents, incidents, and near misses, because of any energy source.

PLANNING

The Energy Control Program provides guidelines for planning efforts and meetings that are required to effectively procure, implement, maintain, and audit the ECP.

TRAINING

The ECP outlines the minimum required training for all individuals based on roles and responsibilities associated with projects. ECP focused training includes all site-specific training, Arc Flash Awareness, Reading Arc Flash Labels, Arc Flash PPE, MOP Expectations, LOTO Policy, Live-Dead-Live, NFPA 70E in-person training repeated every three years, and OSHA-related Electrical Safety Training.

SITE SETUP

The ECP encompasses all site set-up and temporary services for project sites. It requires and provides guidelines for early implementation of the ECP and reinforces both underground utility avoidance and assured grounding. Subcontractors are expected to follow the Ryan Companies Energy Control Program and Subcontractor's LOTO processes must comply with Ryan Companies guidelines.

CONSTRUCTION AND COMMISSIONING

The ECP outlines best practices during critical stages of construction and establishes required procedures to ensure safe work practices. All energy control activities will be required to fill out either a Method of Procedure (MOP) or an Action Plan (AP) based on the exact activities taking place. These two required documents will evaluate the energy hazard present and identify the specific means to reduce, mitigate, or eliminate the associated risks.

FUNCTIONAL AND FIRST COMMISSIONING

Functional and first commissioning inherently introduce new or additional parties to project sites. The ECP outlines that during this time all commissioning activities will be required to meet the guidelines established within the ECP.

COMPLIANCE AND AUDITING

The ECP establishes both the individuals and the frequency for necessary auditing of the ECP to ensure proper implementation and documentation. It also requires regular meetings to discuss the findings of the audits being performed and the necessary corrective actions to be taken.

LOCKOUT/TAGOUT POLICY

The ECP establishes a Lockout/Tagout (LOTO) policy as a minimum expectation for implementation on all project sites. Trade contractors are required to meet or exceed this requirement with their site-specific LOTO policy.

STAFFING

The ECP establishes an Energy Marshall Program with specific staffing requirements for this role for Ryan Companies and trade contractors working at project sites.

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SECTION 1: PURPOSE

This Energy Control Program (ECP) and Lockout/Tagout (LOTO) Program has been established to support both worker safety and the protection of property by safeguarding workers from exposure to electrical hazards and to prevent injury due to the unexpected release of hazardous energies. Active or stored energy sources include, but are not limited to, electrical, battery systems, mechanical (hydraulic, pneumatic, etc.), chemical, thermal (steam), and/or gravitational elements.

This document establishes the minimum expectations for LOTO procedures of energy sources to protect personnel who must approve or participate in the practical safeguarding of equipment while performing service, maintenance, inspection, start-up, or operational activities on Ryan Companies project sites. These activities may include but are not limited to, erecting, installing, constructing, repairing, maintaining, adjusting, inspecting, setting up, troubleshooting, testing, cleaning, and dismantling machines, equipment, or processes.

It is Ryan Companies' intention to provide a safe work environment in compliance with NFPA 70E, OSHA Standards, and specific owner requirements. Ryan Companies intend to perform work in a de-energized/zero-energy state to provide the greatest level of protection for people and property while complying with all applicable governmental regulations.

OBJECTIVES

The objectives of the Ryan Companies ECP are the following:

- Establish a safe and effective means of equipment start up, shutting down equipment and electrical circuits, and maintaining safe work practices around sources of energy.
- Prohibit unauthorized personnel or remote-control systems from starting equipment while work is being performed.
- Establish responsibility for implementing and controlling lockout/tagout procedures.
- Ensure that only approved locks, standardized tags, and fastening devices will be utilized in the lockout/tagout procedures.
- Train and educate individuals to recognize unsafe actions and unsafe conditions and provide a means of reporting these to appropriate levels of leadership.
- Ensure compliance through audits, inspections, evaluations and employee engagement and involvement in the ECP.

ACRONYMS

AP – Action Plan

ATPV – Arc Thermal Performance Value (Rating)

ECP – Energy Control Program

IST – Integrated Systems Testing

LDL – Live-Dead-Live

LOTO – lockout/Tagout

MOP – Method of Procedure

NETA – International Electrical Testing Association

NFPA – National Fire Protection Association

OSHA - Occupational Safety and Health Administration

PPE – Personal Protective Equipment

QA/QC – Quality Assurance/Quality Control

SCCAF – Short Circuit, Coordination, and Arc Flash Studies

SOP – Standard Operating Procedure

VLF – Very Low frequency

DEFINITIONS

Affected Employee – Any personnel directly or indirectly involved in the work effort where lockout/tagout procedures are used.

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Arc Blast – The supersonic shockwave produced when the conductors and surrounding air are heated by the arc, becoming a rapidly expanding plasma.

Arc Blast Hazard – A source of possible injury or damage to health from the energy deposited into acoustical shockwave and high-velocity shrapnel.

Arc Flash – The light and heat produced as part of an arc fault (sometimes referred to as an electrical flashover), a type of electrical explosion or discharge that results from a connection through air to ground or another voltage phase in an electrical system.

Arc Flash Hazard – A source of possible injury or damage to health associated with the release of energy caused by an electric arc.

Arc Flash Boundary – When an arc flash hazard exists, an approach limit from the arc source at which incident energy equals 1.2 cal/cm².

Authorized Employee – Any qualified personnel who is directly responsible for the supervision or execution of any work scope involving LOTO procedures.

Cal Rating – The level of arc exposure is referred to as the ATPV and is measured in calories/cm² (often called the cal rating). This requires employees to wear arc-resistant clothing with a PPE category rating equal to or greater than the determined arc hazard.

Capable of Being Locked Out – The source of energy can be isolated and controlled to provide safeguarding for personnel. Isolation and control must be achieved without the need to dismantle, rebuild, or replace the energy-isolating device or permanently alter its energy-control capability. Any equipment that has integral LOTO accessories or can accept supplemental LOTO accessories to facilitate isolation of energy sources shall be considered capable of being locked out.

Controlled Room Access – As equipment is energized in the building and on the site, the rooms and/or equipment houses will be locked, and access will be controlled by Ryan Companies. For certain rooms, a physical security guard may need to be put in place to control access.

Control Locks – Used to maintain access control to equipment. Control locks may not be of the combination type and must be keyed alike. Control locks are to be managed in the same manner as LOTO locks. Control locks are not allowed to be used for purposes of LOTO.

Dry Run – The act of physically verifying the accuracy of the work script for the proper sequence of activities, equipment designations match the work script and supporting documents, to identify unforeseen issues, verification of positive communications among teams performing the work, etc. The dry run must be performed in the field and verified against equipment in the field.

Disconnecting Means – Any device, or group of devices, by which the conductors of a circuit can be physically disconnected from their source of supply. All disconnecting means must be capable of being locked in the open position.

Electric Shock Hazard – A source of possible injury or damage to health associated with current through the body caused contact or approach to exposed energized electrical conductors or circuit parts.

Energized – Connected to an energy source or containing potential energy.

Energy Marshall – A qualified person responsible for overseeing all energy management activities.

Energy Source – Any source of potential, kinetic, thermal, chemical, electrical, or nuclear energy. Examples include, but are not limited to, electricity, hydronic, hydraulic, or pneumatic systems.

Energized Work – Any work scope that exposes personnel to hazardous energy sources.

Energized Work, Diagnostic – Inspection, testing, voltage and/or current measurements, phase alignment, troubleshooting, circuit and signal tracing, thermal imaging, etc., that are performed on or near exposed live parts within the Limited Approach Boundary.

Energized Work, Manipulative – Making, tightening, or breaking of energized electrical connections or the replacement, removal, or addition of electrical or mechanical components.

Energy-Isolating Device – A mechanical device that physically prevents the unintentional release of energy.

Hazardous Energy – Any source of energy that exists at a level or quantity that could be harmful or could cause injury to personnel through inadvertent release or start-up of equipment.

Limited Approach Boundary – An approach limit at a distance from an exposed energized electrical conductor or circuit part within which a shock hazard exists.

Lock Out Device – Any device that uses positive, physical means to hold an energy-isolating device in a safe position, thereby preventing the energizing of machinery or equipment.

Locked Out/Tagged Out – The combined placement of a lock out device and a tag out device on an energy-isolating device, in accordance with an established procedure, to ensure the energy-isolating device and the equipment it controls cannot be operated until the lockout device is removed.

Qualified Person – One who has demonstrated skills and knowledge related to the construction and operation of electrical equipment and installations and has received safety training to identify the hazards and reduce the associated risk.

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Restricted Approach Boundary – An approach limit at a distance from an exposed energized electrical conductor or circuit part within which there is an increased likelihood of electric shock, due to electrical arc-over combined with inadvertent movement.

Risk – A combination of the likelihood of occurrence of injury or damage to health and the severity of injury and damage to health that results from a hazard.

Risk Assessment – An overall process that identifies hazards, estimates the likelihood of occurrence of injury or damage to health, estimates the potential severity of injury or damage to health, and determines if protective measures are required.

Shock Hazard – A dangerous condition associated with the possible release of energy caused by contact or approach to energized electrical conductors or circuit parts.

Tag Out Device – A prominent warning sign at the point of lock out that can be securely fastened to an energy-isolating device to indicate that the energy-isolating device and the equipment it serves is unavailable for normal operation and cannot be ready for operation until the tag out device is removed.

Working On (energized electrical conductors or circuit parts) – Intentionally coming into contact with energized electrical conductors or circuit parts with the hands, feet, or other body parts, with tools, probes, or with test equipment, regardless of the personal protective equipment (PPE) a person is wearing. Entering the Restricted Approach Boundary is considered Working On (energized electrical conductors or circuit parts). There are two categories of “working on”: Diagnostic (testing) is taking readings or measurements of electrical equipment with approved test equipment that does not require making any physical change to the equipment. Repair is any physical alteration of electrical equipment (such as making or tightening connections, removing, or replacing components, etc.).

SECTION 2: PLANNING AND PROGRAM STRUCTURE

PRECONSTRUCTION

The Energy Control Program is integrated into Ryan Companies’ preconstruction process to ensure that expectations for the Energy Control Plan and the LOTO Program are fully understood by manufacturing, electrical and plumbing trade partners and associated sub tiers and equipment vendors and all subcontractors on-site. This ensures that project teams are established to prevent incidents at the onset of project work.

ENERGY CONTROL PROGRAM SITE TASK FORCE

For each job site an Energy Control Program Site Task Force will be established. The Energy Control Program Site Task Force will generally consist of Energy Marshalls, Deputy Energy Marshalls, LOTO Coordinators, and Safety Representatives from Ryan Companies, the involved Electrical Contractor, Mechanical Contractor, Fire Protection/Plumbing Contractor, and a Safety Representative from the owner/client. The Energy Control Program Site Task Force will coordinate and communicate the Energy Control Method of Procedure (MOP) and Action Plan (AP) processes. Ryan Companies will provide oversight of the Energy Control Program for the project site.

The Energy Control Program Site Task Force responsibilities include coordination and communication of the following:

- Comprehensive oversight of all energy management duties
- Maintain drawings with current state of each system
- Review all MOPs and APs, ensuring there is a proper review by appropriate parties (utility, specialists, design engineer, etc.)
- Administer dry run of MOPs physically in the field
- Maintain logs of all locks and locations including ground clusters for electrical
- Conduct audits of the Energy Control Program and LOTO for the site
- Provide training to all required personnel of the Energy Control Program and LOTO Program
- Report all deficiencies, near-misses, and incidents to proper personnel
- Communicate and reinforce the Stop Work Authority granted to all workers on the site
- Perform verification of intended lockout locations
- Verify that all energized rooms and systems have a controlled access process in place
- Verify that qualified and competent persons are designated for all high-risk activities

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- Coordinate a LOTO chat to communicate the prioritization, timing, and execution of LOTO requests and to provide LOTO schedule updates

ENERGY CONTROL PROGRAM ROLES AND RESPONSIBILITIES

ROLE	RESPONSIBILITIES	MOP	AP
Chief Energy Marshall	• Lead the ECP Task Force • Manage MOPs and MOP reviews • Manage overall governance of the ECP • Execute all medium voltage and high voltage switching • ECP Compliance Subject Matter Expert • Train subcontractor Energy Marshalls and Ryan Company Coordinators and LOTO Managers • Ensure safe conditions for MOP/AP processes • Facilitate and lead MOP and ECP status meetings to ensure the process is being executed as planned • Ensure pre-energization paperwork for MOPs and APs are reviewed and approved before execution of the MOP and/or AP • Develop MOP weekly work plans to support project management schedules • Lead root cause analysis investigations	• Manage the daily schedule of MOPs • Ensure 3rd party representatives have MOP/LOTO plan, etc. • Review MOPs • Sign-off and Dry Run • Final Approval • Leadership of MOPs execution • Verify proper LOTO and project status/one-lines are updated • Support MOP development • Review MOPs • Manage MOP Log • Issue final approval MOP to subcontractor for execution • Lead daily and weekly MOP management meetings	• Manage the daily schedule of APs • Ensure 3rd party representatives have AP/LOTO plan, etc. • Oversight of AP execution if required
Deputy Energy Marshall	• Daily auditing of LOTO activities • Oversee energy management activities • Accountability and authority to challenge the process • Subject Matter Expert for ECP	• Audit the process • MOP development resource • Sign-off and Dry Run • First Approval • Oversee MOP Execution for low voltage energizations	• Audit the process • First Approval

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	• 1st approval to MOPs and APs (sending it to Ryan Companies for final approval) • Perform dry run of MOP with front line lead/coordinator • Perform regular audits to ensure project status is updated and all documentation is complete • Verify the LOTO process • Attend office and field meetings to provide project status updates and support • Update and audit project status daily with the LOTO Manager • Oversight for all low voltage energizations • Oversight of Action Plans • Interact with LOTO team members on LOTO request form approvals • Perform auditing of processes • Be present for any switching activities as directed during a MOP execution and available throughout the entire MOP duration		
LOTO Manager	• Manage project status, lock out and tag out devices, and LOTO procedures • Participate in MOP and project status meetings to ensure the process is being executed as planned • Oversee all LOTO activities • File completed MOPs for record	• Ensure proper lock and tag placement and execution of personal lock and group/complex lock procedures • Accountable for all LOTO devices • Updates project status	• Ensure proper lock and tag placement and execution of personal lock procedures • Issue and maintain inventory of Lockout/Tagout devices • Update project status

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LOTO Coordinators	Facilitate the execution of LOTO in their area of responsibility	- Ensure proper lock and tag placement and execution of personal lock and group/complex lock procedures - Accountable for all LOTO devices in their area of responsibility	- Ensure proper lock and tag placement and execution of personal lock procedures - Issue and maintain inventory of Lockout/Tagout devices in their area of responsibility
Coordinators/Leads	- Ensure work instructions comply with the ECP requirements - Maintain point of contact communications for all energy related activities in that person's area of responsibility - Respond to questions regarding energy management activities for their area - Support compliance - Assist the auditing process - Participate in root cause analysis investigations as appropriate	- Assist in MOP process - Understand all MOPs being executed within their area - Support execution of MOPs within their area if required	- Comply with the management of the sign-off/final approval of APs - Oversight of AP execution - Ensure pre-energization paperwork is completed and approved, including QA/QC - Ensure proper processes are followed to ensure safety
Worker/Tradesperson	- Performs, executes, and maintains a safe working environment - Adheres to ECP and other safety policies - Exercises Stop Work Authority	- Assists in MOP development - Participates in Sign-off and Dry Run processes	- Assists in AP development - Participates in Dry Run AP - Participates in execution of AP

GENERAL CHIEF ENERGY MARSHALL EXPECTATIONS

1. Complete NFPA 70E training on Standard for Electrical Safety in the Workplace and other required training.
2. Train and supervise Deputy Energy Marshalls and the LOTO Manager.
3. Confirm all individuals working on energized or de-energized and locked out equipment are Authorized and/or Qualified Workers based on NFPA 70E, OSHA, or an equivalent qualified electrical safety training standard.
4. Organize and schedule pre-energization meetings in advance of energizing or de-energizing major pieces of equipment, making certain that all critical steps and safety concerns shall be identified and used to generate daily work plans.
5. Provide final approval and accountability prior to all energizations.
6. Supervise the work of the Deputy Energy Marshalls and the LOTO Manager.
7. Confirm existence of and review LOTO plans for all jobsite work and ensure that only approved LOTO plans are incorporated into daily work plans and work activities.
8. Review single-line drawings and ensure all workers involved in LOTO activities are educated on the involved systems.
9. Ensure completion of all owner and equipment manufacturer pre-energization and L3 inspections.

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10. Verify all Foreign Objects and Debris (FOD) inspections are completed prior to beginning any energization process and are in compliance with the Owner's and Contractor's FOD requirements. All re-entry into previously energized equipment shall be approved and managed by the Energy Marshall or Deputy Energy Marshall, who shall ensure appropriate FOD re-inspections and re-tamper tagging are completed prior to re-energization.
11. Ensure tamper tags are installed and adhered to correctly from pre-energization through the appropriate milestones. Any tags that fail shall be replaced immediately.
12. Restrict access to electrical equipment during energization and commissioning activities through proper safety barriers such as cones and bones and other means.
13. Verify removal of temporary protective grounding/earthing equipment.
14. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.
15. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with the energization plan.
16. Conduct audits and inspections to ensure compliance with the ECP and LOTO Program.

PRE-ENERGIZATION KICK-OFF MEETING

A pre-energization kick-off meeting should be held sixty days prior to initial energization. The target audience includes Ryan Companies, Owner/Client, Commissioning Provider, Electrical Contractor, Mechanical Contractor, and Equipment Vendors. This meeting will focus in part on the safety requirements of the project. Some of the topics covered will include room and equipment access, training, dry runs, energization execution, post-energization and future meeting schedules.

METHOD OF PROCEDURE (MOP) KICK-OFF MEETING

A Method of Procedure (MOP) kick-off meeting should be held near the time of the project kick-off. The target audience includes Ryan Companies, Owner/Client, Electrical Contractor, Mechanical Contractor, Equipment Vendors, and any other additional party involved in the MOP process (Facilities, Security, Commissioning Partner, Design Engineer, etc.). Prior to holding this meeting, the tools and processes should be vetted and established with the client. At this meeting it is important to establish the site-specific process (criticality levels, review times, work windows, etc.) to the trades along with review and training of the MOP templates and regularly used tools. Discussion topics will include MOP development best practices and procedures, dry runs, and future meeting schedules.

METHOD OF PROCEDURE (MOP) RECURRING MEETING

A Method of Procedure (MOP) recurring meeting should be held weekly after the initial kick-off meeting is completed. The frequency of this meeting can vary based on the demands of the project. The target audience includes Ryan Companies, Owner/Client, Electrical Contractor, Mechanical Contractor, Equipment Vendors, and any other additional party involved with the MOP process (Facilities, Security, Commissioning Partner, design Engineer, etc.). At this meeting it is important to review tracking log progress against the operations schedule, ensure the accuracy of the tracking log, approval of MOPs to support onsite work and communicate any restrictive change periods. Among other topics for discussion will be revisions to facility procedures, site logistics, access, specific MOP procedures, and future meeting schedules. This meeting may spur requirements for other "one-off" meetings such as a formal MOP review with the Client or informal MOP development meetings with trades & vendors.

COMMISSIONING (CX) ORGANIZATIONAL/KICK-OFF MEETING

A Commissioning (CX) organizational meeting should be held thirty days prior to the first factory witness test (FWT) effort. The target audience includes Ryan Companies, Owner/Client, Design Engineer, Commissioning Provider, Electrical Contractor, Mechanical Contractor and others as deemed necessary. This meeting is designed to level set expectations during the CX process from the leaders of the project site. A Commissioning (CX) kick-off meeting should be held sixty days prior to equipment energization. The target audience includes everyone from the CX organizational meeting, and any others involved in the commissioning process. At this meeting it is important to review the CX plan including roles and responsibilities, project schedule, outline procedures and documentation requirements, site inspections, as well as to establish a communication method for all the parties involved in the CX process. It is also important to reinforce safety requirements, and necessary project specific training requirements. A future meeting schedule should be established for weekly and/or daily CX meetings.

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COMMISSIONING (CX) RECURRING MEETING

A commissioning (CX) recurring meeting should be held daily/weekly depending upon project requirements. This meeting should begin during pre-functional testing. Weekly meetings should start once the equipment arrives on-site and turn into daily meetings two weeks prior to energization. The target audience includes Ryan Companies, Commissioning Provider, Electrical Contractor, Mechanical Contractor, equipment Vendors and any other appropriate parties. During this meeting it is important to discuss the daily activities, high-priority items, and a review of the commissioning calendar. Among other topics for discussion are entry permits, training classes, milestone reviews, and overall commissioning progress.

SYSTEM/PROJECT STATUS MEETING

A System/Project Status meeting should be held daily prior to the morning CX meeting. This meeting should begin prior to equipment safety control measures (LOTO) being required. The target audience includes Ryan Companies, Electrical Contractor, Mechanical Contractor, Equipment Vendors, LOTO Crew, and any other additional parties requiring or isolating an energy source. At this meeting, coordination and communication are critical. Work is coordinated in the field to support multiple start-up activities based upon start-up requirements and schedule milestones. Accurately relaying the status of all energy sources on the project and verifying the means and methods to be utilized for source isolation is necessary. It is imperative for the project to determine a method to accurately track and communicate the status of the project elements.

SECTION 3: TRAINING

The Energy Control Program requires specific training related to the control of hazardous energy including, but not limited to electrical, mechanical, hydraulic, thermal, pneumatic, gravitational, etc. This Energy Control Program does have a special emphasis placed upon electrical safety and electrical safe work practices as referenced in NFPA 70E, OSHA Subpart S, OSHA Standard 29 CFR 1926.960, and OSHA Standard 29 CFR 1910.147, and other applicable codes, standards and regulations. The training matrix below is specific to the safety training requirements of the Energy Control Program and does not reflect additional safety training which may be required by Ryan Companies on job sites.

ECP TRAINING REQUIREMENTS

STAFF	FIRST AID/CPR/AED TRAINING (Every 2 Years)	ARC FLASH AWARENESS TRAINING (Every 3 Years)	NFPA 70E QUALIFIED WORKER COMPREHENSIVE ARC FLASH/ELECTRICAL SAFETY TRAINING (Every 3 Years)	MOP/AP EXPECTATIONS & PROCESS TRAINING	LOTO PROCEDURES/LDL TRAINING
ENERGY MARSHALLS/LOTO TEAMS	X	X	X	X	X
SWITCHING CREWS	X	X	X	X	X
AUTHORIZED EMPLOYEES	X	X	X	X	X
MEP TRADES SAFETY LEADS AND SUPERINTENDENTS	X	X	X	X	X
MEP TRADES START-UP AND SERVICE TECHNICIANS/EQUIPMENT MANUFACTURERS	X	X	X	X	X
CONTROLS/MONITORING/SPECIAL SYSTEMS TRADES	X	X	X	X	X
QA/QC PERSONNEL	X	X			X
MEP RELATED TRADES	X	X			X
CX AGENT	X	X			X

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SAFETY PROFESSIONALS	X	X	X	X	X
NON-MEP TRADES/SHEETROCK/ FINISHING/ HOUSEKEEPING/CLEANING	X	X			

ARC FLASH PERSONAL PROTECTIVE EQUIPMENT (PPE) TRAINING

Both the Arc Flash Awareness Training and the NFPA 70E Qualified Worker Comprehensive Arc Flash Electrical Safety Training will cover the NFPA 70E Personal Protective Equipment Categories in the table below.

Personal Protective Equipment [NFPA 70 E Table 130.7 (C)(15)(c)]	
Arc-Flash PPE Category	PPE
1	Arc-Rated Clothing, Minimum Arc Rating of 4 cal/cm² (16.75 J / cm²)^a Arc-rated long-sleeve shirt and pants or arc-rated coverall Arc-rated face shield ^b or arc flash suit hood Arc-rated jacket, parka, high-visibility apparel, rainwear, or hard hat liner (AN) ^f Protective Equipment Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) ^c Heavy-duty leather gloves, arc-rated gloves, or rubber insulating gloves with protectors (SR) ^d Leather footwear ^e (AN)
2	Arc-Rated Clothing, Minimum Arc Rating of 8 cal/cm² (33.5 J / cm²)^a Arc-rated long-sleeve shirt and pants or arc-rated coverall Arc-rated flash suit hood or arc-rated face shield ^b and arc-rated balaclava Arc-rated jacket, parka, high-visibility apparel, rainwear, or hard hat liner (AN) ^f Protective Equipment Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) ^c Heavy-duty leather gloves, arc-rated gloves, or rubber insulating gloves with protectors (SR) ^d Leather footwear ^e
3	Arc-Rated Clothing Selected so That the System Arc Rating Meets the Required Minimum Arc Rating of 25 cal/cm² (104.7 J / cm²)^a Arc-rated long-sleeve shirt (AR) Arc-rated pants (AR) Arc-rated coverall (AR) Arc-rated arc flash suit jacket (AR) Arc-rated arc flash suit pants (AR) Arc-rated arc flash suit hood Arc-rated gloves or rubber insulating gloves with protectors (SR) ^d Arc-rated jacket, parka, high-visibility apparel, rainwear, or hard hat liner (AN) ^f Protective Equipment Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) ^c Leather footwear ^e
4	Arc-Rated Clothing Selected so That the System Arc Rating Meets the Required Minimum Arc Rating of 40 cal/cm² (167.5 J / cm²)^a Arc-rated long-sleeve shirt (AR) Arc-rated pants (AR) Arc-rated coverall (AR) Arc-rated arc flash suit jacket (AR) Arc-rated arc flash suit pants (AR) Arc-rated arc flash suit hood Arc-rated gloves or rubber insulating gloves with protectors (SR) ^d

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	Arc-rated jacket, parka, high-visibility apparel, rainwear, or hard hat liner (AN) ^f Protective Equipment Hard hat Safety glasses or safety goggles (SR) Hearing protection (ear canal inserts) ^c Leather footwear ^e
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AN: As needed (optional). AR: As required. SR: Selection required.

a Arc rating is defined in Article 100.

b Face shields are to have wrap-around guarding to protect not only the face but also the forehead, ears, and neck, or, alternatively, an arc-rated arc flash suit hood is required to be worn.

c Other types of hearing protection are permitted to be used in lieu of or in addition to ear canal inserts provided they are worn under an arc-rated arc flash suit hood.

d Rubber insulating gloves with leather protectors provide arc flash protection in addition to electric shock protection. Higher class rubber insulating gloves with protectors, due to their increased material thickness, provide increased arc flash protection.

e Footwear other than leather or dielectric shall be permitted to be used provided it has been tested to demonstrate no ignition, melting or dipping at the minimum arc rating for the respective arc flash PPE category.

f The arc rating of outer layers worn over arc-rated clothing as protection from the elements or for other safety purposes, and that are not used as part of a layered system, shall not be required to be equal to or greater than the estimated incident energy exposure.

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SECTION 4: SITE SETUP

Temporary utilities set up for the project site is critical to implement an efficient site plan from start to finish. The temporary utilities setup includes power, water, storm sewer, sanitary sewer, and network among others. While some of these may already be present on the project site it is imperative to take note of what exists and understand the overall needs of the project. The Energy Control Plan will focus on power, water, and network (access control, internet, infrastructure, etc.).

Power

Temporary power can be utilized for trailer setups, temp lighting and receptacles, cranes, specialty tools, NETA testing, and early energization of equipment. It is necessary to plan early for all the expected uses of temporary power for the site to get an expected demand on the temporary service. Once the project team understands the demand requirements, communication with the utility provider is needed to see what is available both in voltage delivery and overall power to be provided. Based on the timeline of the utility providing temporary power it may be necessary to discuss a temporary power generator while the temporary utility service is under construction. Once finalized, a temporary power plan drawing should be produced as required for any permitting requirements and for use at the site. This drawing should include an overall site drawing showing where equipment is located, and feeders are routed. In addition, a single line drawing should be produced to illustrate how everything is connected. It is important that at the temporary power service entrance a breaker or switch isolating device is present to properly control power to the site for each temporary power service. All underground services should meet the requirements of an underground avoidance plan and include a tracer wire for utility locating as well as final as-builts provided once the service is complete.

Water

Water will be used for restroom/cleaning facilities, concrete/paint washout, tire cleaning, flush and fill operations, etc. It is important to identify where your water comes from and where you need to distribute it. If utilizing above ground piping understand the climate and utilize insulation or heat trace as required. Protect all exposed piping from foot/vehicle traffic and miscellaneous damage. Tie in locations should be protected from damage, easily accessible, and isolation valves be present to be able to control water at the project site. The water plan should include a temporary water utility drawing that is continuously updated as conditions change. The drawing should be a site plan that shows general pipe sizes along with isolation valve shut off locations and pipe routing information. All underground services should meet the requirements of an underground avoidance plan and include a tracer wire for utility locating as well as final as-builts provided once the service is complete.

Internet

Project sites often need internet service early on. Internet service needs and options should be developed at the beginning of the project. Understanding service needs at the trailer and in the field will help develop a plan to fulfill your needs by the internet service provider. All temporary network and LV systems installations are required to follow the guidelines set forth within this ECP and LOTO Program. All infrastructure, network equipment, and WIFI devices should be documented on plan view drawings and one-lines to communicate where everything is located and how the systems are configured. All underground services should meet the requirements of an underground avoidance plan and include a tracer wire for utility locating as well as final as-builts provided once the service is complete.

TEMPORARY POWER

Standard Job Safety Requirements for Electrical Temporary Power

1. All temporary power activities must adhere to the requirements of the Energy Control Program and LOTO Program.
2. Compliance with OSHA Standards and the National Electrical Code. Where conditions or requirements are addressed in both these sources the more stringent requirement shall be observed.
3. Energized electrical panels will not be accessible to anyone other than qualified electricians unless all "dead-fronts" and panel covers are installed/locked. Access will be controlled by qualified electricians using locking doors, badging systems, guards and signage. Any unattended panel must be properly covered/protected if not actively being worked on.
4. All openings in temporary electrical panels must be properly covered to prevent accidental contact with electrical current. This includes missing breakers and missing/unused breakers and knockouts. Tape over these openings is not acceptable.
5. All temporary and permanent power outlets (electrical boxes), such as outlets and switches, must have the cover plates installed before they are energized and available for use by construction personnel.

AWS - ATL089 Procedures Manual

6. All branch circuits shall be protected by ground fault circuit interrupters (GFCI) and an Assured Grounding Program.
7. All power cords are to be 3 wires, a minimum of #12 conductors, and designed for hard or extra hard usage. Flexible cords for portable lights will be designated for hard or extra hard usage.
8. At no time will any construction power hand tool be used without ground fault protection. When temporary power is removed, and permanent power is available, subcontractors shall provide ground fault protection for their own cords, lights, and tools.
9. All construction circuits for power hand tools, temporary lighting, elevators, welders, pumps, etc., will be identified at each end and at all junction points by a numerical and/or alphabetical system. Circuits to lights and tools will be dedicated and not shared one with the other.
10. All temporary power equipment and devices exposed to the weather will be made rain tight.
11. Temporary lighting levels will be maintained at a minimum of 5 fc per square foot throughout rough construction and at a minimum of 50 fc per square foot (or permanent lighting) for finishing applications.
12. Employee parking lot lighting and access/egress lighting to the building or work site will be maintained at a minimum of 2 fc per square foot.
13. Construction areas, aisles, stairs, ramps, runways, corridors, offices, shops and storage areas where work is in progress shall be lighted with either natural or artificial illuminations. The illumination shall meet or exceed the minimum OSHA Subpart D 1926.56 Illumination standard.
14. Temporary lighting should be equipped with guards to prevent accidental contact with the bulb. The guard shall be of non-conductive metal. Hanging devices shall be non-conductive material and approved by manufacturer.
15. At least one full-time electrician per building will be assigned to the upkeep and recordkeeping of the temporary power system.
16. When installing underground temporary power lines need be properly marked to identify their location, buried per specifications, location noted in GIS coordinates to provide model comparison prior to backfill, and as-built drawings updated with the newly installed temporary power lines.

Electrical Power Cords

1. Visually inspect the cord daily / before each use for damaged and exposed conductors. If the cord shows signs of damage or excessive wear, take it out of service and do not use it.
2. Inspect to make sure the grounding prong is in good condition and the cord provides a satisfactory ground for the electrical tools being used.
3. When not in use, the cords should be neatly coiled and safely stored.
4. Cords are to be strung overhead using non-conductive material or secured out of walkways.

GROUND FAULT PROTECTION

GFCIs

Ground Fault Circuit Interrupters (GFCIs) are intended to prevent electrocution by quickly interrupting the circuit in the event of a ground fault. They operate independently of the equipment grounding circuit.

1. All 120-volt, single-phase, 15- and 20-ampere receptacle outlets on construction sites, which are not a part of the permanent wiring of the building or structure, and which are in use by employees, shall have approved ground-fault circuit interrupters for personnel protection. Receptacles on a two-wire, single-phase portable or vehicle-mounted generator must be equipped with GFCIs.
2. All 240-volt equipment such as welders, saws, grinders, etc. shall be protected by GFCIs or by implementing assured grounding connector program by the subcontractor/vendor. If assured grounding connector program is utilized by the subcontractor/vendor for their equipment, inspections shall be conducted monthly by a qualified electrician and inspection records maintained by the subcontractor's competent person meeting all the requirements of 29 CFR 1926.404 (b)(1)(iii).
3. All grounding requirements of the National Electric Code still apply.

Testing and Inspection

All equipment GFCIs used during construction activities both temporary / permanent are to be tested monthly by the electrical contractor and testing records be readily available

AWS - ATL089 Procedures Manual

SECTION 5: FACTORY WITNESS, COMPONENT VERIFICATION, PRE-FUNCTIONAL COMMISSIONING

FACTORY TESTING

Factory Testing

This is a concentrated effort to perform testing and/or inspections at the factory to confirm that manufactured equipment and assemblies conform to project requirements before being shipped to the site. This can be accomplished through Factory Acceptance Testing (FAT), Factory Witness Testing (FWT), and Factory Inspections (FI).

Factory Witness Testing

This is performed at the factory and includes testing and inspections. Testing scripts are generally developed by the manufacturer and approved by the Engineer of record (EOR), Commissioning Provider, and Owner. During this test, the team goes to the factory to physically witness the test and to inspect the equipment. This is generally performed on a first of a kind (FOAK) equipment.

Factory Acceptance Testing

This is typically the same as an FWT, just not witnessed by the project team. This is typically done on all equipment.

Factory Inspections

This is typically in addition to a FAT or FWT and is when the team visits the equipment manufacturer's factory to perform a physical inspection of the equipment prior to it being delivered to the site. This specific effort does not include any testing of the equipment during the visit.

Factory Virtual Testing

FVT is a method of inspection or testing that is witnessed by the project team virtually. This testing can replace any of the other testing and can be limited to just an inspection or performing an entire test script.

Factory Pre-Test Meeting

Two weeks prior to the test date, a pre-test meeting is to be held. This pre-test meeting shall be organized by Ryan Companies and include participation from the Owner, Design Team, Commissioning Partner, Equipment Sales Team, and Factory Representative. The factory safety requirements should be reviewed. The factory representative should confirm what PPE is required while within the factory. A copy of the manufacturer's ECP and LOTO policy should be reviewed. A safety orientation should be given to the attendees. The safety orientation should include a review of the required PPE, approach boundaries and all power sources within or near the test area. Review of the test script should be performed, and any potential destructive tests should be flagged and discussed with the entire project team.

Factory Visit

At the beginning of the factory visit all individuals shall participate in the safety orientation. The factory representative is to point out all power sources and any other potential safety hazards. All hazards should have appropriate boundaries in place prior to energization of any power source. OSHA and 70E guidelines and all necessary and appropriate safety practices will be followed.

Component Verification

This includes the verification that systems and equipment have been delivered and installed in accordance with the project requirements. Ryan Companies, installation contractors, and equipment vendors are primarily responsible for this level with potential involvement from the Commissioning Provider, Facilities, and Client teams.

Pre-functional Testing

PFT is the time when the contractors/vendors program, energize, operate, and confirm the equipment functions per the contract documents. This phase is very systematic and includes everything from de-energized functionality checks to fully energized verification of operational sequences. This process is documented with the use of standard checklists as well as any required manufacturer documentation. Issues discovered during this phase are the responsibility of the contractors/vendors to resolve prior to functional testing readiness.

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METHOD OF PROCEDURE (MOP) AND ACTION PLANS (AP)

Method of Procedure

A written MOP will be required for all construction-related activities involving manipulation of energy sources or paths. The party responsible for these activities shall provide a task-specific MOP to Ryan Companies for review 30 days prior to the requested start date of the activity. All dry run comments must be addressed, and the revised MOP sent to Ryan Companies one week prior to the requested start date. The written MOP must identify the scope of work and all impacts (planned and potential) on the equipment, plant, system, etc., to safely and orderly isolate the system, perform the required scope of work, and restore the system to the as-found condition. The MOP must be supported with the following procedural and verification requirements:

1. A step-by-step work script that identifies the necessary sequences of tasks to support the required switching, valving, operation, isolation (or any combination thereof) of the equipment, system, or plant. The work script must be followed in sequential order as written and approved. Should deviation become necessary, the work effort must pause, and the requested deviation is to be escalated to Ryan Companies.
2. Any necessary documents, including but not limited to, floor plans, single-line diagrams, flow diagrams, process and instrumentation drawings, panel schedules, etc., that may be required to clearly indicate where the work will take place, how the energy sources will be controlled, and what equipment systems will be affected.
3. Supplemental items including, but not limited to, site-specific safety policies, standard operating procedures, safety data sheets (SDS), etc., to further demonstrate intent to execute the scope of work safely.
4. Documenting the as-found and as-left positions of all manipulated devices, including, but not limited to, circuit breakers, valves, and control switches (physical and soft).
5. The MOP must be "Dry Run" with Ryan Companies and the team performing the scope of work prior to submitting the MOP for approval.
6. Designated Authorized Employees (Trade partners) who are involved with any complex/group lockout and/or tagout procedures must always follow the task-specific written MOP.
7. The written work scope procedure must account for all required source verification and/or visual and physical inspection of the equipment or system to ensure safe operating conditions for all parties.
8. For electrical switching activities, the MOP shall identify the incident energy value, PPE Class and Arc Flash Boundary distance for each disconnecting means that is to be operated.
9. For mechanical activities, the MOP shall identify the potential hazard and a mitigation plan to reduce or eliminate exposure.
10. MOPs for like (or very similar) equipment can be updated with the correct equipment nomenclature and implemented without re-performing dry run requirements or the approval process. Proper notification prior to implementation and oversight during the execution of the MOP are still required.

Action Plans

An Action Plan (AP) is a reduced version of a MOP. The action plan represents simple, straightforward work efforts. All APs for moving permanent power through the original path of the power distribution system will require an AP to be reviewed and approved by the submitting Trade Contractor's Energy Marshall. The AP will then be sent to Ryan Companies for review and approval.

METHOD OF PROCEDURE (MOP) OR ACTION PLAN (AP)

METHOD OF PROCEDURE	ACTION PLAN
• > 600V present • Major OFCI Equipment (including but not limited to): Switchgear, generators, Battery/UPS Systems, Bus Duct, Chillers, Cooling Towers • CFCI Equipment • Multiple Sources Present • Multiple Shifts Required • Multiple Trades Involved • Mechanical Equipment Start-Up • Load banking of Systems	• Energization/De-Energization of LV Transformers, LV Panelboards & Circuits $\leq$ 200A • Energized Control Panel • Energization to Line Side Disconnect • Pressure System Manipulation • Valve Manipulation • Temporary Construction Power Panels, Circuits, Spider Boxes • NETA testing/Megger Testing

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• Commissioning Script Execution • Test, Fill & Flush • Initial Fill Pressurized Systems • Pipe Pressure Testing • Group/Complex LOTO Required • Temporary Power Utility Tie-In • Temporary Commissioning Power • Testing, Adjusting, and Balancing • Work in Active or Turned-Over Spaces	
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MOP Dry Run

A MOP dry run is a work effort that starts with a complete initial draft of a MOP. The dry run should begin in the office with at minimum the person who wrote the MOP and a qualified Energy Marshall. The effort starts with a complete review of the MOP and all the supporting attachments. Once the initial review has taken place, the MOP should be taken into the field where the work effort will take place. The MOP is read step-by-step (including the backout procedure) with the reviewers going to each room, equipment, breaker, valve, etc., and taking note that the MOP matches the equipment and device labeling, as-found conditions matching expected states in the MOP, and reviewing the area to understand if other hazards are present that need to be addressed within the MOP. At the completion of this effort, all notes for revision should be passed to the MOP author for implementation. The MOP dry run section should be completed and initialed at the end of this effort.

Pre-Task Planning

PTP exists to ensure workers stop and think about hazards involved with that day's work. Each hazard should be identified, along with the mitigation strategy and methods to remove or reduce the hazards to an acceptable level. A PTP must be filled out daily by the foreman of the work crew and all individuals working under this document are required to be listed on the PTP and have reviewed, discussed, and signed the PTP for the day. Under this Energy Control plan MOPs and APs do not replace the PTP documentation but are supplemental to the PTP as applicable.

MOP Tracking Log

A MOP tracking log is a tool utilized to communicate all status of MOPs. It should be used to accurately depict what MOPs have been approved, completed, submitted to Ryan Companies, awaiting approval, and MOPs that are yet to be drafted. In addition, the MOP tracking log will depict the milestone dates that need to be met for submission to ensure proper review time is given prior to the requested start date of the MOP. The tracking log should be updated by each individual trade contractor with the Ryan Companies Chief Energy Marshall providing overall oversight.

PRE-ENERGIZATION INSPECTION

Prior to the energization effort all required inspections have been completed and deficiencies corrected including but not limited to the trade contractor, QA/QC team, Ryan Companies and Client/Owner. At this time one final inspection (pre-energization inspection) will be performed by the Chief Energy Marshall or their qualified designee, designated switching crew, and any other required personnel. After the final inspection, tamper seals will then be applied and recorded on all required equipment, and all requisite inspections completed.

Torque Verification

Torque Verification is the common practice of utilizing a properly calibrated torque wrench to check that all fasteners are secured properly. Reference the fastener type or manufacturer data to determine the required torque value. All bolts are required to be torqued and verified per the manufacturers and project specifications. These should be recorded, and torque seal/markings applied to serve as a visual indication this has been performed. It is also common practice for quality control to perform a torque validation and a different color torque seal to be applied to signify this additional validation.

Megger/hi-Pot/VLF Testing

Megger/hi-Pot/VLF Testing are common tests performed on conductors. VLF (Very Low Frequency) or hi-Pot (High Potential or Dielectric Withstand) testing are commonly performed as part of NETA testing on MV conductors. This testing validates the conductor is safe to energize and place in operation. The NETA testing agency will provide the results on a standard NETA form for documentation. The hi-Pt test makes sure that no current "leaks" to any other point (wire) while a VLF test ensures the same thing

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at a much lower frequency and thereby lower power. Megger (Megohm or Insulation Resistance) testing is commonly by the installing contractor to ensure the conductor is safe to energize and place in operation. This test looks to detect breakdowns of the insulation of the conductor. This should be documented on a megger log and provided to Ryan Companies. Testing instruments should be calibrated per the manufacturer's recommendations or project specifications.

Tamper Seals/Tape

Tamper seals and tape provide a way to guarantee conditions within the protected area have not changed. Installing a form of tamper seal can ensure that no work has been performed from the time of the final inspection to the point of energization. Should a seal be broken, it is expected that a new visual inspection be performed prior to energization. Tamper seals can be utilized for, but not limited to, ensuring coordination settings, equipment inspections, and group/complex LOTO scenarios.

NETA Testing

NETA testing is a set of standardized electrical testing to perform on electrical equipment. Typically, contract specifications include NETA testing. NETA has identified all standard tests that should be performed on each piece of equipment. The NETA testing agency will perform the testing and produce the standard reports to ensure equipment is safe to be energized.

INITIAL ENERGIZATION PROCEDURE

Initial Energization/Mechanical or Start-Up

For the initial start-up of any piece of equipment, the trade contractor will be required to submit all necessary documentation to Ryan Companies at least 72 hours in advance of the scheduled work effort. This documentation includes, but is not limited to:

1. Electrical: NETA Testing Reports, Insulation Resistance Tests, and Torque Verification Reports and point-to-point verification of circuits (continuity checks/ring out of circuits).
2. Mechanical: Pressure Test reports, Flush Reports, Torque Verification Reports.
3. QA/QC: All QA/QC requirements up to this point must be reviewed including Tamper Tape Log and completed Component Verification Checklists.
4. First time breaker rack-in is not to be done manually and will be done using a remote breaker rack-in device.

Operation of Disconnecting Means

Generally, breakers may not be racked in or racked out to an energized bus. When available, disconnecting means should be operated via control devices or remote operators. Remote rack-in tools to rack breakers in and out to energized bus may be necessary under certain circumstances. In these cases a risk assessment should be performed inclusive of the use of PPE.

Control Power Energization

All equipment with an external control power source and equipment with an internal power source with easy access are required to have the control power energized prior to permanent power being brought to the equipment. During this time, it is expected that all control circuits, safety mechanisms, interlocks, protective settings, etc., will be proven and validated as operational where possible. Documentation of this effort is required to be submitted to Ryan Companies.

Equipment Power Identification

Labeling is to be provided and affixed to equipment detailing all external power sources including external control power, 125vDC station battery power, and/or temporary power.

ARC FLASH ANALYSIS, TIME CURRENT COORDINATION STUDY & SHORT CIRCUIT ANALYSIS

Prior to energization a Power System Study shall be completed and implemented. Arc Flash Warning Labels will be printed and affixed to the associated equipment and all protective settings will be implemented to relays, trip units, etc. In the absence of a completed power study, NFPA 70E Tables 130.7 (C)(15)(a)/(b)/(c) shall be used to conduct an arc flash risk assessment to determine the personal protective equipment required, provided that protective settings are set at the lowest safest setting, the Energy Reduction Maintenance Switch (ERMS) is activated, and notification is provided to Ryan Companies.

EQUIPMENT PROTECTION & CLEANING

Reasonable equipment protection should be provided to prevent any physical damage and foreign object/dust/debris infiltration into the equipment. Equipment protection can include filter media, magnetic/plastic sheeting, room air scrubbers, heaters, room

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climate control, etc. Equipment protection shall be implemented at the time of equipment acceptance. Data (temperature and relative humidity) loggers should be used to verify required storage and operation conditions as noted in the equipment submittals are being met and maintained. Cleaning of the equipment shall be performed by the installation contractor using methods and materials accepted by the manufacturer. A complete equipment protection and cleaning plan shall be submitted to Ryan Companies for review four weeks prior to the equipment acceptance. A thorough foreign objects and debris inspection shall be performed with reinspection as necessary.

ESTABLISHING A SAFE WORK CONDITION

Pre-Task Activities

The following activities must be completed prior to initiating any isolation or method of procedure:

1. Method of Procedure (MOP) will be fully developed and have all required approvals.
2. Notify all affected employees that a LOTO procedure is required and the reason for its use.
3. Pre-Task Plan and MOP will be discussed and reviewed with all employees and supervisors engaging in the LOTO activity. After initial review, should any employees be added to this work task that were not previously present, those employees must review the MOP, Pre-task Plan, and walk down to verify the current status of the MOP.
4. All required personal protective equipment (PPE), as identified in the Pre-Task Plan, will be fully inspected by the employee and supervisor before use.
5. Barricades, boundaries, signage, switching tags, mechanical valves, etc. are to be installed.
6. Planned impacts to affected systems, i.e., temporary cooling, power, lighting, etc. are in place and operational and re-fueling has been accounted for.
7. Record the as-found conditions for any building systems that will experience a planned impact due to the work effort.

Barricades

All work areas where system manipulation activities are to take place are to be properly barricaded and signage to deter unqualified/unauthorized access to the space shall be established. Barricading is required to be included in all MOPs and APs.

Boundaries

Work area boundaries are to be established for the purpose of identifying the hazard zone within the workspace. These boundaries are to be identified using the approved SCCAF study for the site. If the SCCAF study is not available, the contractor is to use the NFPA 70E Tables for determining the arc flash boundaries. Only the designated switching crew and authorized personnel will be allowed within this space. Boundaries must be included in all MOPs and APs.

Signage

All equipment and/or spaces must have signage affixed that clearly identifies the state of the equipment for the following conditions:

1. Energized equipment
2. Pressurized systems
3. Multiple sources (nominal and control)
4. When spaces become restricted, allowing only qualified personnel to enter
5. Other project requirements

Safe Working Environment Steps

The following steps must be taken to establish a safe working environment:

1. **Identify Sources.** Determine all possible sources of energy supply to the specific equipment. Check applicable up-to-date drawings, diagrams, and identification tags.
2. **Disconnect Power Sources.** After properly interrupting the load current, open the disconnecting devices for each source.
3. **Visually Verify.** Wherever possible, visually verify that all blades of the disconnecting devices are fully open, or that draw-out type circuit breakers are withdrawn to the fully disconnected position.
4. **Positive Control.** Release stored electrical energy. Release and block stored mechanical energy.
 - a. **Residual Energy.** Discharge or disconnect sources of stored electrical energy such as capacitors and battery strings. Discharge and block stored mechanical energy such as breakers, certain types of disconnects, etc. Springs shall be released, or physical restraint shall be applied when necessary to immobilize mechanical equipment and pneumatic or hydraulic reservoirs. Other sources of energy shall be blocked or relieved.

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- b. **Greater than 600 Volts.** All equipment with a nominal voltage greater than 600V shall be grounded out via integral means or supplemental grounding set. Temporary equipment grounding sets or jumpers, also known as “ground clusters,” must be installed to establish a safe equipotential zone where employees are working. Proper PPE is to be worn according to the level of arc flash risk. When work is required on MV or HV systems, the feeder breaker shall be racked out, locked out, and grounded out. Additionally, signage must be installed that indicates “Warning: Ground Clusters Installed.” These signs must be placed on the equipment immediately adjacent to the access door where the ground cluster is installed and at the associated breaker. The LOTO device key for the source breaker shall be placed in a LOTO box. The LOTO Coordinator or LOTO Manager will affix an additional LOTO lock and tag onto the LOTO box for each source breaker key that is placed in the box and the keys will be associated with the ground clusters, thus ensuring that the ground clusters must be removed before getting access to the source breaker keys.
5. **Apply Lockout/Tagout (LOTO) Devices.** Apply LOTO devices in accordance with the LOTO policy. Each Authorized Person who will be working on the equipment must put a lock on the designated lockout location. Each lock must remain on the designated lockout location while the individual’s work is being performed. Only the Authorized Person who placed the lock may remove the lock.
6. **Verify Zero Energy State.** After ensuring all personnel are clear, verify that the equipment or system is isolated by using a properly rated voltage detector to test each phase conductor or circuit part both phase-to-phase and phase-to-ground. Before and after each test, determine that the voltage detector is operating satisfactorily. This is applicable to Low Voltage (<600V), Medium Voltage (>600V), and DC Power Systems. This is to be performed at each point of work and LDL must be performed at any point downstream of the isolated equipment where work will be performed.

Returning Equipment to Service

The following items must be verified prior to returning any equipment to service:

1. Visually inspect the equipment and work area to ensure that no tools or materials have been left behind and that the equipment is safe to operate. Inspection should include any QA/QC requirements.
2. Replace all guards and other safety devices that may have been removed.
3. Safeguard and account for all employees in the area to ensure that everyone is clear of the equipment.
4. Notify affected employees that the equipment is about to be returned to service.
5. Verify that all equipment control devices are in the appropriate position to support re-energization.
6. Perform a phase-to-ground (and phase-to-phase if applicable) continuity test for all conductors that are to be re-energized under this work effort.
7. Follow the approved MOP or AP to re-energize the equipment.
8. After the equipment has been placed in service, ensure the building systems have returned to the as-found condition, are operating as expected, and the associated systems are clear of any unexpected troubles or warnings.

Mechanically Safe Working Environment Steps

The following steps must be taken to establish a mechanically safe working environment:

1. **Identify Sources.** Determine all possible sources of energy supply to the specific equipment. Check applicable up-to-date drawings, diagrams, and identification tags.
2. **Review Sequence of operations.** Prior to isolating energy sources or closing valves, understand if supplemental equipment should be shut down or specific sequences employed to allow for safe isolation of equipment.
3. **Isolate Energy Sources.** After properly isolating the energy sources, verify the rest of the system is stable and that no additional hazards have been created.
4. **Visually Verify.** Wherever possible, visually verify that valve position indicators are pointing at the closed indicator.
5. **Positive Control.** Release or block stored mechanical energy.
6. **Apply lockout/Tagout (LOTO) Devices.** Each Authorized Employee who will be working on the equipment must put a lock on the designated lockout location. Each LOTO boundary must be walked down and verified by each Authorized Employee before applying locks. Each lock must remain on the designated lockout location while the work is being performed. Only the Authorized Employee who placed the lock may remove their lock.
7. **Verify Zero Energy State.** After ensuring all personnel are clear, verify that the equipment or system that has been isolated is in a zero-energy state by checking the necessary pressure gauges and/or opening blow downs to verify no pressure is present.

AWS - ATL089 Procedures Manual

Energized Electrical/Mechanical Work

Although the best practice and preferred method is to work on de-energized equipment, OSHA and NFPA do recognize that in some circumstances it may create additional hazards or be infeasible to de-energize. Electrical tasks such as troubleshooting and testing for the presence of voltage, current, etc., can only be done while equipment is energized. In these instances, work on energized equipment is allowed, but workers must follow safe work practices and use the appropriate personal protective equipment (PPE). All planned energized work must be approved by Ryan Companies. Energized work is strictly prohibited unless one or more of the following conditions are met:

1. The requesting party can demonstrate to the satisfaction of Ryan Companies and the Chief Energy Marshall that de-energizing introduces additional or increased hazards.
2. The requesting party can demonstrate that de-energizing is infeasible due to equipment design or operational limitations.
3. The equipment operates at less than 50 volts AC or 50 volts DC. (2024 NFPA 70E 130.1)

Prior to performing any energized work, the Qualified Person shall have an approved MOP or AP for associated work.

Generator Start-Up and Safety Procedures

All activities related to generators shall follow all requirements set forth in the Energy Control Program (ECP) and shall be performed in accordance with the following requirements:

1. Only Qualified Personnel will be allowed to enter the generator enclosures during testing.
2. All personnel that will be exposed to energized live parts will comply with NFPA 70E.
3. The generator enclosures shall always be clean and free of loose parts and debris.
4. The EC will be required to fill out a generator run time log for all generator running activities.
5. The EC will be required to supply and install all temporary safety signage required including electrical hazards signage and generator running signage.
6. All personnel in the vicinity of the generator enclosures during generator operation will be required to wear appropriate hearing protection.
7. All LOTO procedures shall be followed.
8. All generator activities concerning running time will be coordinated with Ryan Companies and the Electrical Contractor.
9. The following steps must be taken prior to the generator engine being started:
 - a. Ensure that barriers and adequate signage are in place at all entry points to the generator area.
 - b. Keep unqualified personnel out of the generator enclosure during running times.
 - c. Inform nearby personnel that the generator will be starting and that hearing protection is required.
 - d. Keep all doors to the generator enclosure closed during testing.

SECTION 6: FUNCTIONAL AND FIRST COMMISSIONING

Functional Performance Testing is to validate the equipment operates per the sequences of operations and to prove they all interface with other systems. These tests are very detailed and ensure systems and equipment perform at their rated design function with required redundancy and respond to failure conditions according to the design intent and the contract documents.

Load Banking of Functional Systems

Functional testing may require the use of load banks. All activities related to load banks shall follow the requirements set forth in the Energy Control Program (ECP). Although wiring for rental load banks is temporary wiring, the National Electrical Code (NEC) requires the wiring to comply with all NEC wiring requirements. If not properly planned and installed, there is a risk that the temporary load bank cables and their connectors can severely overheat. This Energy Control Plan establishes guidelines which are considered the minimum expectation for the installation of temporary load bank cables on each project.

Load Bank Cable Sizes and Quantities

Load bank cables are usually rented from the load bank rental supplier and are typically sized for free air installations. Due to higher ampacities allowed for free air cable installation, load bank vendors will typically use #4/0 cable for up to 400 amps, versus the standard 225-amp rating for #4/0 building wire installed in conduit. If the contractor chooses to use free air ampacity ratings, then they are required to install the cables in such a way that they do not cross and are indeed in free air the entire route. If the cables cannot be installed in this manner, the Electrical Contractor should use the more conservative ampacities for wire installed in raceways. The Electrical Contractor should also include voltage drop analysis in the calculations.

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Load Bank Installation Checklist Prior to Operation

The installation contractor must submit a plan for each load bank addressing the following:

1. List the number of cables.
2. The means of connecting to the appropriate bus and breaker.
3. The cable routing plan.
4. A safety plan addressing limiting access to cables and connections.
5. The means to disconnect the system in the event of a load bank fire, cable fire, or other safety issues.
6. Verification that the overcurrent protection settings are appropriate for the cables and load bank being tested and utilized.
7. Validation that if the free air rating is used, that the cables will not lay on top of each other and there will be no cable crossing as heat is additive.
8. Verification that finished surfaces or combustible materials are protected from the load bank cables.
9. Perform a review of the arc flash hazard at the exposed connections to determine the arc flash boundary surrounding the exposed live parts.

Initial testing

When performing initial testing of the load bank do the following:

1. Verify that barriers and guards are in place to prevent unqualified personnel from entering the area and the arc flash boundary.
2. Verify that the load cabling is installed correctly to the feeder breakers prior to energization of the load bank.
3. Verify normal operation of the load bank, fan rotation, and impact of the exhaust air on the surrounding area. This is especially a concern for load bank operations inside the building.
4. Apply no more than 25% of the rated load of the load bank. Using a clamp-on ammeter, measure the flow through all leads to confirm that the current flow through each cable is equal. This step is important to prevent overloading of any one cable.

Monitoring During Operation

The following guidelines should be observed when operating load banks:

1. Guard the cabling route and the exposed connections to ensure that unqualified persons do not enter the area.
2. Regularly monitor the temperatures for the cable using a portable IR temperature scanner to verify that the cable temperature does not exceed the cable rating.
3. Monitor the temperature of the output cables. The cable temperature should not exceed 80% of the rated value of the insulation.
4. Check cam-lock connections daily.
5. The Electrical Contractor will monitor the temperature of the room to ensure that the design temperature of the room is not exceeded so damage to the sprinkler heads does not occur.
6. Under no circumstances should a load test be performed without at least one person monitoring the equipment and the load bank continuously.
7. Any increase or decrease in load, except in emergency conditions, needs to be coordinated among the Electrical Contractor, Commissioning Provider, and the project Commissioning Team.
8. The General contractor will ensure that adequate fire extinguishing devices are located inside the test area.
9. It is the responsibility of the Electrical Contractor to ensure safe operation of the equipment being tested and operated. This includes being aware of and mitigating potential electrical hazards and wearing the appropriate PPE.

Integrated Systems Testing is to validate that all systems and equipment interact properly to deliver the performance and reliability that is required. The last functional test will conclude with all systems left in a fully automated state and enabled with final turnover settings. Functional and First Commissioning inherently introduces new or more parties to the project and it is critical to ensure that these individuals are properly trained to adhere to the requirements of the Energy Control Plan including the appropriate use of PPE. Each test script will require a MOP that the commissioning script will be attached to.

Power Distribution Center (PDC) Load Banking

AWS - ATL089 Procedures Manual

Operation minimum requirements for all Power Distribution Center load bank use/operation. Subcontractor(s) competent person is responsible for implementing and documenting the following steps at a minimum into the subcontractor(s) project LOTO policies and procedures.

1. All workers connecting to or operating the PDCs shall be trained in the operation of that equipment, including proper load bank connections.
2. Prior to connecting/disconnecting any load bank cables to a PDC, the PDC main breaker shall be locked in the open position.
3. After connecting load bank cables to a PDC, and prior to closing the PDC main breaker, a 1 over 1 check shall be completed to ensure cables are connected in the correct phase sequence by the subcontractor's competent person.
4. See the attached Method of Procedure required for all Power Distribution Center (PDC) Load Bank Testing in Data Hall in Exhibit.

SECTION 7: ECP COMPLIANCE AUDITING

Compliance with the Energy Control Program and LOTO Program is a requirement of all employees working onsite and applies to all onsite personnel who will be involved with the control, isolation, observation, testing, and supervision of system safeguarding from hazardous energies. Each company is responsible for the implementation and enforcement of this policy. This program defines the responsibilities of the individual companies and contains standard procedures to maintain compliance with this program.

The audit process will be used to verify the proper execution of the ECP and MOPs/APs being implemented properly. The audit will also verify that the proper planning is taking place, dry runs are conducted, the level of detail in the plans is appropriate for the work, and the proper execution of the MOPs/APs are taking place and being documented. Each subcontractor's Energy Marshall will be performing or participating in audits of the ECP and LOTO Program. All LOTO and/or MOP/AP procedures are subject to an audit. Audits can be completed before, during, and after the procedure. Noncompliance issues are to be documented and tracked in a shared spreadsheet or other tool that all parties have access to. Noncompliance issues are to be corrected immediately.

ECP TASK FORCE AUDIT RESPONSIBILITIES.

PARTIES	ROLES	RESPONSIBILITIES
Ryan Companies	Energy Marshals/LOTO Managers/Safety Professionals	• Coordinate with Client, CX Provider, and subcontractors to ensure daily/weekly/monthly audits are performed and documented. • Collaborate with the Client and Subcontractors, at minimum quarterly to perform overall ECP/LOTO Program assessments.
Owner/Client	Energy Marshals	• Provide direction and feedback based on past audits & assessments.
Commissioning Provider	Energy Marshals	• Collaborate with Client and Subcontractors, at minimum quarterly, to participate in and perform overall ECP assessments.
Electrical Contractor	Energy Marshals/Safety Professionals	• Perform daily/weekly/monthly audits and report any trending opportunities/best practices to Ryan Companies.

AWS - ATL089 Procedures Manual

		Collaborate with Ryan Companies and other contractors to perform internal and program assessments at least quarterly.
Mechanical Contractor	Energy Marshals/Safety Professionals	- Perform daily/weekly/monthly audits and report any trending/best practices to Ryan Companies. - Collaborate with Ryan Companies and other contractors to perform internal and program assessments at least quarterly.
Fire Protection Contractor	Energy Marshals/Safety Professionals	- Perform daily/weekly/monthly audits and report any trending opportunities/best practices to Ryan Companies. - Collaborate with Ryan Companies and other contractors to perform internal and program assessments at least quarterly.

SECTION 8: LOTO POLICY GUIDANCE

STANDARD LOTO POLICIES AND PROCEDURES

Standard LOTO Policies and Procedures

All LOTO devices must comply with NFPA 70E. They must also meet the following criteria:

1. In group/complex LOTO scenarios, the keys under group control must be readily visible for verification purposes. For example, group LOTO boxes must have a clear cover.
2. Lockout devices must be reviewed and approved by Ryan Companies.
3. A LOTO Chain of Custody form will be required when transferring LOTO responsibilities to other parties. Allowable types of equipment that may be transferred include, but are not limited to, service disconnects, unit disconnects, or other approved devices. The party accepting the transfer of custody will be required to maintain all applicable codes and standards for safeguarding personnel and equipment from unintentional contact with incident energy.
4. Each trade contractor is required to maintain LOTO records in a format reviewed with Ryan Companies.
5. Combination locks may not be used.
6. Interlocks, selector switches, software controls, and control circuit devices may not be used for LOTO purposes. In some cases. Emergency power off buttons may be used for LOTO if they are specifically designed for this application and have an integrated lockout device attachment.
7. Attachment plugs of electrical devices, equipment, or appliances are permitted to be used as disconnects provided the control of the attachment plug is maintained using a suitable LOTO device.
8. Energy source locks will be identified by the color **BLACK** and reviewed by Ryan Companies. These locks are required to be unique with 1 key and 1 lock.
9. Personal protection locks must be identified by the color **RED**. These locks are required to be unique with 1 key and 1 lock.
10. Control locks will be identified by the color **BLUE** and be reviewed by Ryan Companies.
11. Personal lock procedures require qualified individuals to review the single-line diagram noting where energy sources (LOTO) are present. Visually verify the locks in place at the sources of all energy and confirm the energy sources of all

AWS - ATL089 Procedures Manual

energy and confirm the energy source has been isolated (breaker is open/valve is closed, etc.). record the lock information and verify the keys associated with the locks match. Place the personal LOTO device on the group lockout device, sign the LOTO log and witness or perform a Live-Dead-Live zero energy state verification procedure. Procedure to

12. Two-man crews are required at minimum in all cases requiring energy isolation. If working alone and the work is identified as low hazard, a plan must be approved before proceeding with LOTO.
13. Authorized and affected workers must complete initial LOTO orientation training prior to performing work on site. A roster of authorized LOTO individuals will be kept to assist in controlling access to energized spaces.
14. After completion of work or end of shift, each person is to remove their personal lockout/tagout device and sign out on the LOTO log. All personal LOTO devices must be removed daily.

ZERO ENERGY STATE

Voltage Presence Verification

All electrical energy verifications shall be done utilizing a Live-Dead-Live (LDL) procedure to verify the presence of, or lack of, energy. All energy paths and sources shall be considered energized until proven otherwise. The following criteria must be followed:

1. Non-Contact Voltage detectors (NCVD) may not be used as the sole means for voltage verification on an electrical system. A contact meter or tester must be used.
2. All equipment used for voltage verification must have the appropriate voltage class and the test equipment must have a current calibration certificate.
3. The LDL test shall be conducted with the presumption that all equipment being tested is potentially fully energized at its maximum potential.
4. All energy paths sources shall be considered energized until proven otherwise.
5. LDL testing shall be performed by qualified persons.
6. LDL testing shall be performed using appropriate PPE based on the hazard level present.
7. In addition to the qualified person completing the LDL test, LDL testing shall be witnessed by at least one Energy Marshal to establish an appropriate LOTO device procedure. Each person working under this LOTO device must also perform or witness the LDL in accordance with the personal lockout procedure.
8. A Live-Dead-Live confirmation document shall be completed as required by the MOP or AP. This document shall identify the equipment being tested, the person completing the test, the required witness, and the date and time the LDL was performed.

Step-by-Step Process for Performing a Live-Dead-Live Verification

The LDL verification steps are as follows:

1. Review the Arc Flash Label or the NFPA 70E Standard to determine the applicable PPE Category. All circuit parts shall be considered energized until proven otherwise. PPE must be worn until the absence of electrical energy can be verified.
2. The test instrument must be inspected for working conditions and verified for correct operation prior to performing Live-Dead-Live testing.
3. While wearing appropriate PPE, validate the metering equipment is of appropriate capacity to handle the voltage to be tested using high or low voltage meters as appropriate.
4. While wearing appropriate PPE, measure the voltage of the equipment to be tested to verify a zero-energy state.
5. While wearing appropriate PPE, re-verify that the meter is still functioning properly by going to the same known voltage source used previously to measure the voltage.
6. It is now reasonable to determine that the circuit in question is de-energized, and PPE can be removed.

CERTIFIED DEAD PROGRAM

When a Live-Dead-Live is performed, the Certified Dead Program may be used to facilitate additional work without the need for additional Live-Dead-Live to be performed. Serialized seals should be used to prove the LOTO has not been tampered with. An example of this type of record is appropriate for use when a single breaker or disconnect controls the power to multiple downstream circuits, proving the single source has been de-energized, locked out, tested and sealed, effectively isolating all circuits downstream. Even when an LDL report has completed a sealed, a test-before-touch is still needed. The circuit can be assumed to be de-energized using a non-contact voltage tester for initial approach. If the NCVD does not indicate any voltage

AWS - ATL089 Procedures Manual

present, then using a contact tester to verify zero-energy is appropriate. This program does not supersede the ability of each employee to perform or request a Live-Dead-Live.

Witness

In addition to the qualified person completing the LDL test record, LDL testing shall be witnessed by at least one other qualified person. A Group Lockout Supervisor or LOTO Coordinator shall either perform the test or be a witness.

Documentation

Live-Dead-Live Records shall be completed when documented LDL testing is required. This record shall identify the equipment being tested, the person completing the test, the required witness, the date and time the LDL was completed, and the serial number of the seal. The completed record shall be collected at the end of each day by the LOTO Coordinator/LOTO Manager. Files are to be maintained for the life of the project.

Serialized Seal

A serialized seal may be applied to equipment, lock box, or lockout station to verify that the LOTO after LDL testing has not been tampered with. The number of the serialized seal shall be noted on the LDL Record.

EMERGENCY PROCEDURES AND INCIDENT REPORTING

In the event of an imminent danger, incident, or other similar emergency, all personnel are required to notify a qualified person to assess and take immediate action to remove and abate any potential hazards. Once a safe condition exists, the qualified person will notify Ryan Companies and any other appropriate Energy marshal. The Energy Marshall will ensure appropriate LOTO measures are in place. The Energy Marshal will perform a root cause analysis of the incident and will identify corrective actions to prevent recurrence.

COMPLEX/GROUP LOTO POLICIES AND PROCEDURES

When servicing, maintenance, inspection, start-up, or operational activities are performed by multiple parties or multiple energy sources exist, a procedure shall be utilized which affords the employees a level of protection equivalent to that provided by implementation of a personal lockout/tagout device. In these scenarios, complex lockout/tagout shall be utilized. The primary company will designate an Energy Marshal that is responsible for work effort coordination, verification of LOTO means and methods as well as the MOP or AP execution. The Energy Marshal shall provide oversight for the duration of the work activity. Complex LOTO will be required when one or more of the following conditions are present: Multiple energy sources, multiple crews, multiple crafts, multiple locations, multiple employers, multiple disconnecting means, particular sequences, or multiple shifts.

The Energy Marshal must implement and coordinate the LOTO of all energy sources and verify that the steps taken in accordance with the specific MOP or AP have in fact isolated the equipment effectively from the energy sources. This must be accomplished before employees participating in the Complex/Group LOTO affix their personal lockout device to the group LOTO box and before performing the specific scope of work.

For Complex/Group LOTO scenarios, the Energy Marshal will be responsible for establishing a safe working condition by reviewing the MOP or AP, designating LOTO points, applying energy source LOTO, and verifying zero energy. The Energy Marshal shall secure the energy source LOTO key in a group lockout device allowing employees to lockout to the group lockout device after the energy source LOTO key has been secured.

Prior to individuals beginning work on a piece of equipment the individual must review the single-line diagram with a qualified person, noting where energy sources LOTO are present. Visually verify the locks in place at the sources of all energy and confirm the energy source has been isolated. Record the lock information and verify the keys associated with the locks match. Place their personal LOTO device on the group LOTO device and sign the LOTO log and witness or perform a Live-Dead-Live process to verify zero energy.

AWS - ATL089 Procedures Manual

After completion of work or end of shift, each person is to remove their personal lockout/tagout device and sign out on the LOTO log. All personal LOTO devices must be removed daily.

The party responsible for startup has final energy control. Ryan Companies must be included in the handoff of LOTO processes to confirm paperwork and requirements are complete before energization.

Energization and De-energization activities and schedules and equipment and personnel involved will be discussed in daily energization meetings and commissioning meetings by building before the start of work.

SHIFT/PERSONNEL CHANGES

Shift/Personnel Changes

For Complex/Group LOTO scenarios that span multiple shifts, multiple days of work effort, or changes in personnel, a Chain of Custody (COC) form must be completed at each change in responsibility. The COC form must accompany an approved MOP or AP.

ABANDONED LOTO SCENARIOS

Abandoned LOTO Scenarios

In situations where the LOTO device owner is not present and re-energization is required, a lock may be removed by an Energy Marshal once the following steps have been completed:

1. Every effort must be made to contact the lock owner before the lock can be removed.
2. The attempts to notify or contact the lock owner must be documented.
3. The Abandoned Lock Removal Request Form is completed and signed off on by Ryan Companies, the subcontractor's management, and the subcontractor's safety manager.
4. Validate in the access control system that the individual is not on site, then disable their badge due to an abandoned lock.
5. After the lock is removed, the lock owner is notified within 24 hours or prior to them returning to work, whichever comes first.

DESIGNATED PERSONNEL REQUIREMENTS

Energy Marshal

An Energy Marshal shall be designated by the subcontractor to perform MOP execution. In Complex/Group LOTO scenarios an Energy Marshal may be designated to provide LOTO for employees working under their supervision provided the following criteria are met:

1. The Energy Marshal is identified in the MOP.
2. The MOP vests primary responsibility in the Energy Marshal for employees working under their protection and acknowledges that they will be held accountable for the safe execution of LOTO.
3. The MOP identifies the method to account for all persons working under the protection of the Energy Marshal.
4. Each employee participating in the group LOTO must be informed of their right to verify the effectiveness of the lockout measures. Each employee must be allowed to personally verify that hazardous energy sources have been effectively isolated. Employees will witness this verification only after an initial voltage test is conducted so that an arc flash boundary and PPE are not required for the verification.

Qualifications for a Designated Energy Marshal

The resume of a subcontractor's Energy Marshal must be submitted to Ryan Companies sixty days in advance of the first planned MOP execution. The Energy Marshal must meet the criteria listed below and be approved by Ryan Companies:

1. NFPA 70E Certified with update training every three years
2. Full understanding of the commissioning process
3. Ability to competently execute MOPs and APs
4. Identify the correct PPE for an activity
5. Understand and correctly interpret ECP signage
6. Expert mastery of single-line diagrams and flow diagrams
7. Understanding of when to back out of a MOP

AWS - ATL089 Procedures Manual

8. Understanding of the company and site ECP and LOTO policies
9. Superior organizational and leadership skills

Designated Switching Crew

The Designated Switching Crew (DSC) will be responsible for reviewing and executing procedures for plant manipulation, isolation and switching activities for the purposes of safely distributing energy sources. The DSC shall be present and responsible for verification of all as-found and as-left conditions of the affected systems, regardless of who requested the alteration of the system and performed the work. If conditions are found outside of the coordinated effort or left in a condition that is deemed to be unsafe, the foreman of the DSC is to escalate the condition to Ryan Companies immediately.

The Energy Marshal will supervise the DSC's and ensure compliance with this policy. Each trade contractor will identify all DSC personnel. Requirements for the DSC position include the following:

1. Specialized training on the equipment and means and methods used in installation.
2. Specialized safety training regarding the proper use and methods involving LOTO.
3. Formal NFPA 70E 8-hour training certification
4. Qualifications of the intended DSC personnel identifying their relevant experience must be available for review by Ryan Companies upon request.

An equipment vendor can perform switching activities provided they adhere to the following requirements:

1. Vendor personnel meet the DSC qualifications listed above.
2. Vendor only performs switching on the equipment manufactured by their company.
3. Vendor has communicated all switching activities with the DSC.

Some examples of tasks performed by the DSC for others may include the following:

1. Complex MOP execution.
2. Valve manipulations for the purpose of testing, balancing, and QA/QC activities.
3. Equipment testing by a manufacturer's representative.
4. Any other activity approved by Ryan Companies.

Authorized Employee

Authorized Employees are defined as any qualified person who is directly responsible for the supervision or execution of any work scope involving LOTO procedures. All Authorized Employees shall be responsible for the following:

1. Following LOTO procedures whenever working on applicable equipment.
2. Removing personal protection locks and tags which must occur when the Authorized Employees are not working on the equipment or when the work effort ends.
3. Verifying that people in the area that are not part of the LOTO work team understand that they are Affected Employees prior to the start of the LOTO procedure.
4. Communicating and coordinating LOTO efforts with all Affected Employees, multidisciplinary Authorized Employees, coordinators and managers.
5. Understanding when equipment and systems need to be locked and tagged out for service, testing and maintenance.
6. Understanding when MOPs and APs are required and which to utilize in different scenarios.
7. Ensuring that all incidents and near misses involving the release of hazardous energy are reported to their supervisor and Ryan Companies immediately.
8. Apprentices if properly trained and supervised may perform some LOTO tasks.
9. Electrical Contractors shall provide electrical awareness training during site orientation and will keep a roster of all electrically authorized and qualified trade partners on site.

Qualified Person

A Qualified Person is one who has demonstrated skills and knowledge related to the construction and operation of electrical and/or mechanical equipment and installations and has received safety training to identify hazards, implement controls, and reduce the associated risks.

Qualified persons shall be tracked via certification in the site access and safety systems.

AWS - ATL089 Procedures Manual

Affected Employee

Affected Employees are all other employees in the affected areas. An Affected Employee may also be defined as a person whose job requires him/her to work in an area in which LOTO procedures are being performed. Affected Employees must be made aware when procedures are being performed and what precautions must occur to ensure the Affected Employee's safety.

AWS - ATL089 Procedures Manual

Exhibit 2 - Article 9 – Safety Requirements and Procedures

Ryan Subcontract Agreement – Article 9

Safety Requirements and Procedures

9.1. Contractor's Safety Policies and Procedures

- 9.1.1. Project Safety Manual. Contractor will prepare a Project Safety Manual that shall include any and all requirements of the Owner and/or Exhibit A. Subcontractor and any and all of their sub-tier contractors (hereinafter referred only as Subcontractor), agrees to adhere to all requirements contained therein. The procedures outlined in this manual are minimum requirements and shall not relieve Subcontractor and any of their sub-tier contractors of any of its safety responsibilities set forth in this Subcontract. Failure to adhere to the requirements of the Project Safety Manual shall be a material breach of this Contract.
- 9.1.2. Contractor's Right to Remedy Deficiencies. If Subcontractor or any sub-tier contractor, fails to comply with any safety procedures required by law or by Contractor, Contractor shall have the right, but not the obligation, to take any necessary steps to remedy the deficiency and charge the entire cost thereof to Subcontractor. The failure of Contractor to take any such action shall in no manner relieve Subcontractor of any of its safety responsibilities set forth in this Subcontract. Contractor reserves the right to implement Article 9.2.3 regarding additional safety personnel.
- 9.1.3. Contractor's Safety Disciplinary Program. Unless otherwise described in the Project Safety Manual, Subcontractor and any of their sub-tier contractors, shall be subject to the following disciplinary actions for safety violations by its employees.

9.1.3.1. Non-serious Violations

- 9.1.3.1.1. First violation – Contractor will verbally warn the violator.
- 9.1.3.1.2. Second and third violations – Contractor will verbally warn the violator and send a written warning to Subcontractor. Subcontractor will be required to promptly respond in writing to Contractor's written warning. Such response shall state the actions to be taken by Subcontractor to correct the repeated safety violation and assure that it will not occur again. Acceptable actions may include removal of the violator from the Project.
- 9.1.3.1.3. Fourth violation – Contractor will notify the violator and Subcontractor. Subcontractor will immediately remove the violator from the Project.

9.1.3.2. Serious Safety Violations

- 9.1.3.2.1. First violation – Contractor will verbally warn the violator and send a written warning to Subcontractor. Subcontractor will be required to promptly respond in writing to Contractor's written warning. Such response shall state the actions to be taken by Subcontractor to correct the safety violation and assure that it will not occur again. Acceptable actions may include either removal of the violator from the Project or a review of the problem with the violator and a decision to retain the violator on the Project, with remedial action taken.
- 9.1.3.2.2. Second violation - Contractor will notify the violator and Subcontractor. Subcontractor will immediately remove the violator from the Project

9.1.3.3. General

- 9.1.3.3.1. If any of Subcontractor's employees are removed from this Project as a result of safety violations, they cannot be transferred to another project being constructed by Contractor.
- 9.1.3.3.2. The removal of any of Subcontractor's employees from the Project shall in no manner affect Subcontractor's obligation to perform its Work in accordance with the Project Schedule.
- 9.1.3.3.3. Nothing contained in this disciplinary policy shall prohibit Contractor from immediately removing from the Project any employee of Subcontractor whose conduct constitutes a serious violation of any safety policy, regulation or requirement which could result in serious injury to any person or serious damage to any property.
- 9.1.3.3.4. Contractor shall, in its reasonable judgment, determine which actions are serious and which are non-serious.

9.2. Subcontractor's General Safety Responsibilities

- 9.2.1. Compliance with Rules, Regulations and Laws. Subcontractor shall take all necessary safety precautions with respect to its Work, shall comply with all safety measures initiated by Contractor, or required by the Owner or this Subcontract, the Project Specific Safety Manual and the expectations set forth in regular jobsite meetings, including, without limitation, foreman meetings, progress meetings and site safety meetings. Subcontractor shall comply with all applicable laws, ordinances, rules, regulations and lawful orders of any public authority for the safety of persons or property. Any legal requirements that are more stringent than Contractor's requirements will take precedence. Failure to attend required safety meetings or briefings, including but not limited to Site Orientations, Safety committees, etc., set by Contractor without prior to approval from the Contractor will be considered a breach of contract. Failure to submit required safety paperwork by the project, including but not limited to daily permits, pre-task plans, etc. is also a breach of contract.

AWS - ATL089 Procedures Manual

- 9.2.2. Responsibility for Subcontractors. Subcontractor shall be responsible for the safety of its employees and the employees of all of its tier subcontractors. Any reference herein to "Subcontractor's employees" or "its employees" shall apply equally to Subcontractor's employees and to any employees of sub-tiers. Subcontractor is responsible for ensuring any/all tier-subcontractors attend a Preconstruction Subcontractor Safety meeting in which a representative of the subcontractor is in attendance. If the sub-tier is operating under the Subcontractors safety program, the Subcontractor must submit (something to show that the sub-tier has been trained on the subs programs), if the sub-tier is operating under their own safety program, sub-tier must submit their own paperwork and have their own Preconstruction Subcontractor Safety form.
- 9.2.3. Additional Safety Personnel. Contractor reserves the right to require Subcontractor, at Subcontractor's own cost, to add one or more people to its crew who are dedicated to the safety of Subcontractor's employees at the Project, if Subcontractor's safety measures are deemed substandard as documented in Contractor's regular safety audits. Safety Representative is defined as one who meets the definition of OSHA's competent person and has a valid OSHA 30-Hour construction safety course certificate, possess a current FIRST AID/CPR/AED from a nationally recognized program and has a minimum of 3 – 5 years of construction management experience.
- 9.2.4. LOTO Coordinator / Manager All subcontractors working with energy sources as defined by Federal OSHA on the project shall designate a trained LOTO Coordinator/Manager to oversee the subcontractors LOTO program. Subcontractor(s) and any tier contractors shall each designate a LOTO Coordinator on the project who will be responsible for the implementation, training and regulatory compliance of the subcontractor's LOTO policy and procedures. The LOTO policy shall meet the minimum requirements as listed in the Ryan Project Safety Manual LOTO Policy or any Client project requirement related to LOTO
- 9.2.5. Energy Marshall(s) All electrical subcontractors working on a Mission Critical project shall provide a dedicated on-site Energy Marshall(s). The Energy Marshall will be responsible for the safe energization / de-energization activities for the project. Additional Energy Marshalls may be required by the contractor at the subcontractor's cost. The Energy Marshall will be responsible for the training, coordination, compliance, emergency response, inspections, LOTO policy/procedures, permitting, and Method of Procedures (MOP) of the subcontractor. The Energy Marshall will provide oversight and final submission to the contractor for all the subcontractor(s) incident investigation(s), root cause, and corrective actions related to energization / de energization activities.
- 9.3. Subcontractor's Safety Responsibilities Prior to Commencing Construction
 - 9.3.1. Training on Project Safety Manual. Subcontractor shall designate an employee to receive orientation and training from Contractor on the Project Safety Manual. Such designated employee shall then train all employees of Subcontractor who will be working on the Project on the safety procedures contained in the Project Safety Manual. Subcontractor warrants that employees of Subcontractor will not be allowed to commence work on the Project until they have received such training. After Subcontractor completes the training of its employees, it will have each employee sign a form provided by Contractor indicating that the employee has been trained on the procedures in the Project Safety Manual. Subcontractor shall submit the executed form to Contractor's superintendent prior to allowing its employees to commence work.
 - 9.3.2. Competent Person. Subcontractor and all sub-tiers shall provide a properly trained "competent person," as defined by OSHA regulations. Such competent person shall be an employee of managerial level unless Subcontractor notifies Contractor to the contrary and shall be on the Project site at all times while their scope of work is being performed. OSHA Standards – 29 CFR 1926.32(f) define a competent person as one who is capable of identifying existing and predictable hazards in their scope of work, surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees and who has authorization to take prompt corrective measures to eliminate them."
 - 9.3.3. Job Hazard Analysis. Subcontractor shall prepare a properly completed job hazard analysis (JHA) addressing the complete scope of work and submit it to Contractor's superintendent prior to commencing any work on the Project. The Subcontractor's competent person shall train all employees and their sub-tiers on the job hazard analysis prior to beginning any work on the Project. The JHA shall identify and evaluate hazards, safe methods of hazard control and personal protective equipment (PPE) required to safely perform the work. The JHA must be updated and personnel re-trained whenever changes in facilities, equipment, materials, or processes have occurred. Additional JHAs may be required to be submitted daily .by the competent person.
 - 9.3.4. Safety Data Sheets (SDS). Subcontractor shall submit hazard communication information along with all applicable SDS documents to Contractor's superintendent prior to commencing any work on the Project. SDS information must be submitted both electronically and in hard copy and be shared with any non-construction onsite if they may be affected by (or exposed to) said material (i.e. tenants, visitors, owners' representatives, etc.). 15-minutes portable eyewash stations must be provided by Subcontractors if required by SDS for any product used on site by Subcontractor's personnel.
 - 9.3.5. Subcontractor's Safety Program. Subcontractor shall submit an electronic and hard copy of its site-specific safety program to Contractor's superintendent prior to commencing any work on the Project. Subcontractors are required to submit the Site-Specific Safety Program for all of its sub-tier contractors. If sub-tier employees are working under the subcontractor's safety program, the subcontractor shall be responsible to train their sub-tier contractors on the subcontractor's safety requirements/policies.
 - 9.3.6. Subcontractor's Drug Testing Program. Subcontractor shall submit a copy of its drug testing program to Contractor's superintendent if requested by Contractor or Owner.

AWS - ATL089 Procedures Manual

9.3.7. OSHA 30 Training. At all times when Subcontractor is working on the Project site, Subcontractor shall have at least one managerial level employee (competent person, lead person, foreman, superintendent or jobsite manager) who holds a valid OSHA 30- Hour construction safety course certificate. Subcontractor agrees to provide documentation of such training promptly upon request of Contractor. Subcontractor has to have a representative on site when sub-tier is performing work.

9.4. Subcontractor's Safety Responsibilities During Construction

9.4.1. Project Safety Manual. Subcontractor shall be responsible to enforce compliance with the policies and procedures in the Project Safety Manual by all of its employees and tier subcontractors.

9.4.2. Erection of Safeguards. Subcontractor shall erect and maintain, as required by Federal/State OSHA regulations, the Project Safety Manual, and the Project site conditions and the progress of its Work, all necessary safeguards for the safety and protection of persons who may be affected by Subcontractor's Work. Such safeguards shall include posting of "danger" signs and other warnings, erecting warning lines, notifying any individuals who might be adversely affected by Subcontractor's Work and promoting and enforcing safety regulations at all times. Any guardrails removed must be re-installed immediately following completion of task (100% tie-off and hard-rail barricade system must be in place). See also 9.4.11

9.4.3. Structural Overloading Prevention. Subcontractor shall not load or permit any part of its Work or the Project to be loaded in a manner that would create a safety hazard. Subcontractors shall not load equipment and/or materials which exceed structural limitations or conduct any loading operations that would create a safety hazard. Engineering verification approval may be required.

9.4.4. Altering Safety Measures Installed by Others. Subcontractor shall not remove, alter or otherwise impair the effectiveness of safety measures installed by Contractor or others without the approval of Contractor. Where safety measures must be removed or altered to allow Subcontractors to perform work such as but not limited to cables, barriers, floor hole covers, guardrails, the subcontractor shall provide additional controls or safety measures required to maintain safe working conditions.

9.4.5. Safety Meetings. Subcontractor shall hold weekly safety "toolbox" meetings with its employees and submit a copy of the agenda and meeting minutes (as available) to Contractor's superintendent. Subcontractor shall require the Competent Person or the Safety Representative to attend the Monthly Safety Committee Meeting. Subcontractor shall require all of its employees to attend Contractor's All Job Safety meeting held at least once a month. It is the Subcontractor's responsibility to communicate all safety and other important information to its employees in whatever language they primarily speak. If Subcontractor is utilizing non-English speaking employees, Subcontractor shall, at all times, have at least one English speaking employee on the Project Site who can interpret for its non-English speaking employees. Language barriers will NOT be an excuse for lack of safe practices and additional translators may be required by the contractor at the subcontractor's cost.

9.4.6. Reporting Accidents. Subcontractor shall promptly report all incidents, accidents, minor injuries, damage to property, or near misses to Contractor's superintendent. Subcontractor shall continue to report all information that pertains to the accident/injury/property damage until a resolution is reached with the injured person(s) and the person(s) is back to work and/or the damaged property is repaired or replaced. Subcontractor shall submit their report promptly (including a Root-Cause Analysis and Corrective Action(s) plan to prevent reoccurrence) as requested by the Contractor or Owner/Client requirements.

9.4.7. Forklift Operators. If Subcontractor and/or its sub-tier utilizes forklifts/powered industrial trucks to perform its Work, they shall employ only currently certified forklift operators to comply with the 29 CFR 1910.178 Forklifts/Powered Industrial Trucks standard and manufacturer specifications/recommendations. All attachments shall be approved by the manufacturer and rated with an identification for the use. Operators may be subject to additional training requirements when specific attachments are used in a manner that would reclassify the equipment as a crane. Refer to 29 CFR Subpart CC 1926.1400 (c)(8).

9.4.8. Crane Operators. If Subcontractor utilizes cranes to perform its Work, only certified crane operators will be used when required by the Owner or Federal, State or local regulations. Subcontractor(s) will be required to provide the Crane Operator Evaluation documentation and a lift plan. Crane Erection Plan shall be submitted prior to assembly / disassembly of the crane. Additional documentation to be provided by the subcontractor(s) shall also identify the qualified person who has performed the annual inspection of the crane(s), qualified rigger(s), qualified signal person(s), competent person performing the daily inspections and the assembly/disassembly director. If the crane will be moved (traveled) around the jobsite, a detailed job hazard analysis must be completed and turned in specifically addressing safe traveling operations.

9.4.9. Helicopters. Subcontractor represents that all use of a helicopter shall be in accordance with all laws, rules, regulations and guidelines relating thereto, including but not limited to any regulations of the Federal Aviation Association. Subcontractor shall provide a lift safety procedures/plan including the items listed in the Helicopter Crane Procedures Form at least 30 days prior to operation on the Project.

9.4.10. Erection Equipment. If Subcontractor and/or its sub-tier utilizes Erection Equipment to perform its Work, only certified operators will be used when required by the Contractor, Owner/Client, or Federal, State or local regulations. The Subcontractor shall provide a lift plan; if any of the lifts taking place meets any of the conditions referenced in 5.22, Lift Plans, Safe Lift Permit, and a critical lift plan shall also be submitted prior to start of work.

AWS - ATL089 Procedures Manual

- 9.4.11. Personal Protective Equipment. Subcontractor and Sub-tier employees shall report to the Project site with proper **jobsite required** apparel and safety equipment. The subcontractor's competent person shall determine if additional PPE is required for their task(s). Subcontractor's employees shall:
- 9.4.11.1. Wear a hardhat or ANSI Type 1 safety helmet (see jobsite requirements) at all times while on the Project site.
 - 9.4.11.2. Wear ANSI approved safety eyewear at all times while on the Project site. ANSI Approved Prescription glasses require side shields. The use of additional eye protection/face shield must be worn as recommended by the equipment or tool manufacturer.
 - 9.4.11.3. Wear adequate work shoes (above ankle height) at all times on the Project site. Tennis shoes or casual shoes will not be permitted.
 - 9.4.11.4. Wear proper clothing (not less than long pants and a short-sleeved shirt) and wear additional protective clothing as required by the work being performed by the employee.
 - 9.4.11.5. Wear ANSI rated Class 2 or 3 high visibility apparel at all times on the Project Site unless otherwise approved by the Ryan Corporate Safety Director.
- 9.4.12. Tie-off and Personal Fall Arrest System. Subcontractor and all Sub-tiers shall provide fall protection at the point of work and going to and from the point of work whenever one of its employees is exposed to a potential fall of six (6) feet or greater (Washington State's requirement is four feet). Subcontractor and sub-tiers shall comply with the following requirements:
- 9.4.12.1. CFR Subpart M Fall Protection Standard 1926 requires fall protection at six (6) feet. It is Contractor's policy that structural steel erection, installation of metal decking, and detail work, wood frame construction, precast concrete, and scaffolding will follow the requirements of Subpart M which states that fall protection is required at six (6) feet.
 - 9.4.12.2. Each employee performing any of the work described in 29 CFR 1926.501(b)(1) through 1926.501(b)(15) and 6 feet or more above a lower level shall be protected from falling with the use of guardrail systems, safety net systems, or personal fall arrest systems.
 - 9.4.12.3. Guardrails and safety nets shall be used in compliance with 29 CFR 1926.502(b) and 29 CFR 1926.502(c).
 - 9.4.12.4. Personal fall arrest systems shall be used in compliance with 29 CFR 1926.502(c).
 - 9.4.12.5. Subcontractor shall use full body harnesses as part of a personal fall arrest attached to a shock absorbing lanyard or self-retracting lanyard (SRL) equipped with double locking snaphooks and then attached to an anchorage point capable of supporting 5,000 pounds or shall be designed, installed, and used as part of a complete personal fall arrest system which maintains a safety factor of at least two; and under the supervision of a qualified person.
 - 9.4.12.6. Subcontractors using personal fall arrest systems and self retracting lanyards shall maintain compliance with ANSI/ASSP Z359.14-2021.
 - 9.4.12.7. 29 CFR 1926.502(d)(16) Personal fall arrest systems, when stopping a fall, shall be rigged such that an employee can neither free fall more than 6 feet, not contact any lower level.
 - 9.4.12.8. Personal fall arrest systems and components subjected to impact loading from a fall shall be immediately removed from service and shall not be used again for employee protection. 29 CFR 1926.502(d)(19)
 - 9.4.12.9. When subcontractor uses telescoping/aerial lifts, the employee shall be tied off at all (29 CFR 1926.453(b)(2)(v) and the sub-contractor shall ensure the employee using the personal fall arrest systems while working on telescoping/aerial lifts comply with 29 CFR 1926.502(d) of Subpart M. If the employee's feet must leave the lift platform, the employee shall use a method of fall protection that is approved by the lift manufacturer.
 - 9.4.12.10. When an employee is working in a "scissors lift," the employee shall close the gate and/or chain at all times. If the employee's feet must leave the lift platform, the employee shall use a method of fall protection that is approved by the lift manufacturer.
 - 9.4.12.11. When subcontractor's workers are using personal fall protection equipment, the subcontractor shall provide for prompt rescue of employees in the event of a fall or shall assure that employees are able to rescue themselves. 29 CFR 1926.502(d)(20) .
 - 9.4.12.12. The subcontractor shall provide a training program for each employee who might be exposed to fall hazards. 29 CFR 1926.50(a)(1) The subcontractor shall verify compliance with a written certification record and the latest training certification shall be maintained on the jobsite and available for inspection at all times.
- 9.4.13. UAS. If Subcontractor utilizes drones in connection with its Work, Subcontractor shall comply, or cause its subcontractor to comply, with the following requirements:
- 9.4.13.1 Subcontractor or its subcontractor (the "Operator") holds a valid exemption, operating certificate or other necessary authorizations issued by the Federal Aviation Administration ("FAA") required for the commercial operation of the UAS in connection with the Work. Operator will operate the UAS in accordance with the rules and regulations contained in the FAA regulations and exemption.
 - 9.4.13.2 Operator will notify Contractor in the event that the FAA or other regulatory body with jurisdiction over it takes any action against Operator's exemption, operating certificate or other authority to operate the UAS.

AWS - ATL089 Procedures Manual

- 9.4.13.3 It is the sole obligation of Operator to report any accidents, incidents or other reportable mishaps involving the UAS to the appropriate governmental agency, including but not limited to the FAA and National Transportation Safety Board. A copy of any such report shall be provided to Contractor.
- 9.4.13.4 Operator warrants that all crew, including the pilot in command, visual observer, sensor or payload operator, or other persons necessary for the safe operation of the flight have the qualifications, experience, licenses and certificates required by applicable laws and regulations and that all have the necessary skill required to perform their duties.
- 9.4.13.5. Operator warrants that the operation of the UAS is provided pursuant to an operations manual that is sufficient to operate the UAS safely and in compliance with all applicable laws and regulations. Operator understands Contractor is not in the business of providing aviation services and that Operator is fully responsible for determining if its mission can be flown safely and in accordance with the law. If it cannot, then Operator will not fly such mission.
- 9.4.13.6 Operator will operate the UAS in accordance with any federal, state or local privacy laws and regulations. To the extent, privacy laws and regulations have not been adopted regarding UAS operations, Operator shall operate the UAS in accordance with FAA's guidance on privacy best practices.
- 9.4.13.7. The UAS shall not exceed the lesser of 55 lbs. or the maximum weight permitted by Operator's UAS aircraft insurer.
- 9.4.14 Respirable Crystalline Silica. Subcontractor and Sub-tiers shall comply with the requirements of OSHA 29 CFR 1926.1153. Subcontractor is responsible for educating and training its employees in recognizing, avoiding and controlling respirable crystalline silica generated by construction tasks and operations. A Silica Control Plan/Program shall be submitted in both electronic and hardcopy for to the Contractor's superintendent prior to commencing any work on the Project. The Subcontractor/Sub-tier Silica Competent Person must be present at all times to ensure compliance during performance of silica generating task(s).

AWS - ATL089 Procedures Manual

Exhibit 3 – Preconstruction Subcontractor Safety Meeting **SEE PROCORE FORMS**

AWS - PRECONSTRUCTION SUBCONTRACTOR SAFETY MEETING



Project Name: _____
Project Number: _____
Superintendent: _____
Date: _____
Subcontractor Name: _____
Phone #: _____

MEETING ATTENDEES

Name	Company
_____	_____
_____	_____
_____	_____
_____	_____

Issues covered:

1. Description of subcontractor scope of work:

2. Submittals by Subcontractor (must be received on jobsite before work commences).

- Copy of Subcontractor's Safety Program – both electronic and hard copy.
- Copy of Subcontractor's Hazard Communication Program, including SDS information (electronic and hard copy).
- A detailed project-specific Job Hazard Analysis – this will also be used to train all of your on-site personnel (provide documentation).
 - JHAs must detail fall protection for steel erection, wood-frame and leading-edge work; daily JHA's and pre-task plans are required.
 - Verify additional requirements such as Roof Access Permits, Ground Disturbance Permit, Guardrail Removal Permits, Working at Heights, etc.
- Steel Erector must provide a site-specific Erection Plan.
- Per Article 5 (5.19) of the Ryan Subcontract, Subcontractor must provide registered engineered approved temporary bracing plans prior to commencement of work for Structural Precast, Tilt-up Construction, Masonry and Steel. Wood frame temporary bracing plans may be submitted by a competent person.
- Per Article 5 (5.20) of the Ryan Subcontract, if Subcontractor's work involves demolition of structural elements of an existing structure, the following shall be applicable: Subcontractor shall provide a written detailed fully engineered demolition plan as per OSHA requirements before structural demolition commences. Engineered plan must be signed and stamped by a licensed engineer in the state the work is being performed in. Subcontractor will train all members of their crew on safely executing any demolition plans before work commences. General demolition will require a written demolition plan prepared by the subcontractor's competent person to be submitted to Ryan prior to commencing work.
- Exposure Control Plan for any task generating silica dust; include Objective Data if not using OSHA Table 1. All employers on Ryan jobsites shall comply with OSHA 29 CFR 1926.1153 Respirable Crystalline Silica requirements (see Article 9.4.12) and/or applicable state OSHA requirements.
- Scope specific safety checklists for demo, steel erection, precast/tilt, electrical/MEP, roofing, underground utilities, and wood frame are available in Procore Forms.

AWS - ATL089 Procedures Manual

3. **It is the Subcontractors' responsibility to convey all safety information to their employees in whatever language they may primarily speak. Language barriers will NOT be an excuse for lack of safe practices. Translators are required to be on the project at all times.**
4. Review Article 9 of the Ryan subcontract. Also view the Ryan Jobsite Safety Orientation video and review the Site-Specific Project Safety Manual.
5. ANSI approved safety eyewear is required at all times.
6. Type 1 Class E-rated Safety Helmet that meets (1) ANSI z89.1 and/or EN 397 Certification, (2) Rated Type 1 and (3) Rated Class E, (4) Equipped with Manufacturers-installed 'chinstrap' is required at all times, for all site personnel, including visitors.
7. Structural Overloading Prevention. Subcontractor shall not load or permit any part of its work or the project to be loaded in a manner that would create a safety hazard. Subcontractors shall not load equipment and/or materials which exceed structural limitations or conduct any loading operations that would create a safety hazard. Engineering verification approval may be required.
8. Review any site-specific issues:
 - a. Emergency Management Program/Response Plan and communicate muster points to your team. Subcontractors will be responsible for the creation and training to their employees of all project-specific emergency response plans for their scope of work including rescue plans. Examples are working at height, excavations, electrical emergencies, confined space(s), etc.
 - b. Environmental Issues (Note: Ryan Environmental feedback is required for the Project Safety Manual).
 - c. Jobsite Specific Interruption Prevention (call before you dig – utility locates)
 - d. SDS information must be shared with any non-construction personnel onsite if they may be affected by said material (i.e. tenants, visitors, owner's representatives, etc.).
 - e. Review the use of hand-held communication equipment, i.e.: distracted equipment operation; see page 12.
 - f. Use of UAS/Drone(s) ☐ Yes or ☐ No (If "Yes", See Exhibit 1-Article 9- Section 9.4.11)
 - g. Any person who operates an ATV (single passenger) or UTV (side-by-side/multiple passengers) on Ryan jobsites must be trained to safely operate their machine, including conducting inspections and any necessary maintenance prior to use. **Maximum 10 MPH speed limit enforced.** (See Exhibit 12).
 - h. Vertical and horizontal rebar shall be protected.
 - i. Any floor holes/openings created must be immediately protected by the Subcontractor or tier subcontractor. All columns, holes, and dock pits shall be immediately guarded in accordance with OSHA hole cover/fall protection requirements by the contractor who creates the hazard.
 - j. Any guardrails removed must be re-installed immediately following completion of task. Personnel must be 100% tied-off and a hard rail barricade system must be in place. You must complete required Guardrail Removal Permit.
 - k. Adequate portable toilet facilities are provided and must be used when needed; violators will be dismissed from the project.
 - l. Any surplus chemicals or substances brought on site must be labeled and subsequently removed by the subcontractor at the end of the project or fees associated with disposal may be charged.
 - m. 15-minute portable eyewash stations must be provided by Subcontractor if required by an SDS for any product used on site by Subcontractor's personnel.
 - n. Subcontractor(s) may need to perform work in an occupied space. Subcontractor may be subject to owner's safety and security requirements when working in occupied area(s).
 - o. This project has a strict photography policy. Only authorized users are allowed to take photographs/videos, using Ryan issued digital cameras (see Exhibit 21).
9. Subcontractor personnel information:
 - a. Name of your Supervisor on site: _____
 - b. Name of the on-site Project Safety Representative, who will be responsible for your safety performance (must meet OSHA's definition of a competent person): _____
 - c. Does your company have a full-time Safety Director? ☐ Yes ☐ No
If "yes": Name: _____ Phone#: _____

AWS - ATL089 Procedures Manual

- d. Name of the subcontractor's Lockout Tagout Coordinator(s) _____ (if applicable)
e. Name of the subcontractor's Energy Marshal(s) _____ (if applicable)

How often will this Safety Director be visiting and inspecting this jobsite? ____ times per week; ____ times per month

10. Competent Person/Project Safety Representative Information:

- a. I understand the training and certification requirements that apply to all Subcontractors. I will maintain a list of my company's OSHA 30-hour trained Project Safety Representatives on this site and provide the valid certificate record(s) to Ryan.
- b. Name of your company's designated "competent person/Project Safety Representative" for subcontractor's portion of the work:

- c. Does Subcontractor have more than one "competent person/Project Safety Representative" on site for their portion of work (if so, please name):

11. Is the forklift operator for Subcontractor's portion of work certified and evaluated to meet OSHA standards?
☐ Yes ☐ No **Subcontractor is responsible to provide current training and/or certification records to Ryan. It is your responsibility to resubmit to Ryan, any changes or updates to certifications for your time on the project.** All attachments shall be approved by the manufacturer and rated with an identification for the use. Operators may be subject to additional training requirements when specific attachments are used in a manner that would reclassify the equipment as a crane. Refer to 29 CFR Subpart CC 1926.1400 (c)(8).

12. An ANSI Class 2 compliant high visibility garment/vest will be required at all times on this jobsite and in certain cases ANSI Class 3 (required for Flaggers, working in heavy traffic or in right-of way):

☐	This site has frequent exposure to moving equipment and truck traffic. (ANSI Class 2)
☐	Required for flaggers working in heavy traffic (>than 50 mph) or in right-of-way. (ANSI Class 3)

13. Electrical Subcontractor's Competent Person/Energy Marshal/LOTO Coordinator will ensure that all electrical personnel follow OSHA Electrical Safety-related work practices, **Lock-out/Tag-out procedures**, and NFPA 70E arc flash requirements while working or near any exposed energized electrical parts. Subcontractor's competent person shall ensure their employees are trained on NFPA 70E requirements. Subcontractor's Note: Subcontractor must provide site-specific detailed LOTO plan/procedure – this includes Power Distribution Center (PDC) load bank operations, and all temporary / permanent power sources. Prior to any energization activities, the electrical subcontractor(s) shall conduct a documented meeting with the parties listed below to review all safety, inspection(s) and procedural documents. Work will not commence until the meeting occurs and the electrical subcontractor(s) Energy Marshal(s), LOTO Coordinator(s), Competent Person(s), and Safety Representative(s) / Manager(s) of the subcontractor(s) review and approve all documents related to the activity. Separate documented meetings are required for each piece of equipment to be energized.

14. Does Subcontractor have any tier subcontractors? Yes: ☐ No: ☐

a. If "yes", list/name tier subcontractor(s):

- b. Prior to commencing any work, all subcontractors and their tier subcontractors shall attend the preconstruction subcontractor safety meeting and complete the preconstruction safety meeting form. Subcontractors are still responsible for ensuring that all of their workers and sub tiers follow all project safety requirements. The subcontractor will be responsible to maintain project supervision at all times including times when their sub-tier subcontractors are on the project.

AWS - ATL089 Procedures Manual

15. Do Subcontractor and/or tier subcontractor work activities include **roof work**? Yes: ☐ No: ☐

If “yes”, prior to beginning work, subcontractor’s Competent Person must complete a Daily Roof Access Permit with the Ryan Field Manager. The Daily Roof Access Permit will be closed out at the end of each workday by the Competent Person. Subcontractor’s workers must all be trained on Ryan’s roof access awareness and the conditions of the Daily Roof Access Permit. Workers will not be allowed to access the roof if they are not listed as authorized on the Permit.

16. Do Subcontractor and/or tier subcontractor work activities include **ground penetrations**? Yes: ☐ No: ☐

If “yes”: Prior to beginning work, Subcontractors must call for utility locates and are reminded of their duty to ensure any / all tier subcontractors also call for utility locates if their work includes any earthwork and / or surface/ground penetrations, including excavations of any kind, driving of steel stakes, ground rods, metal rods, or posts, etc. **A Daily Ground Disturbance Permit must be created for any work activities which may disrupt ground surfaces. Hydrovac or potholing required for buried utilities, no hand-digging or mechanical equipment (reference GMR Section 3.6 for additional AWS requirements).**

17. Do Subcontractor and/or tier subcontractor work activities include entrance into any permit-required confined space(s) as defined by 29 CFR 1926.1200 Subpart AA. ☐ Yes ☐ No

If “yes”, Subcontractor/tier subcontractor must provide notice of location to Ryan site supervision for posting purposes. **Permit must be completed with Ryan supervision prior to entry.**

18. Do Subcontractor work activities include **crane operations**? ☐ Yes ☐ No

If “yes”: Is crane: ☐ Owned or ☐ Rented

- a. Is operator: ☐ Subcontractor employee ☐ Rental Company Employee ☐ NCCCO certified

- b. Do you understand that all crane operators must comply with OSHA 29 CFR 1926.1427 Operator Qualifications and Certifications (i.e., NCCCO certified or equivalent), **operator evaluation with documentation**? ☐ Yes ☐ No

In addition to training and certification, all crane operators will need to be evaluated by their employer for the equipment they will be operating. See Ryan [Crane Operator Evaluation Form](#) Exhibit 11. Provide copies of certifications for Certified Crane Operator, Qualified Rigger(s) and Qualified Signal Person(s) to Ryan site supervision)

- c. Name of on-site person responsible for crane operations: _____

- d. Name of competent/qualified person responsible for daily crane inspections: _____

- e. Do you understand that all cranes used on this site must have proof of current annual inspection before work can commence? ☐ Yes ☐ No **Provide a signed copy by the company’s qualified person to Ryan site supervision.**

- f. List names of all qualified riggers and signal persons:

- g. Do you understand this project's requirements for the use of a critical/engineered lift plan? Per Article 5.22 of the contract, refer to the crane section of this manual for details. ☐ Yes ☐ No

- h. If the crane will be moved (traveled) around the jobsite, a detailed **job hazard analysis** must be completed and turned in specifically addressing safe traveling operations.

- i. Will vertical lifts involve helicopter crane operations? If yes, see Helicopter Crane Procedures Form and Inspection Checklist in Procore. **Note: If operations include helicopter crane picks, Ryan Legal and Ryan Safety Director(s) must be notified well in advance.**

AWS - ATL089 Procedures Manual

19. As per Department of Labor (DOL) Standards, and Ryan's Legal Department, no person under the age of 18 shall be allowed to perform work on Ryan jobsites.
20. Ryan strictly prohibits employees, or any other person providing services to the company or located on the company's premises, including offices and work sites, from possessing weapons at the workplace.
21. Harassment Policy: It is the policy of Ryan Companies US, Inc. to maintain a working environment free from offensive or degrading remarks or conduct based on gender, age, race, national origin, religion, disability, marital or public assistance status, sexual orientation, genetics, or any other status protected by applicable state or federal law. Offensive behavior prohibited by this policy includes, as examples, requests to engage in illegal immoral or unethical conduct. Offensive and harassing behavior are prohibited and will not be tolerated for the protection of jobsite employees, public by-standers and local businesses.
22. You acknowledge that you have received the AWS Global Construction H&S Minimum Requirements (GMR). **Refer to attached addendum for the full list of the GMR requirements.**

Comments:

As a representative of the Subcontractor named above, I accept the rules and regulations covered in this meeting, on the attached Article 9, and on the attached AWS Global Construction H&S Minimum Requirements addendum. I understand that non-compliance may result in disciplinary action, and/or stoppage of our portion of the work until compliance is met.

It is also understood that any stoppage of the work due to safety considerations will not affect the scheduled completion date for our portion of the work.

Subcontractor's representative also understands and has verified with the Ryan Jobsite Manager that Subcontractor's current Certificate of Insurance is on site and that Subcontractor is not authorized to commence work at the jobsite without such verification.

Subcontractor representative:

Name (Printed): _____

Title: _____

Signature: _____

Date: _____

RYAN representative:

Name (Printed): _____

Title: _____

Signature: _____

Date: _____

cc: Ryan Site Safety Files and Subcontractor office

AWS - ATL089 Procedures Manual

Exhibit 4 - Job Hazard Analysis (JHA)

SEE PROCORE FORMS

JOB HAZARD ANALYSIS (JHA)				
Subcontractor:				
Job:		Analysis by:	Received by:	Date:
Crewmembers:				
Activity:				
Principal Steps in An Activity	Potential Hazards	Required Recommended Controls		

AWS - ATL089 Procedures Manual

Exhibit 5 – Site Specific Orientation – Special Safety Regulations

SITE SPECIFIC ORIENTATION – SPECIAL SAFETY REGULATIONS



Safety is the responsibility of every person involved with the construction of the AWS - ATL089 site.

Violation of any of these rules will be cause for immediate disciplinary action, including dismissal.

1. Alcohol and/or drug use, fighting or horseplay will result in immediate termination.
2. Offensive and harassing behaviors are prohibited and will not be tolerated for the protection of jobsite employees, public bystanders and local businesses. Should there arise any instances of unacceptable conduct, Ryan Companies US, Inc. will call for immediate removal of offending persons from the jobsite.
3. All accidents, injuries and unsafe conditions must be reported to Ryan site supervision immediately.
4. All persons working in an area with a **six feet or greater* fall exposure** must be tied off to a substantial structural member or static line designated for the sole purpose of a lifeline. (Note: 4 feet requirement in Washington State)
5. All fall protection equipment shall be inspected prior to each use, with special care given to the retractable **device**, anchorage points, and continuity of perimeter cables.
6. Disciplinary program:

Non-Serious Violation:

- 1st Violation: Verbal warning
- 2nd Violation: Verbal & written warning to your company
- 3rd Violation: Verbal & written warning to your company
- 4th Violation: Removed from project

Serious Violation:

- 1st Violation: Verbal & written warning to your company
- 2nd Violation: Verbal & written letter to remove you from site

Nothing in this policy prohibits the immediate dismissal or removal from the jobsite of any employee whose conduct constitutes a serious violation of the safety requirements, or which could cause serious danger to him/her-self, co-workers, property, equipment, or the employees of others.

7. Always be aware of your surroundings. Avoid distractions. Know your muster points. Keep your head on a swivel. You have the right to stop work if you observe an unsafe condition or act during work activities on the jobsite. Report any unsafe conditions to your supervisor or Ryan Field Supervision immediately.
8. Each contractor who creates a hazard is responsible for guarding that hazard. Each contractor who removes any guarding is responsible for reinstalling the guarding properly.
9. Safety harnesses, lanyards, safety glasses, hearing protection, respirators and all other personal protective equipment must be worn where hazards exist. Hard hats and ANSI approved safety eyewear must be worn on the jobsite at all times – ANSI Class 2 high visibility garments/vests will be worn at all times.
10. All employees must dress properly while working. Minimum attire includes long pants, shirt, and work boots. Work boots should provide sturdy support of the ankle. No high-top athletic type cross-training shoes are allowed. **Tank tops, tennis shoes and shorts are not acceptable.** Loose clothing, jewelry, or radios (other than those authorized for communication) are prohibited. Use of earbuds / headphones for listening to music is prohibited.
11. Each employee, as well as each Contractor, is responsible to practice good housekeeping.
12. All scaffolds must be secured when in use and tied to the structure as determined by the Contractor's "competent person". Prior to the work-shift, a "competent person" must inspect the Contractor's scaffolds.
13. All ladders shall be used in accordance with manufacturer specifications. Ladders used as access to other levels shall be secured to a substantial structural member.
14. Whenever welding, cutting, or burning operations are taking place, an approved fire extinguisher must be present. Whenever welding, cutting, or burning operations with fire-causing sparks takes place at elevated heights, each contractor is responsible to have a **ground spotter**. Torch heads shall be removed at shifts' end. Hot Work permits may be required – see your jobsite supervisor.

AWS - ATL089 Procedures Manual

15. Oxygen cylinders shall be stored a minimum of 20' away or separated by a 1-hour fire barrier from gasoline, acetylene, propane and any other combustible liquid. Compressed gas must be secured and stored in an approved container. An approved fire extinguisher shall be no closer than 25' away from any flammable/combustible storage area(s).
 16. Ground fault circuit interrupters are required. All extension cords and power tools must be properly grounded, or they will be removed from the site.
 17. Each employee will comply at all times with the federal, state, local OSHA and Ryan regulations and policies pertaining to the AWS - ATL089 jobsite to ensure a safe and healthful workplace. Right to know / SDS information is available from your employer, and all SDS information from all Subcontractors is available through the Ryan Jobsite Manager. All chemical containers must meet the GHS standards for labeling.
 18. All Ryan ALL JOB Safety Meetings are mandatory and will be held every **Second Wednesday** of the month at 7:00 am.
 19. Employees exposed to and involved in earth-moving operations shall be trained on the safe approach to equipment and wear ANSI Class 2 compliant high visibility garments/vests.
 20. While performing work, there will always be at least one OSHA 30 trained person on-site at all times for each Subcontractor.
 21. Graffiti of any kind will not be tolerated.
 22. I have been made aware of the environmental condition of this site. For environmental information, contact Ryan's Jobsite Manager or Ryan's Environmental Specialist, Jonathon Blaha, at 612-492-4430.
- Note: The following policies and procedures have been added since the Jobsite Orientation Video was produced:**
23. Use of hand-held communication devices while operating equipment on this project must be approved by your foreman/employer; this includes any **mobile phones or devices**, two-way radios, CB radios, and/or walkie-talkies.
 24. No person shall enter a **permit-required confined space** without authorization by their employer and/or Ryan site supervision.
 25. Prior to commencing any work generating respirable crystalline silica dust, you must be educated and trained by your employer and provided with the proper tools, PPE, and safety equipment to perform your work safely in accordance with the Respirable Crystalline Silica standard (OSHA 29 CFR 1926.1153).
 26. Any person who operates an ATV (single passenger) or UTV (side-by-side/multiple passengers) on Ryan jobsites must be trained to safely operate their machine, including conducting inspections and any necessary maintenance prior to use. 10 MPH speed limit is enforced. (Exhibit 12)
 27. Prior to the start of work, all activities requiring roof access will require a Daily Roof Access Permit to be completed in advance by the subcontractor's competent person. All authorized roof workers must be trained on the permit requirements and Ryan's Daily Roof Access Permit procedures.
 28. Daily Ground Disturbance Permits are required to be completed prior to the start of any work activity which will move dirt, pavement, sod or rock, or any activity that penetrates the ground at any depth by any means, including: digging, excavating, trenching, drilling/boring, grading, driving stakes/posts or form pins. Note: Competent person must call for locates.
 29. Ryan strictly prohibits employees, or any other person providing services to the company or located on the company's premises, including offices and work sites, from possessing weapons at the workplace.

- ☐ I have read and understand these "Special Safety Regulations" (Exhibit 4) for this jobsite/ #7389-001.
- ☐ I recognize and understand my responsibility to comply with these regulations as well as other safety regulations for this project.
- ☐ I understand this project is under surveillance to document the safety, security, and progress of the workers and the project.
- Monitoring of this surveillance is done by both human and artificial means.

I have viewed the Ryan Jobsite Safety Orientation video: ☐ Yes - today ☐ Yes - at a previous Ryan jobsite ☐ No

Name (please print full name):		Phone #	
Name (signature):		Date:	

Company & Supervisor: _____

Emergency Contact Name & Phone Number: _____

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Exhibit 6 (Reference Appendix A & B from Jobsite Emergency Management Manual)

EMERGENCY RESPONSE PLAN FOR CONSTRUCTION SITES



1. In the case of an Emergency, notify the following:
 - **911**
 - Ryan Superintendent: _____ #: () _____
 - Back-Up Name: _____ #: () _____
2. Whoever calls “**911**” must stay on the phone line with the emergency operator until directed to hang up by the operator.
3. _____ is the on-site emergency coordinator and will notify the Ryan Regional Emergency Task Team (RETT) of the incident and will follow / refer to the Ryan Jobsite Emergency Management Manual (see Tools and/or Greenway).
4. _____ will be in charge of the accident scene.
5. _____ will administer first aid (refer to list provided during Preconstruction Subcontractor Safety Meeting).
6. At the time of an emergency, no unauthorized personnel are allowed to enter, and no personnel should be allowed to leave the site until the “**ALL CLEAR**” is given.
_____ will coordinate this.
7. At the time of the emergency someone will be assigned to show emergency crews to the accident site.
_____ will coordinate this.
8. Refer all questions and any media to the job Superintendent _____.
9. In the event the jobsite needs to be evacuated, all personnel will meet at: _____
10. _____ will do a head count of employees.
11. _____ will secure and investigate the accident scene.
12. _____ will give the “**ALL CLEAR**”.
13. _____

Post a copy of this completed form on-site.

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Exhibit 7– Jobsite Emergency Contact List

JOBSITE EMERGENCY CONTACT LIST

[illegible]

Post a copy of this form on site

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Exhibit 8 AWS INCIDENT REPORT

SEE PROCORE INCIDENTS

NEAR-MISS / INJURY / ILLNESS / PROPERTY/EQUIPMENT DAMAGE / THEFT / HARASSMENT

E-MAIL ALL PAGES OF THIS INCIDENT REPORT **WITHIN 2 HOURS TO:**

- Claims.Administrator@RyanCompanies.com
- John.Gaddini@RyanCompanies.com
- Bill.OConnell@RyanCompanies.com

*The purpose of this report is to provide necessary information to all parties involved in the event of any incident, near miss, injury, illness, property damage, equipment damage theft, or violation of Ryan's Anti-Harassment Policy. **Complete all sections of this report that pertain and email all pages of the report to the email addresses listed above within 2 hours of the incident.** Include any related paperwork, pictures and/or video if possible.*

COMPLETED BY:					
DATE:		PHONE			
AWS IDENTIFIER (ie: MSP6)		RYAN PROJECT #:			
FIELD MGR:			PHONE:		
PROJECT TYPE:	☐ AMZL(TES) ☐ AMXL (TES) ☐ OXD/IXD or ATS (TES) ☐ ARS (NACF) ☐ SORT/TNS/HAZMAT (NACF) ☐ WAGON WHEEL ☐ UNKNOWN ☐ OTHER				
WHO REPORTED THE INCIDENT TO RYAN?		PHONE:		E-MAIL:	

1. INCIDENT INFORMATION

INCIDENT TYPE:	☐ NEAR MISS ☐ BODILY INJURY ☐ ILLNESS ☐ VEHICLE/EQUIP/PROPERTY ☐ THEFT/ VANDALISM ☐ OFFENSIVE BEHAVIOR/ANTI-HARASSMENT POLICY VIOLATION (See Section #9)				
INCIDENT DATE:		INCIDENT TIME:	☐ AM ☐ PM ☐ EST ☐ CST ☐ PACIFIC		
SHIFT INFORMATION:	☐ NORMAL DAYTIME SHIFT ☐ NIGHT SHIFT ☐ WEEKEND ☐ HOLIDAY ☐ OVERTIME ☐ OTHER				
SPECIFIC LOCATION OF INCIDENT ON JOBSITE:					
WAS PROJECT SITE SHUT DOWN DUE TO THIS INCIDENT? ☐ YES ☐ NO					
DESCRIBE WHAT HAPPENED, WHEN, HOW INCIDENT OCCURRED: (use Section 11 if more room needed)					
DESCRIBE WEATHER /ENVIRONMENT AT TIME OF INCIDENT:					
RYAN REPRESENTATIVE RESPONSIBLE FOR FOLLOW-UP:					
RYAN CONTACT INFORMATION FOR FOLLOW-UP:		EMAIL:	PHONE:		

2. INJURED or MAIN PARTY INVOLVED - INFORMATION

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MAIN PARTY NAME:		PHONE:	
MAIN PARTY ADDRESS (IF KNOWN):		CITY/STATE/ZIP	
IS MAIN PARTY: ☐ RYAN EMPLOYEE ☐ SUBCONTRACTOR ☐ MHE ☐ MHE TIER ☐ OWNER CUSTOMER VISITOR / OTHER			
SUBCONTRACTOR NAME:		MHE TIER NAME:	
MHE NAME:		PHONE	
JOB TITLE & TRADE:		SUPERVISOR:	
Approximate age of main party ☐ 18-25 ☐ 26-30 ☐ 31-35 ☐ 36-40 ☐ 41-45 ☐ 46-50 ☐ 51-55 ☐ 55-60 ☐ 61 or older			
Approximate number of years of experience of main party: ☐ LESS THAN 1 YEAR ☐ 1 YEAR ☐ 2 YEARS ☐ 3 YEARS			
☐ 4 YEARS ☐ 5-10 YEARS ☐ 11-15 YEARS ☐ 16-20 YEARS ☐ MORE THAN 20 YEARS			

3. BODILY INJURY / ILLNESS INFORMATION

TYPE OF MEDICAL RESPONSE: ☐ EMERGENCY ☐ CLINIC VISIT ☐ FIRST AID ☐ UNKNOWN/OTHER:	
INJURY / ILLNESS TYPE:	☐ STRAIN ☐ PUNCTURE ☐ LACERATION ☐ FRACTURE ☐ BRUISE ☐ ILLNESS/ OTHER:
WAS POST-ACCIDENT DRUG TESTING PERFORMED FOLLOWING INCIDENT? ☐ YES ☐ NO	
BODY PART(S) INJURED:	CHECK ONE: ☐ RIGHT ☐ LEFT ☐ BOTH ☐ N/A
WAS INJURED PARTY HOSPITALIZED? ☐ YES ☐ NO ☐ UNKNOWN	WAS INJURED PARTY PLACED ON MODIFIED DUTY? ☐ YES ☐ NO ☐ UNKNOWN
OBJECTS, MATERIALS OR EQUIPMENT INVOLVED IN THE INJURY/ILLNESS:	
WHAT TASK WAS THE INJURED PARTY PERFORMING WHEN THE INJURY/ILLNESS OCCURRED?	
WHAT PERSONAL PROTECTIVE EQUIPMENT (PPE) WAS INJURED PARTY USING AT THE TIME OF THE INCIDENT? ☐ BOOTS ☐ GLOVES ☐ HI VIS APPAREL ☐ SAFETY GLASSES ☐ HARD HAT ☐ OTHER (EXPLAIN BELOW)	

4. NEAR-MISS INFORMATION

DESCRIPTION OF NEAR MISS:	
CORRECTIVE ACTION:	
DATE CORRECTED:	

5. PROPERTY DAMAGE

(*If Vehicle, please complete the Motor Vehicle Accident Report form as well)

WHAT WAS DAMAGED: ☐ VEHICLE ☐ EQUIPMENT ☐ BUILDING ☐ PERSONAL PROPERTY ☐ OTHER:	
MAKE:	MODEL:
YEAR:	COLOR:
OWNER NAME:	SERIAL OR VIN #
OWNER ADDRESS:	PHONE:
WAS POLICE REPORT FILED / REQUESTED?	☐ YES ☐ NO
POLICE REPORT #	
DESCRIPTION OF PROPERTY	
DESCRIBE DAMAGE:	
OTHER MATERIAL / OBJECTS INVOLVED:	

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PRELIMINARY ESTIMATED VALUE \$:		PRELIMINARY ESTIMATED DAMAGE \$:	
WHO WITNESSED THE PROPERTY DAMAGE INCIDENT? (See Witnesses section #7)			

6. THEFT or VANDALISM

DESCRIBE EQUIPMENT / PROPERTY STOLEN OR VANDALIZED:			
ADDRESS & LOCATION THEFT / VANDALISM OCCURRED AT:			
TIME THEFT / VANDALISM OCCURRED: (if known):		☐ AM ☐ PM	
WAS POLICE REPORT FILED OR REQUESTED?	☐ YES ☐ NO	POLICE REPORT#:	
WHO IS THE OWNER OF THE STOLEN OR VANDALIZED ITEMS / PROPERTY?			
DESCRIBE KNOWN DETAILS OF THE THEFT OR VANDALISM:			
PRELIMINARY ESTIMATED VALUE \$:		PRELIMINARY ESTIMATED DAMAGE \$:	
WHO WITNESSED THE THEFT OR VANDALISM? (See Witnesses section #7)			

7. WITNESSES (IF ANY)

WITNESS #1 NAME:		POSITION:	
WITNESS ADDRESS:		E-MAIL:	
EMPLOYER:		PHONE:	
WITNESS LOCATION AT TIME OF INCIDENT:			
WITNESS STATEMENT:			
WITNESS #2 NAME:		POSITION:	
WITNESS ADDRESS:		E-MAIL:	
EMPLOYER:		PHONE:	
WITNESS LOCATION AT TIME OF INCIDENT:			
WITNESS STATEMENT:			

8. PREVENTATIVE OR CORRECTIVE ACTION TAKEN (MUST BE COMPLETED)

WHAT PREVENTATIVE/CORRECTIVE ACTION WAS TAKEN TO PREVENT A SIMILAR INCIDENT FROM OCCURRING?

9. OFFENSIVE BEHAVIOR / ANTI-HARASSMENT POLICY VIOLATION / ALLEGATION

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PLEASE DESCRIBE THE REPORTED ALLEGATION(S) BELOW:

10. PROVIDE AN AERIAL-VIEW DIAGRAM AND/OR PHOTOS OF THE INCIDENT SITE

WERE PHOTOS OR VIDEO OF THE INCIDENT TAKEN OR AVAILABLE ☐ YES ☐ NO # OF IMAGES:

ARE PHOTOS OR VIDEO OF THE INCIDENT INCLUDED WITH THIS REPORT? ☐ YES ☐ NO

IF NOT, WHO MAY BE CONTACTED FOR PHOTOS OR VIDEOS?

NAME

PHONE

Drag/drop an image file, or copy/paste a diagram or photo from another application illustrating the scene at the time of the incident. Suggestion: draw an aerial view diagram of the incident site on a piece of paper and take a photo of the diagram, email the photo to yourself and attach it here. **Please include labels and / or directional information describing photos.**

NOTE: IF MORE INFORMATION IS OBTAINED AFTER THIS REPORT IS SUBMITTED, PLEASE UPDATE THIS FORM AND RE-SUBMIT IT SO THAT THE CLAIMS ADMINISTRATOR CAN UPDATE THE FILE. THANK YOU!

11. ADDITIONAL DESCRIPTION OR INFORMATION (IF NEEDED)

SEE PROCORE FORMS

[illegible]

Exhibit 10 – Safety Violation Notice **SEE PROCORE FORMS**

SAFETY VIOLATION NOTICE

RYAN

Complete this notice and attach in Procore to an “Observation – Safety Violation” for your Project.

Worker / Employee Name: _____

Subcontractor / Employer: _____

Occupation / Craft: _____

Project Name & Number: AWS - ATL089 #7389-001

Project Address: 11145 Alcovy Road Covington GA 30054

Work Performed: _____

Location on Project: _____

Describe Violation: _____

Action taken: ☐ Warning
(check one) ☐ Suspension (enter # of days) Days (minimum 24 hours)
 ☐ Termination

Notifications made to:
(signatures not required)

- ☐ Worker / Employee
- ☐ Subcontractor / Employer
- ☐ Employee's Supervisor
- ☐ Ryan Superintendent
- ☐ Ryan Safety Department

Issued by (print name): _____

Title: _____

Date: _____ Time: _____

Distribution: Worker/Employee, Ryan Superintendent, Ryan Safety Department, Subcontractor/Employer

Note: If this Safety Violation is part of an incident, injury, or near miss, etc., attach a copy of this notice to the Incident Report

Exhibit 11 – Written Verification for the Steel Erection Contractor See Procore Forms

**WRITTEN VERIFICATION
FOR THE STEEL ERECTION CONTRACTOR**



Date: _____
Job Number: _____
Job Name: _____
Supervisor: _____
Steel Subcontractor: _____

☐ The concrete or masonry has met the strength design to commence steel operation on this project.

Define area of project released for steel erection: _____

Date(s) of Concrete Placement: _____

Testing Agency: _____

(Cylinder break information available in Ryan office)

☐ Repair or replacement of the anchor bolt has occurred with approval of the project structural engineer.

Location of anchor bolt: _____

Date Anchor Bolt Modified: _____

☐ The perimeter safety cable has been inspected by Ryan site supervision. Ryan agrees to take control of this cable, inspect and maintain until removed from use.

Date of Perimeter Safety Cable Inspection: _____

Inspector(s) Name: _____

Location or Floor Number: _____

Will the perimeter cable be used as part of a fall protection system? Yes ☐ No ☐

Signature of Ryan Supervisor: _____

Signature of Steel Erection Supervisor: _____

Note: this form may be issued multiple times on a project.

Original – File in Safety File System on Site
Copy – Project Manager
Copy – Steel Erector

Exhibit 12 – Crane Operator Evaluation Form (Sample) See Procore Forms

CRANE OPERATOR EVALUATION FORM - Ryan Employees



This form must be completed for any Ryan-employed crane operator.

Date of Evaluation: _____

Project Number: _____

Project Name: _____

Ryan Field Manager Name: _____

=====

Operator's Name: _____
(print name)

Operator's Employer: _____

Evaluator's Name: _____
(print name)

Evaluator is: (check one): ☐ Ryan Employee ☐ Acting as Ryan Agent

Crane Make: _____ Crane Model: _____

Configuration of Crane Equipment: Please provide a description of the equipment's configuration at the time of evaluation such as attachments, boom length, counterweight set-up, etc.).

Task(s) Performed: Please provide a description of the task(s) performed by the operator during the evaluation such as hoisting formwork, personnel basket, concrete bucket, multi-crane pick / lift, blind pick / lift, etc.

With respect to equipment that the candidate will operate, I have evaluated the candidate and have determined that he/she:

- (i) has the skills necessary to operate the equipment safely for the assigned task(s);
- (ii) has the skills and knowledge as well as the ability to recognize and avert risk; and
- (iii) can apply the equipment's load charts and the manufacturer's procedures.

Evaluator's Signature

Ryan Field Manager Signature

Date: _____

Date: _____

Note: Retraining must be provided in relevant topics for each operator when the performance of the operator or an evaluation of the operator's knowledge indicates that retraining is necessary. When retraining is required, the employer must re-evaluate the operator with respect to the subject of the retraining.

Exhibit 13 – ATV / UTV Use Checklist / Form

ATV / UTV USE CHECKLIST / FORM



- All persons who operate an ATV (single operator) or UTV (side-by-side/multiple passengers) on Ryan jobsites must be trained by their employer to safely operate their specific machine
- Read the Owner's manual safety section for your machine and document this has occurred before operation commences
- View the safety video for the make/model you will be operating; these can be found online or via You Tube
- Inspect the machine daily for any issues that could affect safe operation and read the Owner's manual for specific information
- Operator/driver is responsible for notifying passengers about safety requirements and will not allow passengers on single operator ATVs
- If the machine is equipped with seatbelts, they must be worn by the operator and any passengers; appropriate personal protective equipment must be worn
- ATVs/UTVs may only be operated in authorized areas / locations within the jobsite. Be aware of any site-specific hazards such as excavations, trenches, uneven/unstable terrain and/or inclines; complete a job hazard analysis prior to use
- Operators may not use excessive speed at any time; 10 MPH maximum speed limit enforced.
- Note: Contact your State DMV, Public Safety, or DNR for any specific requirements or registration obligations

The list below is a general checklist of safety items to look for; note that your machine may have additional safety requirements:

- ☐ Trained on safe operation of the specific ATV/UTV
- ☐ Trained on inspection, including steering, suspension, frame, hitch, headlights, taillights, other controls, etc., ensuring all are undamaged and working correctly
- ☐ Brakes work correctly; **throttle moves freely**
- ☐ Tires in good shape, psi correct
- ☐ Seat belts undamaged and working correctly
- ☐ Refueling safety reviewed
- ☐ Know the criteria for removing the machine from service and/or report damage to your supervisor
- ☐ **Regular maintenance performed** which meets manufacturer's recommendations, no modifications without approval from the manufacturer.
- ☐ If pulling a trailer, it must be inspected, maintained, safely attached to ATV/UTV, never over-loaded; latched to the correct size ball, safety pin used, and safety chains attached; do not haul excessive loads and ensure loads are evenly distributed;
- ☐ Windshield cleared for optimal visibility, example: clear off frost, ice, dust, dirt and/or bugs
- ☐ Review possible need for attaching strobe light and/or flag for enhanced visibility.

Exhibit 14 – Daily Roof Access Permit (See Procore Forms or hard copy)

Project Name		
Project Number	Date Click or tap to enter a date.	
Subcontractor Requesting Roof Access (Name)		
Ryan Companies On-site Contact (Name)		
Designated Competent Person (Name)		
Area of Roof or Building to be accessed		
Description of Work to be Performed		
Dates of Access:	Start Click or tap to enter a date.	Finish Click or tap to enter a date.
☐ Attach a detailed Job Hazard Analysis and identify controls and a detailed Rescue Plan		
☐ A signed list of trained personnel authorized for roof access is included below		

The designated competent person (named above) has performed a risk assessment on this project and has identified the potential hazards and controls associated with roof work in a detailed Job Hazard Analysis, which is attached. The competent person agrees to monitor and comply with all requirements of the JHA as well as OSHA 29 CFR: 1926.500-503 Subpart M – Fall Protection Standards, State Plan OSHA requirements and utilizing Ryan/s Roof Access Permit.

Method of Access	
☐ Stairs	☐ Portable Ladder
☐ Stair Tower	☐ Roof Hatches equipped with guardrail system
☐ Fixed Ladder	☐ Other:

Exposure Assessment – Identification of Potential Hazards*

☐ Skylights	☐ Snow / Ice / Water / Slippery	☐ Leading Edge Falls
☐ Roof Openings	☐ Pitched / Sloped Roofs	☐ Change in Roof Elevation / Parapet Walls < 42" (inches)
☐ Access Hatch/ Holes	☐ Motorized Equipment (Snowblowers, Rock Haulers, etc.)	☐ Loading & Unloading Areas (Material Hoisting, Signaling, Staging)

Other hazards:

• The above is not inclusive of all potential hazards

Safety Precautions - (Check all that are to be taken)

All Trades
☐ Guardrails will be used ☐ Fall Arrest equipment will be used (5,000 lb. anchor point, harness, retractable, roof cart, lanyard, etc.) ☐ Complete warning line system (500 lb. tensile strength) to be installed a minimum of 15' (fifteen feet) back from all changes in elevation greater than 6' (six feet). Warning line to be maintained at 34"-39" and flagged every six feet (6') with high visibility material. ☐ Roof Access points will be provided and properly maintained (Ladders, access hatches, or stairs / stair towers) ☐ All crew members have been trained in Fall Protection and the requirements of this Roof Access Permit prior to roof access and prior to work being performed (see list below)

Note: The most stringent Federal or State Plan OSHA regulation will apply i.e. WA State's 4-foot (4') fall protection rule.

Roof Openings / Penetrations

Are roof openings / penetrations going to be created in the work to be performed? ☐ Yes ☐ No

If "yes", at no time will roof penetrations be allowed without: (1) this written permit, (2) a JHA provided covering all tasks, including method(s) for providing protection and including workers signatures, and (3) Competent Person reviewed permit and JHA.

At no point will penetrations be left unattended/unprotected. Any/all roof penetrations shall have guardrails or be covered, secured, and labeled at the end of the shift with the words "Cover" or "Hole" and "Do Not Remove".
Competent Person Initial Here

Protection from falling objects per 1926.502(j): to protect workers below from falling objects, all material and debris must be kept clear of the working area by removal at regular intervals. Tool tethering will be used for tools and equipment that may fall to the lower level.

Competent Person Initial Here: _____

Note: Safety Monitoring Systems in combination with warning lines, per 1926.502(h) are prohibited for roof penetration work. Competent Person Initial Here

Fall Protection

☐ Full Body Harness	☐ Leading Edge Retractable	☐ Guardrail System
☐ Mobile Roof Tie Off Carts	☐ Tie Off Anchors	☐ Other
☐ Shock Absorbing Lanyard	☐ Hole Covers	☐ Other

Roof Access Permit and JHA submitted by Subcontractor's Competent Person:

Print Name:	Signature:
Date:	

COMPETENT PERSON WILL CLOSE OUT PERMIT AT THE END OF EACH DAY.

Project Boundaries:

Provide a sketch below of roof work areas, including access points, leading edges, designated anchor points, loading areas, warning line locations, restricted areas, rooftop units, designated walk paths, and electrical hazards, etc.

A COPY OF THIS PERMIT SHALL BE POSTED / MAINTAINED AT THE RYAN OFFICE.

[illegible]

TRAINED PERSONNEL AUTHORIZED FOR ROOF ACCESS

[illegible]

Exhibit 15(a) – AWS Construction Zone (CZ) Daily Ground Disturbance Safety Permit

(Please complete in Procore or return hard copy to Ryan Site Supervision)

Project Name/Number: _____ Today's Date: _____

Contractor Requesting Ground Disturbance _____ Competent Person: _____

Ryan Companies US, Inc. on-site Contact: _____

Description and Location of work to be performed: _____

Dates of Disturbance (Approximate): Start: __/__/__ Finish: __/__/__

The designated competent person listed above has made documented notification to all owners/performed a risk assessment/JHA on the named project and has identified the potential hazards and controls associated with ground disturbance. Additionally, the competent person agrees to monitor and comply with all requirements of this plan as well as OSHA 1926.650-652 Subpart P – Excavations, State Plan, OSHA, and AWS Requirements.

Disturbance Type: ☐ Hand Digging ☐ Boring ☐ Excavator ☐ Vac Truck ☐ Rock Excavation ☐ Trencher ☐ Grading <6" (15 cm) ☐ Grading >6" (15 cm) ☐ Driving Stakes/Post/Form Pins			
	YES	NO	N/A
1. Confirm all private and public utilities have been located. Ticket(s) #	☐	☐	☐
2. The requesting contractor's competent person has walked the jobsite to look for signs of utilities/services, and has reviewed all available as-builts, contract drawings, current BIM models, and other available information (including owner's rep) to identify utilities, and other services in the area of the excavation?	☐	☐	☐
3. All known and assumed underground utilities in the dig zone and search zone have been correctly identified? Utilities identified: Fiber: ☐ Electrical: ☐ Gas: ☐ Plumbing: ☐ Storm/Sewer/Water: ☐	☐	☐	☐
4. Utility providers' contact information has been obtained – check applicable: Fiber: ☐ Electrical: ☐ Gas: ☐ Plumbing: ☐ Storm/Sewer/Water: ☐	☐	☐	☐
5. Potholing/Hydrovac is required for exposing underground utilities within 30 (thirty) feet of ground disturbance have been (or will be) used to verify exact location of all known underground utilities. All excavations within 10' (3 m) of located known live utility need to be completed by vacuum truck method only. Subcontractor(s) Competent Person:	☐	☐	☐
6. Method of protecting/supporting exposed utilities has been determined and will be utilized on exposed utilities.	☐	☐	☐
7. Precautions have been taken to prevent contact with overhead lines, structures, or other utilities. Preventative Measures:	☐	☐	☐
8. Provisions have been made to address unattended open excavations to ensure safety of the general public, construction employees, and wildlife until excavation is complete.	☐	☐	☐
9. Excavation design and construction work plan has been developed based off known construction conditions. Changes to the plan require notification to Ryan Field Management.	☐	☐	☐
A. Access Considerations: (Excavation 4' or greater) Ramps, Ladders, Stairways, Etc.	☐	☐	☐
B. Sloping, Benching, or Shoring Required: (Excavation 5' or greater; in WA State 4' or greater)	☐	☐	☐
C. Professional Engineering Required: (Excavation 20' or greater)	☐	☐	☐
10. Has the trench / excavation been assessed as a confined space or a Permit Required Confined Space? If a Confined Space Permit is required, a rescue plan must be in place.	☐	☐	☐
11. Could a hazardous atmosphere exist? If yes, complete a Confined Space Entry Permit and monitor for air quality and/or flammable or toxic gases.	☐	☐	☐
12. Has a Job Hazard Analysis (JHA) been prepared for this excavation and/or ground disturbance.	☐	☐	☐
13. Has the Competent Person held a pre-dig safety meeting to review and discuss this Daily Ground Disturbance Permit with all personnel involved prior to all work being performed? Date: _____ Time: _____ Location: _____ (Note: personnel must sign acknowledgement below)	☐	☐	☐
14. Perform a manhole survey of existing conduit/pipe to confirm all utilities are accounted for.	☐	☐	☐
15. Fiber team notified if work will occur within 6' of a fiber pathway.	☐	☐	☐
16. Work in the public right of way, all applicable laws and notification requirements are followed.	☐	☐	☐
17. Is the disturbance in an Operational Zone or potentially impacting and Operational Zone? If yes, stop and see additional Operational Zone Excavation permit requirements per AWS.	☐	☐	☐

Additional Comments, Instructions, or Requirements:	
Backfill materials around existing utilities shall be the same or similar to what was removed, or as required by the project documents.	☐
Ensure proper and safe compaction around existing utilities.	☐
Protect all existing tracer tape/wire and replace any removed as part of the project.	☐

THIS DOCUMENT CONFIRMS THE CONTRACTOR’S COMPETENT PERSON HAS ACKNOWLEDGED THEY HAVE LOCATED AND WILL PROTECT KNOWN AND SUSPECT UTILITIES. THIS DOCUMENT IN NO WAY CHANGES THE CONTRACTOR’S SOLE RESPONSIBLTY FOR LOCATING AND PROTECTING ALL UNDERGROUND UTILITIES, COSTS ASSOCIATED WITH DAMAGED UTILITIES, AND ENSURING THE EXCAVATION COMPLIES WITH THE APPLICABLE OSHA (local/state) REGULATIONS AS REQUIRED BY THE CONTRACT.

Requesting Contractor	Competent Person	Signature	Date

Made Safe/Closed-Out By Competent Person: _____ Date:

All personnel working under this permit MUST sign that permit requirements are understood.

Name (print)	Role/Position	Sign	Date

Exhibit 15(b) – AWS Operational Zone (OZ) Daily Ground Disturbance Safety Permit

(Please complete in Procore or return hard copy to Ryan Site Supervision)

Project Name/Number: _____ Today's Date: _____

Contractor Requesting Ground Disturbance _____ Competent Person: _____

Ryan Companies US, Inc. on-site Contact: _____

Description and Location of work to be performed: _____

Dates of Disturbance (Approximate): Start: __/__/__ Finish: __/__/__

The designated competent person listed above has made documented notification to all owners/performed a risk assessment/JHA on the named project and has identified the potential hazards and controls associated with ground disturbance. Additionally, the competent person agrees to monitor and comply with all requirements of this plan as well as OSHA 1926.650-652 Subpart P – Excavations, State Plan, OSHA, and AWS Requirements.

Disturbance Type: ☐ Hand Digging ☐ Boring ☐ Excavator ☐ Vac Truck ☐ Rock Excavation ☐ Trencher ☐ Grading <6" (15 cm) ☐ Grading >6" (15 cm) ☐ Driving Stakes/Post/Form Pins			
*Any work using heavy equipment operation requires a separate "Operational Zone Heavy Machinery Permit" to be completed and submitted in addition to the Ground Disturbance Safety Permit.			
Risk Assessment/JHA is provided and includes an Excavation map showing all known utilities, excavation area, and excavation types for the entire excavation area.			
	YES	NO	N/A
1. Confirm all private and public utilities have been located. Ticket(s) #	☐	☐	☐
2. The requesting contractor's competent person has walked the jobsite to look for signs of utilities/services, and has reviewed all available as-builts, contract drawings, current BIM models, and other available information (including owner's rep and adjacent property owners) to identify utilities, and other services in the area of the excavation?	☐	☐	☐
3. All known and assumed underground utilities in the dig zone and search zone have been correctly identified? The use of ground penetrating radar (GPR) or other approved methods to mark utilities has been performed. Utilities identified: Fiber: ☐ Electrical: ☐ Gas: ☐ Plumbing: ☐ Storm/Sewer/Water: ☐ Other: ☐	☐	☐	☐
4. Utility providers' contact information has been obtained – check applicable: Fiber: ☐ Electrical: ☐ Gas: ☐ Plumbing: ☐ Storm/Sewer/Water: ☐	☐	☐	☐
5. Potholing/Hydrovac is required for exposing underground utilities within 30 (thirty) feet of ground disturbance have been (or will be) used to verify exact location of all known underground utilities. All excavations within 10' (3 m) of located known live utility need to be completed by vacuum truck method only. See additional work requirements in additional comments below.	☐	☐	☐
6. Model Change Management or Method of Procedure is created and approved, including approval from the fiber team prior to work taking place.	☐	☐	☐
7. Method of protecting/supporting exposed utilities and/or monuments has been determined and is in place.	☐	☐	☐
8. Precautions have been taken to prevent contact with overhead lines, structures, or other utilities. Preventative Measures:	☐	☐	☐
9. Provisions have been made to address unattended open excavations to ensure safety of the general public, construction employees, and wildlife until excavation is complete. A traffic control plan is provided to the owner 48 hours before starting work in the public right-of-way.	☐	☐	☐
10. Owner shall be notified when excavating within 10' (3 m) of a live utility.	☐	☐	☐
11. Excavation design and construction work plan has been developed based off known construction conditions. Changes to the plan require notification to Ryan Field Management.	☐	☐	☐
A. Access Considerations: (Excavation 4' or greater) Ramps, Ladders, Stairways, etc.	☐	☐	☐
B. Sloping, Benching, or Shoring Required: (Excavation 5' or greater; in WA State 4' or greater)	☐	☐	☐
C. Professional Engineering Required: (Excavation 20' or greater)	☐	☐	☐
12. Has the trench / excavation been assessed as a confined space or a Permit Required Confined Space? If a Confined Space Permit is required, a rescue plan must be in place.	☐	☐	☐
13. Could a hazardous atmosphere exist? If yes, complete a Confined Space Entry Permit and monitor for air quality and/or flammable or toxic gases.	☐	☐	☐
14. Has a Job Hazard Analysis (JHA) been prepared for this excavation and/or ground disturbance.	☐	☐	☐

15. Has the Competent Person held a pre-dig safety meeting to review and discuss this Daily Ground Disturbance Permit with all personnel involved prior to all work being performed? Date: _____ Time: _____ Location: _____ (Note: personnel must sign acknowledgement below)	☐	☐	☐
16. Perform a manhole survey of existing conduit/pipe to confirm all utilities are accounted for.	☐	☐	☐
17. Fiber team notified if work will occur within 6' of a fiber pathway.	☐	☐	☐
18. Work in the public right of way, all applicable laws and notification requirements are followed.	☐	☐	☐
19. Is the disturbance in an Operational Zone or potentially impacting an Operational Zone? If yes, stop and see additional Operational Zone Excavation permit requirements per client(s). A final stakeholder meeting is to be held to review all the information on this permit prior to the work being performed.	☐	☐	☐
Additional Comments, Instructions, or Requirements:			
Backfill materials around existing utilities shall be the same or similar to what was removed, or as required by the project documents.			☐
Ensure proper and safe compaction around existing utilities.			☐
Protect all existing tracer tape/wire and replace any removed as part of the project.			☐
To perform machine excavation in a live area, the area that is to be dug shall be completely circumscribed by vacuum truck excavating a perimeter around the dig area, to a depth of > or = to 2' (60 cm) below the lowest known utility in the area, or until bedrock is encountered. If no existing utilities exist in the circumscribed area, the vacuum truck perimeter shall be completed to 2' (60 cm) deeper than the requested maximum machine excavation. Once the Owner Representative for the project verifies no utilities cross this boundary, machine excavation will then be allowed in the area.			☐
Is bedrock encountered (Y/N)? ____ If yes, the excavation will be completed by vacuum truck to bedrock depth. After confirming no crossing utilities, rock excavation may occur directly below and within the width of the vacuum truck excavation above. Adjacent known utilities to the rock excavation that aren't directly below the vacuum truck trench width but are within 5' (1.5 m) of the rock excavation shall be potholed and special precautions developed to protect any utility within 2' (60 cm) of the rock excavation.			☐
Is excavation in a public corridor (Y/N)? ____ All excavations completed in a public corridor where utilities exist that support the OZ shall be completed by vacuum truck.			☐
Any encountered utilities not shown in the current as-built drawings will be noted for location, depth, and description.			☐

THIS DOCUMENT CONFIRMS THE CONTRACTOR'S COMPETENT PERSON HAS ACKNOWLEDGED THEY HAVE LOCATED AND WILL PROTECT KNOWN AND SUSPECT UTILITIES. THIS DOCUMENT IN NO WAY CHANGES THE CONTRACTOR'S SOLE RESPONSIBILITY FOR LOCATING AND PROTECTING ALL UNDERGROUND UTILITIES, COSTS ASSOCIATED WITH DAMAGED UTILITIES, AND ENSURING THE EXCAVATION COMPLIES WITH THE APPLICABLE OSHA (local/state) REGULATIONS AS REQUIRED BY THE CONTRACT.

Requesting Contractor	Competent Person	Signature	Date

Made Safe/Closed-Out By Competent Person: _____ Date: _____

All personnel working under this permit MUST sign that permit requirements are understood.

Name (print)	Role/Position	Sign	Date

Exhibit 16 - Silica Exposure Plan (See Procore Forms or Tools for fillable form)

Note: Personnel considerations will be dependent on both the time required to complete the task and the requirements of OSHA Table 1.		
Date of Task:	Duration of Task(s) per 8 hr sh	Description of the silica dust-producing task to be performed:
	☐ Single Task <4 Hours* ☐ Single Task > 4 Hours* ☐ Multiple Tasks > 4 Hours*	
Project Name:	Perimeter Precautions: (Control measures to keep unauthorized personnel out of the work area)	
Project Number:	Location of task on Jobsite: Choose indoor or outdoor and provide details below.	
	__Indoor __Outdoor	
Competent Person's Name (Print):	RYAN Superintendent Name (Print):	
Competent Person's (Signature):	Other subcontractors affected and notified of the silica dust generating task:	
STEP 1: Will you generate silica dust while working with any of the following materials? Check all that apply. (Note: Consult the SDS Sheet if needed)		
☐ Asphalt ☐ Brick ☐ Concrete ☐ Concrete Block ☐ Drywall and/or Joint Compound ☐ Fiber-Cement Products	☐ Grout ☐ Gunite/Shotcrete ☐ Mortar ☐ Rock ☐ Rock Tile (Concrete) ☐ Sand	☐ Stone/Granite/Quartz/Lmstn/Sndstn/Shale/Slate ☐ Precast Materials (Concrete) ☐ Stucco (EIFS) ☐ Terrazzo ☐ Tile (Clay or Ceramic) ☐ Other:
STEP 2: Select the task you will be performing and the dust control measures you will be using with the materials selected above.		
Note: All workers engaged in any of the tasks below requiring the use of a respirator MUST BE clean shaven.		
☐ Abrasive Blasting ☐ Blasting System with Water ☐ Respiratory Protection ☐ Other:	☐ Drilling/Coring ☐ Hand held drill with dust collection ☐ Hand held drill with vacuum ☐ Hand held drill with water ☐ Drill Press with water or vacuum ☐ Respiratory Protection ☐ Other:	☐ Polishing ☐ Tool with vacuum ☐ Tool with water ☐ Respiratory Protection ☐ Other:
☐ Bushhammering ☐ Tool with vacuum ☐ Tool with water ☐ Respiratory Protection ☐ Other:	☐ Jackhammering ☐ Tool with vacuum ☐ Tool with water ☐ Respiratory Protection ☐ Other:	☐ Sanding/Sacking/Patching ☐ Hand-held sander with vacuum ☐ Hand held sander with water ☐ Walk-behind with vacuum ☐ Walk-behind with water ☐ Respiratory Protection ☐ Other:
☐ Cutting/Sawing ☐ Masonry Saw with water ☐ Hand-held grinder with vacuum ☐ Walk-behind saw with water ☐ Walk-behind (Soft Cut saw with HEPA vac system) ☐ Hand-held power saw with water ☐ Respiratory Protection ☐ Other:	☐ Milling ☐ Shroud with vacuum dust collection ☐ Portable mixing station with water ☐ Silo dispensing system ☐ Respiratory Protection ☐ Other:	☐ Scarifying/Scraping/Scabbling ☐ Walk behind tool with vacuum ☐ Walk behind tool with water ☐ Respiratory Protection ☐ Other:
☐ Demolition/Disturbing/Grinding ☐ Duct Fan Unit ☐ Water Applied at task ☐ Heavy Equipment with enclosed cab ☐ Hydraulic breaker ☐ Respiratory Protection ☐ Other:	☐ Mixing/Pouring ☐ Shroud with vacuum dust collection ☐ Enclosure with dust collection ☐ Enclosed operation with water ☐ Respiratory Protection ☐ Other:	☐ Wet Sweeping/Clean-Up/Housekeeping ☐ Sweeping compound ☐ Vacuum ☐ Respiratory Protection ☐ Other:
STEP 3: What precautions will be taken to limit/restrict access to the work area. These precautions must be adequate to protect nearby personnel from being exposed to airborne silica dust. CIRCLE all that apply.		
Preconstruction meeting/pre-planning Signs/barricades Tape/rope Advance notice HEPA air machines Ventilation Plastic enclosures / negative air / local exhaust Other:		
STEP 4: What type of training has been provided to all workers engaged in the task(s) listed above. CIRCLE all that apply.		
Silica Hazard Training Verbal Communication Toolbox Talks PreConstruction Meeting Videos SDS/HazComm Job Hazard Analysis Other:		

STEP 5: Which housekeeping measures will be used on the project to limit employee exposures to silica? Note: appropriate controls during the operations will prevent unnecessary exposure and clean-up after the task(s) have been completed. CIRCLE all that apply.

Equipment with HEPA Vac System, Wet Sweeping, Poly Enclosure, HEPA Air Filtering Machines, No Compressed Air for Cleaning; No Dry Sweeping; Wet Compound Sweeping; Area Inspection by Competent Person Upon Completion of Task; Hand Wash Station/Hygiene Facility

STEP 6: Have workers been medically cleared to perform the task(s) described in this Exposure Control Plan? See OSHA 29 CFR 1926.1153 (h)(1) for Medical Surveillance requirements.

☐ Yes, Medically Cleared per OSHA 29 CFR 1926.1153 (h)(1) **Hard Hat Sticker**
☐ No, Not Medically Cleared (If "no", contact Superintendent/Safety Director/Field Coordinator)

Engineering controls including local ventilation, equipment specifications, deploying wet methods; Administrative controls including written ECP, housekeeping, accident prevention signs, washing/hygiene facilities, safety and health program, job hazard analysis, limit time of worker exposure and implementing rotation of tasks, coordination of job activities to minimize silica exposure to workers; personal hygiene and personal responsibility for clean eating areas, personal contamination and proper PPE; (respiratory, eye, face, hand, head and foot protection); Monitor and evaluate effectiveness of engineering and administrative controls for unforeseen changes.

STEP 8: When a situation or condition arises that has not been addressed in this ECP, work shall **STOP and will not proceed until a safe plan of action is created and added to this ECP. The revised ECP shall then be reviewed with the crew before proceeding. Added tasks shall be logged in the space provided below and the crew shall initial in the space provided by their signatures to acknowledge the change has been communicated. More than two changes will require a new ECP for the shift.**

Change #1: Describe the change, reason and steps taken to control dust.

Change #2 Describe the change, reason and steps taken to control dust.

SAFETY AUDIT, HOUSEKEEPING AND CLEAN-UP (INSPECTION OF WORK AREAS DURING AND AFTER THE SHIFT)

	TIME	PERFORMED BY (Sign Name)	ADDITIONAL HAZARDS FOUND
Audit #1			
Audit #2			

CREW SIGNATURES (ALL MEMBERS OF THE CREW INVOLVED IN THE TASKS INCLUDED IN THIS ECP MUST SIGN)

	PRINT NAME	SIGNATURE	PLAN CHANGES
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

POST-TASK REVIEW (Competent person must fill out the following at the end of each work shift)

	Y	N	
Did today's tasks go as planned?	☐	☐	If NO , why not?:
Did workers report any concerns?	☐	☐	If YES , what was the concern?:
Did others report any concerns?	☐	☐	If YES , what happened?:
Work area cleaned at end of the shift?	☐	☐	If NO , why not?:

Copies of this plan shall be retained at the work site for review. DO NOT DISCARD.

**Workers engaged in more than one task with a combined duration of greater than 4 hours may need their respiratory protection measures re-evaluated. Any worker engaged in more than one task during their 8 hour shift must wear the highest level of respiratory protection during ALL tasks performed in the 8-hour shift. (i.e. if ANY of the tasks require use of a respirator for over 4 hours performing that task, then the respirator must be used during all tasks for the day regardless of duration of the task).*

Exhibit 17 - Monthly Jobsite Safety Committee Meeting Agenda (See Procore Meetings)

1. The Monthly Jobsite Safety Committee Meeting will be led / conducted by the Ryan Field Manager or Ryan Safety representative and shall include the following:
 - Ryan Safety Representative/Coordinator and/or Safety Consultant
 - Ryan Field Manager or site supervision
 - Subcontractor Safety Representative / Competent Person and/or site supervision
2. The AWS - ATL089 Monthly Jobsite Safety Committee Meeting will be held on the **First Wednesday** of each month at **10:00 am** at the Ryan jobsite office. Note: hold this meeting shortly in advance of your project's ALL JOB Meeting so results from the Monthly Jobsite Safety Committee Meeting can be shared / discussed.
3. Jobsite Safety Committee members will inspect the AWS - ATL089 jobsite prior to the monthly Jobsite Safety Committee Meeting.
4. Safety-related items to be reviewed:
 - a. Intro Safety Committee members
 - b. Create a Sign-In Form (see Procore Forms) and have all committee members sign-in.
 - c. Review any previous months' issues or outstanding action items needing follow-up
 - d. Conduct a Safety Audit/Inspection walk-through with committee members utilizing the Procore Safety Inspections
 - e. Review any safety issues observed during the Safety Inspection and create "Observations" for each safety issue observed
 - f. Review any Safety Violations issued over the past month
 - g. Incidents, injuries, near-misses, and corrective actions – review from past month and ensure reporting is complete in Procore
 - h. Ryan Best Practices – identify Safety best practices that can be developed or deployed – document Best Practices in the Monthly Jobsite Safety Committee Meeting Minutes
 - i. Review any submitted Safety Suggestions / Safety Enhancements
 - j. Look ahead - identify high risk scopes / potential hazards in consideration of upcoming construction phase(s) or staffing / weather or seasonal conditions – utilize Pre-Task Planning
 - k. Subcontractor Safety – review any deficiencies or acknowledge / recognize improvements / achievements; include Ryan Self-Perform in subcontractor reviews
 - l. Safety Consultant – obtain feedback or reports (if any)
 - m. Record Monthly Jobsite Safety Committee Meeting minutes – see Procore Meetings
5. Documentation of meeting minutes and attendees shall be created, distributed and kept in Procore Meetings.

Exhibit 18 - Overhead Power Line Awareness Permit (See Procore Forms)

RYAN COMPANIES US, INC.

Today's Date _____

Job Number: _____

Contractor Name: _____

Temperature: _____

Wind: _____

Note Workers with existing heart conditions will not be allowed to work with cranes, high-reaching equipment, material handling equipment, hauling equipment, and earthmoving equipment, i.e., pacemakers and defibrillator devices.

Questions

- Have you been orientated regarding power line issues on this site? ☐ yes ☐ no
- Will your work involve equipment that can get near the 20' "safe zone" of the Energy Provider's power lines? ☐
yes ☐ no (Refer to the [Overhead Power Line Awareness Procedures](#) tables A and T to establish your project specific "safe zone")

Identify

After finding all of the power lines at your site, identify the activities you'll be doing that may put you or your workers at risk. Mark one or more of the following:

- | | |
|--|--|
| ☐ Cranes (mobile or truck mounted) | ☐ Aerial lifts |
| ☐ Drilling rigs | ☐ Dump Trucks |
| ☐ Backhoes / Excavators | ☐ Ladders |
| ☐ Long-handled tools | ☐ Material Handling & Storage |
| ☐ Other tools/high-reaching equipment | ☐ Scaffolding |
| ☐ Concrete pumper | ☐ Other |

Eliminate or Control

After identifying the power line and high-risk activities on the jobsite, you must determine how to eliminate or control the risk of electrocution. Mark one or more of the following:

- | | |
|---|---|
| ☐ Move the activity | ☐ Use non-conductive tools |
| ☐ Change the activity | ☐ Use an observer / competent person |
| ☐ Have the utility de-energized power line | ☐ Use warning lines with flags |
| ☐ Use of protective technology (list): | |
| ☐ Other: | |

REMINDERS:

- **Communicate with your crew regarding power line safety on a constant basis.**
- **Always maintain your minimum 20-foot safe clearance distance from the power line.**
- **When in doubt, STOP and talk to your supervisor.**

Jobsite Sketch - Not to scale

(Highlight approximate areas of your daily working activities covered by this permit.)

A large, empty rectangular box with a thin black border, intended for a jobsite sketch. It occupies the central portion of the page.

Note: Add North arrow; Power line locations and their distances from work areas

Completed by: _____ Date

Title

Approved by: _____ Date

Title

Exhibit 19 – Cattle Gate Procedures

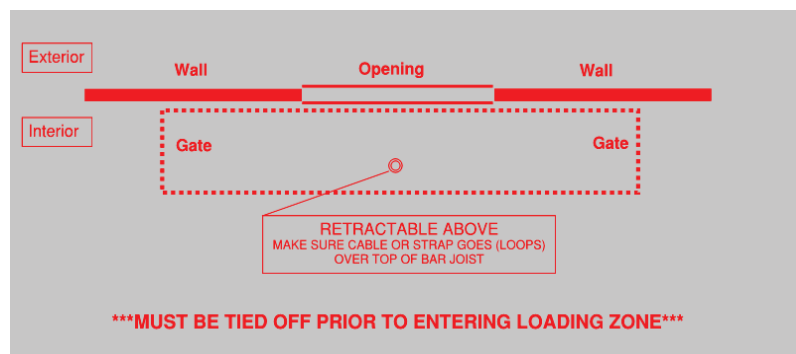
Side Loading Gates:



1. No person shall be permitted to enter a corralled loading area without first being tied-off:
 - a. Inspect self-retracting lifeline, anchor attachment point, and full body harness prior to use.
 - b. After donning full body harness, attach self-retracting lifeline to harness D ring behind shoulders.
 - c. Tie-off to an anchor attachment point with the self-retracting lifeline must be completed prior to entering the Cattle Gate corralled area.

Anyone inside the corralled area MUST be tied off at all times and before any gates can be opened!

2. All gates must be in place and closed at **ALL times when not in use.**
3. Both side gates must be in place and closed before opening the exterior gate for importing or exporting materials / equipment.
4. Enter the corralled Cattle Gate area from one side gate only and close the gate behind you. The other gate must remain closed.
5. While importing or exporting equipment / materials from the Cattle Gate area with the exterior gate open, both interior side gates must be closed.
6. Use hand signals and/or verbal communication for hoisting, and/or landing materials or equipment within the loading zone. Close the exterior gate immediately following use. While remaining tied off, move the load away from exterior gate and open the opposite side gate for exiting the Cattle Gate corral. This minimizes fall exposure for anyone working outside of the Cattle Gate opening.
7. Shut the interior gate, **do not walk away until this is complete.**



Long-loading / Backing:



THIS IS A FALL HAZARD AREA!

1. No person shall be permitted to enter a corralled loading area without first being tied-off:
 - a. Inspect self-retracting lifeline, anchor attachment point, and full body harness prior to use.
 - b. After donning full body harness, attach self-retracting lifeline to harness D ring behind shoulders.
 - c. Tie-off to an anchor attachment point with the self-retracting lifeline must be completed prior to entering the Cattle Gate corralled area.
 - d. Enter the corralled area, shutting Gates 1 and 2 behind you.
2. Weighted-base guardrail system must be in place with railings inserted into weighted bases with pins installed.
3. All interior Gates (Gates 1 and 2) must be closed prior to opening Exterior Door.
4. While loading/hoisting materials or equipment, interior Gate 1 (see diagram) must be closed/secured at **ALL** times.

Reminder: You MUST be tied off before Exterior Door can be opened!

5. Open Exterior Door.
6. Receive material(s) and move material(s) as needed to avoid blocking the Exterior Door.
7. Close and secure Exterior Door.
8. Open interior Gates (Gates 2 and 1) to remove material.
9. Shut interior Gates (1 and 2) and replace posts, weighted bases and pins if needed.
Do not walk away until this is complete.

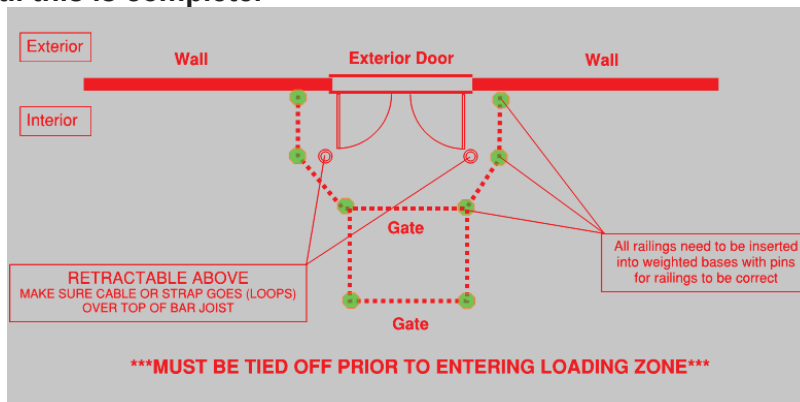


Exhibit 20

TEMPORARY GUARDRAIL SYSTEMS REMOVAL PERMIT



No work will be permitted, by any Contractor and/or it's employees, on any guardrail systems until this permit is completed.
(This document will serve as compliance with Federal OSHA 1926.500 thru 1926.503 (c)(3) of OSHA's Subpart-M regulations, and State Plan OSHA Requirements)

Date of Request: _____ Project Name/Number: _____
Start Date of Work: _____ Anticipated Completion Date: _____
Contractor Name: _____
Competent Person (s) Name: _____ Phone Number: _____
Ryan's Onsite Contact: _____

Location(s) of Guardrails Systems Removal: (list detailed area description below)

Type of fall protection being removed (wood, cable, weighted gates/base)	Location (North/South/East/West)	Elevation/Floor	Grid/Column Line(s)	Length of guardrail system being removed (# of feet, total)

Employee(s) Performing the Work: Any employee tasked with guardrail system(s) removal shall be properly trained in all applicable OSHA Subpart-M regulations. The competent person, named above, certifies that the following employee(s) have been properly trained for all tasks associated with guardrail system(s) removal requirement, personal fall protection systems required for their safety and all fall protection systems necessary for ensuring the safety of others exposed to any hazards created during the removal of existing guardrail systems.

Employee Name:		Employee Signature:	
1.		1.	
2.		2.	
3.		3.	
4.		4.	
5.		5.	

Removal Criteria: Prior to the removal of any perimeter fall protection the contractor shall be responsible for providing applicable fall protection equipment for those performing the task as well as protecting all others in the area. The following protections shall be taken by the contractor performing the work:

Fall Protection Plan:	Yes	No
Warning lines will be erected at a minimum of 15-ft from any unprotected sides or edges prior to removal of the guardrail system(s)? (Warning lines must meet OSHA Subpart M requirements and State Plan OSHA Requirements)	☐	☐
Do warning lines have flagging and safety signage posted ("Danger" Fall Protection Beyond This Point Restricted/Controlled Access Zone)?	☐	☐
Warning lines will be maintained at a proper distance & height throughout the entire work shift?	☐	☐
Do all floor holes, roof holes, & shaft ways have a hard rail barricade system erected at a minimum of 6' from the opening? "Danger" Fall Protection Required Beyond This Point signage must posted. NOTE: Warning lines cannot be used as barrier protection for openings.	☐	☐
Have employees been provided with fall protection systems & training prior to removal of the guardrail systems.	☐	☐
Are all other trades protected from any fall hazard prior to removal of guardrail systems?	☐	☐
All guardrails will be re-installed upon completion of the task as well as at the end of each work shift?	☐	☐

****Answering “No” to any of the above conditions will be grounds for denial of perimeter fall protection system removal****

I, _____, (Print Name) as a representative/competent person of _____, (Print Company Name) agree to all fall protection requirements as outlined in this Permit, applicable Federal OSHA's *Subpart-M* regulations, State Plan OSHA Requirement and/or Ryan site specific policies. In addition, I agree to all temporary measures for protecting both employees and/or other trades, prior to the removal of existing guardrail systems. Upon completion of the task, but not to exceed one shift, I agree to properly re-install all guardrail systems in accordance with Federal OSHA's *Subpart-M* regulations, State Plan OSHA Requirement and/or Ryan site specific policies.

(Signature of Contractor Representative/Competent Person)

(Date)

Completion Inspection: Guardrail systems shall comply with the following requirements before being turned over as general work areas for other trades on this site. Once the following conditions have been completely met, the work area may be turned back over for access by others. Should any unsatisfactory conditions or safety hazards be identified, immediate actions must be taken by the contractor's competent person that removed the guardrail systems.

Points of Inspection: Top rails must be able to withstand, at a minimum, 200 pounds of force at all points along the system. Mid-rails must be able to withstand, at a minimum, 150 pounds of force and all toe-boards must be installed and capable of withstanding 50 pounds of force at all points along the system. Note: If applicable, mesh netting system must be reinstalled.

Name of Competent Person Inspecting Guardrail Systems:	Contractors Name:
Floor/Area/Grid Lines Inspected:	
Top Rail: Top rail height 42 inches. NOTE: from finished floor. Yes: _____ No: _____	Mid Rail: Mid rail height 21 inches. NOTE: from finished floor. Yes: _____ No: _____
Intermediate Post: Intermediate supports 8 feet on center. NOTE: can be omitted if slack plus/minus 3 inches can be constantly maintained. Yes: _____ No: _____	Cable Guardrail Deflection: Not more than 3 inches with 200 pounds of force applied to top rail and maximum 3 inches deflection on mid rail with 150 pounds of force applied. Yes: _____ No: _____
Cable Clamps: Cable clamps installed correctly. NOTE: never saddle a dead horse. Minimum of 3 Drop Forged clips used. Minimum of 3 inches between clips. Cable splices formed by eyes only. Yes: _____ No: _____	Cable Guardrail System: Cable diameter is at least 1/4 inch. NOTE: guardrail system only – NOT FOR TIE OFF Cables are flagged at 6 ft. intervals, (top rail). Yes: _____ No: _____
Made Safe/Closed-Out by Competent Person: Signature: _____ Date: _____	

Exhibit 21 - Safe Lift Permit Plan



Additional documentation is required if the lift is classified as a “critical” lift or if the lift is performed by a helicopter. If the lift involves temporary works, please see the full list of the GMR Risk Mitigation Requirements. All lifts require a JHA, PTP and Safe Work Plan to be added to this permit.

Title of Lift:

Project Name:

Date of Lift:

Location:

Date Form Completed:

Subcontractor Name:

Competent Person:

Characterize the Load(s)

This Plan Covers:

Single Load Only ☐ Variety of similar loads (plan for largest) ☐

Source of Load Weight: (Nameplate, Weighed, Calculated)

Length:

Width:

Height:

Diameter:

Load Weight: (add 10% if not directly weighed)

Crane/Hoist Configuration

Crane and/or Hoisting Equipment to be utilized:

Rated Gross Capacity:

For Hoists using a PIT attachment ONLY:

PIT Make/Model:

Attachment:

Attachment listed on Data Plate: ☐ Yes ☐ No

Boom Length:

Max Fork Height for Lift:

PIT Capacity with Attachment:

Characterize the Task (Include directions for lifting, rotation, travel. Barricading with signage of the lift area is required to ensure unexpected traffic are aware of the lift in progress. Safe clearance distance or protection is required when lifting near or over unprotected equipment or services. Protection systems shall be designed by a qualified person and installed by a competent person. Documented Inspections of the protective system(s) are required by a competent person if left in place for extended durations.)

Evaluate the Hazards

(Sharp corners, nearby equipment, load twisting, PIT, mezzanine load)

Plan the Rigging

(Sketch or photo rigging plan on grid. Show the load rigging, lifting device, and the type of gear to be used.)

Include all information required to determine that the load is properly rigged, and that appropriate rigging gear is selected. Include, as applicable, sling angles, eyebolt orientation, padding points, center of gravity, type of sling hitch, and any other pertinent information.

Define Rigging Gear Requirements

1. List each piece of rigging gear (such as load hook, shackles, slings, eyebolts). List weight if component >10 lbs.
2. Label the sketch or photo using the corresponding letter for the gear.
3. Draw sling angles and the resulting load reduction factors for slings and eyebolts.
4. Calculate the force on each piece of rigging gear. Show that angles are accounted for in determining forces.
5. Determine the required rigging gear capacity and size. Indicate if this is an exact specification or a minimum.

Total Rigging Weight:

Type	Weight	Force on Rigging Gear	Capacity/Rating/WLL	Size Specification
A				
B				
C				

D				
E				
F				
G				
H				
I				
Total Lift Calculations				
Gross Capacity of Crane/Hoist = B				lbs.
Gross Load Weight = Load Weight from A + Rigging Weight from F (MUST BE LESS THAN 1000 LBS)				lbs.
Percentage of Lifting Device Capacity = Gross Load Weight/Gross Capacity (MUST BE LESS THAN 90%)				lbs.

Plan Approval (A documented communication plan is required between the crane operator, signal person(s), and rigger(s).)		
<u>Plan Preparer / Competent Person</u>		
Print Name:	Signature:	Date:
<u>Certified Crane Operator / Qualified Rigger(s)</u>		
Certified Crane Operator:	Signature:	Date:
Qualified Rigger:	Signature:	Date:
Qualified Rigger:	Signature:	Date:
<u>Qualified Signal Person(s)</u>		
Print Name:	Signature:	Date:
Print Name:	Signature:	Date:

Pre-Lift Plan Acceptance (All participants prior to lift must sign to acknowledge understanding of plan during pre-lift brief and that they have been trained to perform the task. Subcontractor(s) Competent Person(s) shall perform a personnel training records review prior to the activity(s) and ensure all personnel have been trained on this permit and have been trained to perform their individual tasks associated with the lift or the lift does not commence).		
Employee:	Signature:	Date:

Exhibit 22 – Ryan Photograph Policy



AWS Authorized Photographer:

- Only AWS authorized photographers are allowed to take photos on AWS data center projects.
- All Ryan team members and all trade partner personnel must obtain AWS approval prior to taking photos onsite.
- AWS authorized photographers must also have AWS Procore access.
- All authorized user request forms must be submitted through the Ryan project management team and may take several days to process.

Photography Policy Requirements:

- Photos can only be taken using approved cameras, issued by Ryan Companies.
- Photos can only be taken in the construction zone (CZ), prior to PFHO. Once the PFHO milestone has been achieved, the ROMP/phase is designated as an operational zone (OZ).
- Photos cannot be taken within the operational zone (OZ), without approval from AWS. This AWS approval process could take up to 2 weeks and is not guaranteed.
- Photos cannot provide visibility to the OZ, portions of the live buildings, or anything that would give an indication of location.
- Photos cannot be taken down network aisles where network racks are placed at NEA.
- Photos cannot show any network rack, patch rack, or service rack.
- Photos cannot show BACOP, AMCOP, BOB, or HMI equipment.
- All photos should focus on the item needing a photo (wide-angle shots are not allowed).

Photography Procedures:

- An authorized user can check out an approved camera, by submitting a Ryan Camera Checkout Form to the Ryan administrator.
- The Ryan administrator will log the date, camera number, project, trade partner, authorized username, and reason for photos (designated folder name).
- An approved camera will be issued to the authorized user.
- Cameras must be returned within 4 hours or as otherwise approved by the Ryan administrator.
- The authorized user will return the camera to the Ryan administrator for photo downloads.
- The Ryan administrator will watermark all photos (Amazon.com Confidential), create the subfolder designated by the authorized user, and upload all watermarked photos into the AWS Procore designated folder.
- The Ryan administrator will sanitize the memory card each day and between issuing the camera to another authorized user.
- Authorized users can create additional sub-folders within the AWS Procore folder structure for organizational purposes.

File Sharing:

- Authorized users can include the photos in other AWS Procore project tools (observations, inspections, punch-list, etc.), but photos cannot be removed from AWS Procore.
- The only option for sharing photos is by sending links to the photos through AWS Procore. Only people added to the AWS Procore projects (with signed NDAs) will have access.

By signing below, I agree/acknowledge that I have read and understand the requirements of the Ryan Companies Photography Policy for AWS Data Center Projects.

Print Name: _____ Signature: _____

Exhibit 23 – Working at Heights Permit



Working at heights refers to any work that is done that exposes a person to a fall hazard of 6 feet or greater (4 feet in Washington). Working at heights can also refer to work done near unprotected sides and edges, leading edges, roofs, hoist areas, working on a scaffolding, working on ladders, elevated work platforms, equipment, near holes, shafts, wells, pits, excavations, trenches or any other items listed in OSHA's Subpart M, CFR 1926.500 – 503. **Working at Heights cannot be performed alone!**

Project Name: _____
Project Number: _____
Subcontractor Name: _____
Working at Height location: _____
Building: _____
Date: _____
Time: (Start - End) _____
Ryan Companies On-site _____
Contact Name: _____

Can the work be performed without working at heights by substituting the task(s)?

☐ YES ☐ NO

Can most of the work be performed at ground level away from fall hazards?

☐ YES ☐ NO

Can engineering controls be used / installed?

☐ YES ☐ NO

QUALIFIED TRAINED PERSONNEL PERFORMING THE TASK(S):

PRINT NAME	JOB TITLE	CONTACT PHONE NUMBER

REQUIRED RESCUE PLANS: Provide an attachment to this permit of a detailed rescue plan for working at heights to include at a minimum:

- Type of rescue equipment available (e.g., harness equipped with suspension trauma straps, rescue retrieval equipment, lifts, cranes, rescue basket, stair tower / ladders, etc.).
- Location of rescue equipment.
- Names and contact information of trained rescue personnel.
- Procedures for retrieving a fallen worker safely.
- Name, phone and address of the closest medical care center.
- 911 is to be called immediately and shall not be the primary source listed in the plan as the rescue team.

REQUIRED RISK ASSESSMET (JHA): Provide an attachment to this permit of a detailed Risk assessment / Job Hazard Analysis (JHA) for today's working at heights. Crewmembers are to be trained on the risk assessment / JHA by their employer prior to performing the task(s). In accordance with the CODE of FEDERAL REGULATIONS (CFR) 1926.500 – 503, the competent person has performed a risk assessment for both existing and potential hazards and rescue provisions associated with working at heights.

DESCRIPTION OF WORK: Provide a brief description of the work taking place at heights, including the specific tasks and equipment to be used.

HAZARDS IDENTIFIED: List any potential hazards associated with working at heights, such as falling, falling objects, congested working areas, equipment, moving parts, weather conditions / slippery surfaces, etc.

CONTROL MEASURES: Outline the control measures to mitigate the identified hazards. This may include substitution / elimination, engineering, administrative and PPE. Examples could be scaffolding, guardrails, safety nets, fall protection equipment, etc. A visual inspection of all safety equipment will be done by a competent person daily or before each use. Damaged or defective equipment must be immediately removed from service. All safety equipment shall fit the user and be properly stored when not in use.

ANCHORAGE POINTS: If fall arrest equipment is to be used, list the type of anchorage devices to be used and what they will be attached to. Installation of anchorage points shall be positioned to always maintain 100% tie – off. Anchorage Points shall meet all requirements of OSHA’s Subpart M, OSHA 29 CFR: 1926.502 (d)(15) requirements, or OSHA State Plan requirements.

EMERGENCY PROCEDURES: Describe your specific emergency procedures in the event of an incident, including first aid, CPR, AED measures and evacuation plans where the work is being performed.

APPROVAL:

SUBCONTRACTOR’S COMPETENT PERSON NAME:	SIGNATURE:	DATE:

SUBCONTRACTOR'S SAFETY REPRESENTATIVE:	SIGNATURE:	DATE:

WORKER'S ACKNOWLEDGEMENT: I acknowledge that I have read and understand the working at heights permit and agree to follow the safety procedures outlined in this permit, the attached Job Hazard Analysis (JHA) and Rescue Plan. I also have been adequately trained to safely perform my task(s) by my employer including the use and inspection of all required PPE and fall protection systems.

Working at heights permit requires at least two people to be always assigned to the work area.

Working at height cannot be performed alone!

PRINT NAME:	
SIGNATURE:	
DATE:	

PRINT NAME:	
SIGNATURE:	
DATE:	

PRINT NAME:	
SIGNATURE:	
DATE:	

PRINT NAME:	
SIGNATURE:	
DATE:	

WORK COMPLETED:	
COMPETENT PERSON NAME:	
SIGNATURE:	
DATE:	



PDC LB Testing in Datahall

Method of Procedure (MOP)

Overview

This document establishes a Method of Procedure (MOP) for connecting, energizing, and disconnecting LBs (Load Bbank) to perform LB testing on PDCs in AWS data halls.

The GC team may choose to either utilize this MOP in it’s current format, or create their own MOP that includes the same level of care. The GC remains responsible for the overall safety of the project and should insert additional steps as necessary to ensure safe work practices.

Ref-MOP Documents Information

Version	Date	Author(s)	Approved by	Revision Notes
01	09/16/2024	nimaghad@	smmcgary@	
02	10/15/2024	nimaghad@	smmcgary@	

Ref-MOP Table of Contents

Important Notes2

Safety Requirements.....2

Important Indicators and Acronyms2

Technical and Related Documentation2

Personnel Information2

Affected Equipment3

Required PPE, Tools, and Materials3

Preparation Work Prior to Connecting to PDCs and Mitigation Steps.....3

Procedure for Connecting LB to the PDC4

Procedure for Disconnecting LB from the PDC5

Comments.....6

Important Notes

This MOP outlines the steps for connecting and disconnecting LBs to PDCs in the datahall.

-  Important LB testing should be completed on one source of row power at a time, either primary or catcher, to maintain power to any energized network racks. Single source of power is always needed for powered racks in the space.
-  Important Refer to the LB manufacturers operation manual for LB control wiring and interconnection between multiple LB for the purpose of centralized control.
-  Important Ensure employees have adequate experience, training, and supervision for working with and operating LBs and LB cables.
-  Important Do not perform this procedure if you are not qualified.
-  Important Always work safely and prioritize safety over performance and speed.

Safety Requirements

Requirement	Description
Safety Briefing	Follow the GC/EC safety briefing procedure to identify additional safety hazards and correctly mitigate those hazards to minimize risk to personnel.

Important Indicators and Acronyms

The following indicators identify critical steps in this procedure.

Indicators					
	Critical switching step		Safety alert		Stop, validate/verify, go/no go
	Change of state		Important note		Rollback here, if necessary

Technical and Related Documentation

Title
NFPA 70E and/or any other regional applicable electrical safety standards.
GC/EC Health and Safety Plan
PDC Product Submittal
LB Operation Manual

Personnel Information

List and provide information on personnel participating in this procedure (add rows as necessary).

Company Name	Name / title	Alias/Contact #	Role	Initials
			Supervisor (PIC)	
			Worker #1	
			Worker #2	
			Worker #3	
			Worker #4	

Company Name	Name / title	Alias/Contact #	Role	Initials
Escalation Contacts				
			CM	

Affected Equipment

List all equipment that will be worked on, shut off, or locked out during this procedure (add rows as necessary).

Line #	Equipment name	Location
1	PDC x.xAx or PDC x.xBx	ROMPXX

Required PPE, Tools, and Materials

List the personal protective equipment (PPE), tools, and materials needed to do this procedure (add rows as necessary).

PPE / Tools / Materials	Reason needed
Hard Hat	Required of all personnel present in construction zones
Safety Glasses	Required of all personnel present in construction zones
Work Clothing and Footwear	Required of all personnel present in construction zones
Safety Vest	Required of all personnel present in construction zones
Hand tools	General Construction
Lockout Tagout (LOTO) Kit	Required to complete LOTO on Equipment

Preparation Work Prior to Connecting to PDCs and Mitigation Steps

Section 1: Preparation Work Prior to Connecting to PDCs and Mitigation Steps				Completed by <vendor>
Step	Location	Equipment	Action/Task	Initials
1.1.	Office		Record the time and open the work order. • Time: _____	
1.2.	Office		In the Employee Information section, record the contact information for all personnel involved in this work.	
1.3.	Office		 Important: Ensure the qualified personnel are involved in this procedure. PIC and all people involved in this work confirmed that they have received proper training to connect/disconnect and operate the LB and LB cables. **All involved workers shall initial the box to the right.	
1.4.	Jobsite		 MOP Rehearsal – Prior to beginning the MOP, all employees assigned to execute the MOP must participate in a thorough rehearsal of the MOP. This rehearsal must be led by the PIC. Important: AWS CM should be invited to the walk-through/dry-run, but their attendance is not mandatory.	

Section 1: Preparation Work Prior to Connecting to PDCs and Mitigation Steps				Completed by <vendor>
Step	Location	Equipment	Action/Task	Initials
1.5.	Jobsite		Ensure all tools, materials, and PPE are staged in the work area.	
1.6.	ROMPXX	LB	 Confirm LBs are on site, in good operating condition, and meet the requirements of the LB plan. Check for any physical damage during the offloading.	
1.7.	ROMPXX	LB Cable	 Confirm correct number and size of LB cables are onsite.	
1.8.	ROMPXX	LB	Place LBs and LB cables in data hall per Commissioning team provided LB plan.	
1.9.	ROMPXX	LB Cable	 Verify all LB cables are laid flat without running on top of each other.	
1.10.	ROMPXX	LB Cable	If cable bundling is needed, arrange three phases bundles rather than similar phases. Make sure LB cables are never coiled.	
1.11.	ROMPXX	LB Cable	 Ensure all LB cables are identified with matching phase colors on both ends. Important: Phase tape colors should correspond to the PDC/LB phase coloring scheme.	
1.12.	ROMPXX	LB Cable	 Confirm cables are checked for visible insulation damage, bulging, and Camlock boots are secured properly. Perform tug test on each Camlock boot to ensure tightness.	
1.13.	ROMPXX	LB Cable	Maintain enough slack in the cables to allow switching between Primary and Catcher test ports on the same PDC.	
1.14.	ROMPXX	LB Cable	Connect all cables to the LB following colors matching each phase, neutral, and ground.	
1.15.	ROMPXX	LB	Confirm all camlock connections at the LB are in locked position by twisting them and perform pull test.	
1.16.	ROMPXX		 Confirm that safety signage is installed and occupants have been notified of energized LBs in the datahall.	
1.17.	ROMPXX	LB	 Confirm LB main power switch is in OFF position.	
1.18.	Jobsite		Notify PIC/ Site Superintendent that LB to PDC connection will be starting.	
Summary: Pre-work and mitigation steps are now complete.				

Procedure for Connecting LB to the PDC

Section 2: Connecting LBs to <PDC x.xA/Bx>				Completed by <vendor>
Step	Location	Equipment	Action/Task	Initials
2.1.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 CST REQUIRED Wearing proper PPE, OPEN and LOTO Q1/Q4 breaker. Verify zero voltage reading on the HMI.	
2.2.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Verify 250A Q2/Q5 breaker is CLOSED .	
2.3.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Verify 250A Q3/Q6 breaker is CLOSED .	

Section 2: Connecting LBs to <PDC x.xA/Bx>				Completed by <vendor>																																				
Step	Location	Equipment	Action/Task	Initials																																				
2.4.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 At the PDC, remove Camlock exterior cover and safety caps, if applicable, and perform zero energy check on each phase, neutral and ground port. Important: PIC and workers to initial on this step.																																					
2.5.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Identify the 30A branch circuit breakers feeding live racks and mark them as DO NOT OPEN .																																					
2.6.	ROMPXX	PDC x.xA/Bx Primary/Catcher	OPEN all branch circuit breakers (MCBs) except the ones marked to remain CLOSED in the previous step.																																					
2.7.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Identify Phase, Neutral, and Ground cables and connect them to the matching Camlock sockets on the PDC.																																					
2.8.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Tighten Camlock connections and make sure there is no issue with the locking mechanism or insulation (perform pull test).																																					
2.9.	ROMPXX	PDC x.xA/Bx Primary/Catcher	At the PDC side, using digital ohm meter, check for the continuity between multiple runs of each phase, neutral, and ground. <table border="1"><thead><tr><th></th><th>Phase A</th><th>Phase B</th><th>Phase C</th><th>Neutral</th><th>Ground</th></tr></thead><tbody><tr><th>Phase A</th><td></td><td></td><td></td><td></td><td></td></tr><tr><th>Phase B</th><td>X</td><td></td><td></td><td></td><td></td></tr><tr><th>Phase C</th><td>X</td><td>X</td><td></td><td></td><td></td></tr><tr><th>Neutral</th><td>X</td><td>X</td><td>X</td><td></td><td></td></tr><tr><th>Ground</th><td>X</td><td>X</td><td>X</td><td>X</td><td></td></tr></tbody></table>  Important: Similar phases and N-G shall read 0 Ω and others shall read open circuit.		Phase A	Phase B	Phase C	Neutral	Ground	Phase A						Phase B	X					Phase C	X	X				Neutral	X	X	X			Ground	X	X	X	X		
	Phase A	Phase B	Phase C	Neutral	Ground																																			
Phase A																																								
Phase B	X																																							
Phase C	X	X																																						
Neutral	X	X	X																																					
Ground	X	X	X	X																																				
2.10.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Confirm all above steps are completed, PIC to initial.																																					
2.11.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Remove LOTO from Q1/Q4 breaker and CLOSE .																																					
2.12.	ROMPXX	Rack xx.x	 Important: If there is any rack deployed in this aisle, ensure power is restored to the rack.																																					
2.13.	ROMPXX	PDC x.xA/Bx Primary/Catcher	On LB, turn on control power and confirm proper fan operation. Info: LB is now ready for testing operation.																																					
2.14.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Create duplicate section 2 for each PDC (primary and catcher side) that will be connected to the LB for testing.																																					
Summary: Connecting LB to <PDC x.xA/Bx> Complete.																																								

Procedure for Disconnecting LB from the PDC

Section 3: Disconnecting LBs to <PDC x.xA/Bx>				Completed by <vendor>
Step	Location	Equipment	Action/Task	Initials
3.1.	ROMPXX	LB	Turn off the LB connected to the associated PDC.	

Section 3: Disconnecting LBs to <PDC x.xA/Bx>				Completed by <vendor>
Step	Location	Equipment	Action/Task	Initials
3.2.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Verify zero Amps draw on PDC HMI screen. (If HMI is not available, confirm no load by measuring current on connected cables).	
3.3.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Wearing proper PPE, OPEN and LOTO Q1/Q4 breaker – Verify zero voltage reading on the HMI.	
3.4.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Verify all branch circuit breakers (MCBs) except the ones feeding live racks are OPEN .	
3.5.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Wearing proper PPE, remove Phase, Neutral, and Ground cables from PDC LB ports and secure the ends.	
3.6.	ROMPXX	PDC x.xA/Bx Primary/Catcher	At the PDC, reinstall all Camlock socket.	
3.7.	ROMPXX		 Confirm all above steps are completed, PIC to initial.	
3.8.	ROMPXX	PDC x.xA/Bx Primary/Catcher	 Remove LOTO from Q1/Q4 breaker and CLOSE .	
3.9.			 Confirm branch circuit breakers (MCBs) feeding live racks are still CLOSED .	
3.10.	ROMPXX	PDC x.xA/Bx Primary/Catcher	Create duplicate section 3 steps for each PDC (primary and catcher side) that will be disconnected from the LB.	
Summary: Disconnecting LB from <PDC x.xA/Bx> Complete.				

Comments

Following the procedure, use this section to identify any issues or additional information for future use.

Amazon Data Services, Inc.



Global Construction H&S Minimum Requirements
Version 2.1 – July 2025

Contents

1 Introduction.....	4
2 Purpose.....	4
3 Global Construction Health and Safety Minimum Requirements (GMR)s.....	5
3.1 Contractor Prequalification.....	5
3.2 Construction Safety Management Plan (CSMP)	5
3.3 CSMP Validity and Application	6
3.4 Project Legal Risk Register	6
3.5 Hazard Identification, Permit Programs & Uncommon Work Areas	7
3.6 Risk Management and Mitigation.....	8
3.7 Management of Change	14
3.8 Documentation Control.....	14
3.9 Subcontractor Management	15
3.10 Project H&S Committee Governance	15
3.11 Project Health & Safety Metric Targets and Reporting.....	15
3.12 Site-specific H&S Requirements in the Construction Zone(s)	16
3.13 High Risk Activities (HRA's).....	17
3.14 Violence and Weapons	17
3.15 Ready to Work / Fitness for Duty	18
3.16 Lone Worker Requirement	18
3.17 Incidents and Incident Reporting.....	18
3.18 Emergency Response Plan.....	20
3.19 Traffic Management Plan	21
3.20 Communications	22
3.21 Personal Protective Equipment.....	22
3.22 Training Needs Analysis.....	23
3.23 Critical Equipment Procurement Risk Assessment Process	23
3.24 Simultaneous Operations Management Plans (Inclusive of CZ and OZ).....	23
3.25 Planned Inspections	24
3.26 Mobile Technology.....	24
3.27 GC and Subcontractor Project Safety Officers	25
3.28 Recordkeeping	25
3.29 Stop Work Authority.....	25
3.30 Continuous Improvement Program.....	26

3.3.1	Discipline Policy	26
Appendix A: Health and Safety Metric Reporting		26
Appendix B: H&S Incident Reporting Form		28

**AMAZON DATA SERVICES, INC. GLOBAL CONSTRUCTION H&S MINIMUM
REQUIREMENTS**
Effective July 2025 Version 2.1

1 Introduction

Amazon or Landlord (“Owner”) expects anyone performing construction projects on its behalf to execute their Work in a safe and healthy manner. This document sets out the minimum H&S and reporting requirements that General Contractors (“GCs”, or “you” throughout this document) must comply with for Owner Projects.

Where a Project is to be constructed and delivered by (i) a GC engaged by Amazon Data Services, Inc. and its affiliates (“Amazon campus.”), then Amazon is Owner and all references in this document to ‘Owner’ should be read as such; (ii) a GC engaged by a Landlord, then Landlord is the ‘Owner’ and all references in this document to Owner should be read as such. The Landlord shall comply with and cause its General Contractors to comply with the minimum requirements in this document. Capitalized terms not defined below will have the meaning attributed to them in the Contract, ((i) where Amazon is Owner, in the “Contract” is the Global Construction Agreement between you and Amazon (GCA), For any builds where the GC is engaged by a Landlord, the Landlord will (i) provide any documentation referred to in this document to Amazon on request or, where otherwise noted to be provided to Amazon via Landlord, promptly on receipt; and (ii) where specifically stated in this document, obtain Amazon’s approval.

Safety is a critical component of construction work and a key contractual responsibility for you in managing and performing such work on construction sites. Your ability to establish up-front controls and safety programs to systematically identify, evaluate and eliminate or control anticipated hazards throughout the life of the project, is a critical aspect of health and safety compliance and overall project success. Reducing construction site injuries and fatalities to the greatest extent possible and ensuring compliance with all workplace safety standards is fundamental to building a culture of safety. The interactions developed through these safety programs will also bring benefits in the form of improved communication, documentation, and cost savings through efficiency on the Site ((i) where Amazon is the Owner, the “Site” is as defined in the GCA; (ii) where Landlord is the Owner, the “Site” is Premises as defined in the Lease).

2 Purpose

This Amazon Global Construction H&S Minimum Requirements (“GMRs”) document is neither a substitute for, nor a legal interpretation of, the health and safety regulations that apply to the work that you have been engaged by Owner to manage and/or perform on the Site. These minimum requirements do not in any way relieve your responsibility to ensure the safety of everyone on Sites under your control, including your employees, contract workers, subcontractors, suppliers and their respective personnel, along with all Amazon employees or appointed vendors (collectively, “Site Personnel”), and anyone else on the site including any visitors (collectively, “Visitors”), including but not limited to by controlling safe work practices and correcting health and safety hazards at the Site. These GMRs are a collection of minimum requirements where the intention is to ensure overall safety performance of any Work you do on behalf of the Owner ((i) where Amazon is the Owner, the “Work” is as defined in the GCA; (ii) where Landlord is the Owner, the Landlord’s “Work” is as defined in the Work Letter). You are required to evaluate these GMRs and any applicable regulatory requirements for the Work to be managed or performed on the Site and have robust safety programs in place to meet or exceed these minimum requirements and comply with applicable regulatory requirements. Where there is a difference between the local law and the minimum requirements in these GMRs, the stricter requirement(s)/requirement that affords the greater level of safety must be followed.

1. This document applies to all Work you are performing for Owner on any Site.
2. This document is not all-inclusive and is not intended to address all hazards at Site or within the workplace.
3. You are solely responsible for ensuring that all Site Personnel and Visitors on the Sites are provided with safe means and methods for the Work being performed.
4. Your adoption of the GMRs is not intended to be a substitute for your knowledge and specialized skills. You are required to develop the means and methods necessary to implement the minimum requirements

of the GMRs.

Your acceptance of these GMRs does not indicate that Amazon has taken any responsibility for your processes or the health or safety of your employees, nor does this document relieve you of your obligations to (1) control the means and methods by which you and Site Personnel perform Work at the Site, (2) independently ascertain what health and safety practices are appropriate and necessary for the performance of such Work, and (3) develop, implement, and enforce a comprehensive health and safety program appropriate for the Work performed that complies with all Amazon requirements, rules, regulations and industry standards, including permits, governing you and the Site. By requiring these minimum or core standards, Amazon is not warranting them or assuming any responsibility for the safety of Site Personnel and Visitors or creating any employment or co-employment relationship. The GC Construction Safety Management Plan (CSMP) that you are responsible for preparing applies to you and all Site Personnel performing Work on the Site.

Unless otherwise stated in the Contract, if Laws recognize the concept or role of principal contractor, prime contractor, controlling employer, construction supervisor, or other equivalent responsible for health and safety matters at a site (statutory or otherwise), General Contractor constitutes and is appointed this role for the Project and Work until Substantial Completion unless otherwise provided by Law.

3 Global Construction Health and Safety Minimum Requirements (GMR)s

3.1 Contractor Prequalification

3.1.1 For Amazon Builds (any Project to be constructed and delivered by a contractor engaged by Amazon Data Services, Inc. and its affiliates), you must review the GMRs and confirm conformance to be considered for selection. You agree to submit all qualification documentation to Amazon's prequalification process. If you are successful at tender, the GMRs will be incorporated into your Work Order with Amazon. For builds where the GC is engaged by a Landlord, you must review the GMRs and confirm conformance, the GMRs will be incorporated into your Contract with Landlord.

3.1.2 You agree to comply with the Owner's prequalification process. You must also provide the following prequalification information to the Owner. For builds where the GC is engaged by a Landlord, the below information shall be submitted to Amazon via Landlord:

- Safety performance data including, but not limited to:
 - Total Recordable Incident Rate (TRIR) statistics (previous 3 years, and annual reporting).
 - Any regulatory citations/orders issued by Authority Having Jurisdiction (AHJ) in the country where the Project is located
 - Occupational fatalities over the past 3 years
- Company safety manuals that demonstrate compliance with relevant local laws.
- Variance Request: In circumstances where the Owner identifies concerns regarding historical safety performance metrics through its prequalification compliance monitoring platform, the GC may be required to submit a formal variance request within Avetta. As part of that variance request, you may be required to incorporate a variance corrective action plan, which details how you plan on improving health and safety performance. Amazon reserves the right to request additional documentation or modifications to the variance corrective action plan prior to final determination of prequalification status. All variance request approvals shall be contingent upon strict adherence to the approved corrective action plan and demonstrated improvement in safety performance metrics.

3.2 Construction Safety Management Plan (CSMP)

3.2.1 The CSMP document is intended to ensure that all construction activities are conducted safely, efficiently and in accordance with these GMRs.

3.2.2 The GC shall compile a site specific CSMP to achieve safety and health standards that comply with local laws at a minimum. The GC must submit the CSMP to Amazon via the Procore Tool at a minimum of **4**

weeks prior to mobilizing the Site (for builds where the GC is engaged by a Landlord shall be submitted to Amazon via Landlord). This CSMP should define the appropriate controls to mitigate risks associated with the Work Order Scope and include the GC's:

- Health and Safety Management System (e.g., ISO45001, NSC Journey to Excellence, ANSIZ10, ILO-OSH, VPP Star, or OSHA Recommended Practices (2016 and 1989), or equivalent). Including evidence of certification if achieved.
- Health and safety procedures, processes and programs intended to manage H&S risks during construction activities.
- Processes intended to manage H&S risks during all construction activities.
- Physical location of the construction activities.
- Description of work.
- Key leadership and project personnel responsible for the application and enforcement of the CSMP.
- The company's health and safety Policy/Statements for the Site work activities – signed off by the leader of the organization within the previous 12-month period.

3.3 CSMP Validity and Application

3.3.1 The CSMP must be reviewed by the GC annually or when there are changes about work/design scope or other applicable changes that introduce new hazards that have not previously been identified on the project Legal Risk Register (see section 3.4). In the event of any changes, they shall be incorporated into the CSMP as an update and resubmitted to Amazon within 7 days via Procore (for builds where the GC is engaged by a Landlord, the CSMP shall be submitted to Amazon via Landlord). You must distribute updates of the CSMP to the key leadership and project personnel responsible for the application and enforcement of the CSMP, and the Project H&S Committee.

3.3.2 There are times when certain parts of the construction Site will be handed over to Amazon operations, and delineation between operations and construction should be established. The application of the CSMP is limited to the GC Scope of Work in the physical area identified by the GC as "construction zone." Where there is a need to delineate between the CZ/OZ, the planned boundaries will be reviewed and agreed to by representatives of Amazon Construction and Operational teams, which for builds where the GC is engaged by a Landlord shall be submitted to Amazon via Landlord. The agreed CZ/OZ boundaries shall be shared at the orientation of all Site Personnel, and any updates to the OZ or CZ boundaries, once agreed to by Amazon Construction and Operational POCs (via Landlord), shared with all Site Personnel during the Project.

3.4 Project Legal Risk Register

3.4.1 Regardless of construction project type, the GC and all subcontractors are required to operate in accordance with their contractual obligations and follow all local laws.

3.4.2 The GC must draft and retain a legal register of Regulations, Acts, Compliance Codes and other relevant standards applicable to the project and update when changes to local laws are identified during the course of the Project. At a minimum, this legal risk register will identify applicable legal, regulatory and Code compliance requirements. The GC must refer to the legal risk register to create the projects risk register, to facilitate risk identification, risk description, risk assessment and ranking, and risk response/mitigation strategies.

3.4.3 The Owner and, if the Project is a build where the GC is engaged by a Landlord, the Tenant Representative via Landlord, must be regularly updated by you on any changes to legislation that has the potential to impact health and safety the Project during the Project. The updates are to be communicated during:

- 1) Weekly/Monthly Progress meetings
- 2) H&S Committee Meetings

3.5 Hazard Identification, Permit Programs & Uncommon Work Areas

- 3.5.1 You must identify, assess and control all safety risks associated with each task or activity performed on the Site.
- 3.5.2 You shall ensure that you or any of your employees, contract workers, subcontractors, suppliers and their respective personnel conducts, prior to the work being undertaken, a documented pre task risk assessment (job hazard analysis (JHA), Safe Work Method Statement (SWMS), or any equivalent risk assessment process stipulated by the local laws.
- 3.5.3 The risk assessment should be based on the perceived risk of any work that has the potential to result in serious injuries, fatalities or damage including High Risk Activities.
- 3.5.4 You shall ensure that a pre-task plan (PTP) program with your necessary approvals is in place and implemented for each day's work activities. You shall conduct regular audits of the PTP program to:
- Certify that Site Personnel and their management teams are conducting PTPs for all tasks,
 - Ensure hazards associated with tasks are identified, mitigated, and communicated to relevant Site Personnel, and,
 - Ensure change analysis is performed, and documented on the PTP, whenever a change in facilities, equipment, materials, or processes occurs. The PTP program procedures and expectations shall be incorporated into your site-specific orientation.
- 3.5.5 You shall establish and administer a hazard communication (HazCom) plan to ensure compliance with all local hazardous waste and chemical regulations.
- 3.5.6 Your HazCom plan shall require the submittal of current safety data sheets (SDSs) for all chemicals prior to their arrival on site.
- 3.5.7 You shall have SDSs readily available for review upon request by all Site Personnel. The right to review SDSs and the procedure to do so shall be communicated to Site Personnel in your site-specific orientation training.
- 3.5.8 Some sites will contain "Uncommon Work Areas", which are restricted areas that are not accessible by all Site Personnel (for example, areas that have specific job safety hazards that require specialized training for access). Examples of uncommon work areas include where steel erection, roofing, excavation or trenching are taking place.
- 3.5.9 Uncommon Work Areas must have established policies and procedures for entry.
- 3.5.10 You must establish "Permit to Work Program" prior to any Site Personnel performing the following specific high-risk work activities:
- Hot work,
 - roof access,
 - excavation/ground penetration,
 - guardrail/fall protection removal,
 - confined space,
 - live electrical work, and critical lift
 - work at height
 - permit to load/unload temporary works
- 3.5.11 Your site-specific orientation training shall include what work activities require permits, and the requirements and processes of the Permit to Work Program itself.

3.6 Risk Management and Mitigation

Risk management requirements are designed to ensure safety, compliance, and project success. The GC CSMP must include a detailed risk management strategy to include training, emergency preparedness, communication protocols, compliance monitoring, to ensure consistent safety standards, identify and implement controls to

mitigate identified risks and promote continuous improvement. Specific risk controls for high-risk activities must incorporate detailed permit systems. The following minimum requirements are in addition to all applicable health and safety regulations and apply to all Site Personnel and visitors performing any of the following activities:

3.6.1.1 Temporary Works

Temporary works include the parts of the Project that are needed to enable the permanent works to be built. You agree to the following minimum requirements:

- Temporary works must be designed in accordance with recognized engineering principles and local law. This includes proprietary temporary works equipment. You must demonstrate you have in place effective arrangements for controlling risk arising from the use of temporary work (e.g. access scaffolds, excavation support, formwork etc.). This includes proprietary temporary works equipment.
- Temporary works (including support systems and temporary structures) must be:
- Designed by a qualified designer (professional engineer or competent person)
- Detailed on up-to-date drawings and plans
- Where changes are required to the design, the changes must be authorized by the designer prior to the change being undertaken.
- Designs must be documented in a drawing or plan, kept up to date if approved alterations are made, and communicated to all Site Personnel.
- Temporary works must be installed as per the design and/or installation instructions by a competent person(s).
- Methodology for the installation of temporary works must include controls for the Work to be completed safely. Risks associated with installation must be managed as per the relevant risk management procedures.
- Prior to use, temporary works must be inspected and verified by competent people and installed in accordance with the manufacturers' instructions, any drawings or plans, and relevant local laws.
- Prior to loading/unloading temporary works or associated plant and equipment, a permit must be issued by a competent person to the GC.
- Temporary works must be regularly inspected as per the designer's requirements, manufacturer's requirements, relevant legislation or Codes of Practice.
- Temporary works must also be inspected following any events that may affect its stability (e.g. weather events).
- Prior to removing or dismantling any temporary works, it must be verified by a professional engineer or competent person that the permanent works are constructed, are in a stable condition and self-supporting (i.e. all permanent load paths have been established) and that temporary work is no longer required. The following must also be considered:
- Risks associated with dismantling temporary works must be managed as per the relevant risk management procedures.

3.6.1.2 Fall Hazard Control – Working at Heights

You agree to the following minimum requirements:

- Implementation of a robust fall protection program that is strictly enforced with disciplinary action for first time offenders. Repeat offenders of your fall protection program must be permanently removed from the site.
- All Site Personnel working at height shall be protected from fall hazards by hard barricades/guardrail systems installed and maintained in accordance with all local laws. For EMEA, working at height practices should be compliant with all local laws and Institution of Occupational Safety and Health (IOSH) standards. For AMER, working at height practices should be compliant with all local laws and Occupational Safety and Health Administration (OSHA) standards. For APJC, working at height practices should be compliant with all local laws and IOSH/OSHA standards in the absence of local guidelines.
- When fall hazards cannot be eliminated by barricades/guardrail systems, PFAS connected to an anchor point capable of withstanding appropriate load rating must be used. Warning lines must be used on the roof and controlled decking zones with appropriate signage warning of the hazard, and must be

installed, inspected and maintained by a competent person in accordance with local laws/regional requirements. Fall protection requirements apply to all trades and tasks associated with construction.

- Implementation of a site and task specific fall rescue plan that meets the following minimum conditions:
- Any environmental and site-specific factors which may affect rescue and evacuation operations and how these factors are to be controlled.
- The roles and responsibilities of each team member in an emergency.
- The designated personnel for rescue and a means of contact must be identified.
- Potential rescue paths must be identified.
- The identification of emergency rescue equipment.
- The proper inspection and validation of rescue equipment.
- Site Personnel and equipment must be readily accessible and the primary means for rescue, although provisions may include assistance from outside agencies.
- Perimeter safety cable systems shall be installed by qualified persons and corners must have the appropriate length stiffeners, so that the system can be tightened to specification without deflection of the vertical corner post.
- Perimeter cables and other fall protection guardrail systems shall be frequently inspected.
- Edge protection or similar must be installed to prevent any chance of falling from heights.

3.6.1.3 Mobile Elevating Work Platforms (MEWPs)

You agree to meet the following minimum requirements:

- Documentation of daily pre-operation inspections of all MEWPs. You shall establish controls to ensure equipment inspections are completed and available upon request. Records related to such inspections shall be kept on file until the completion of the project or for one year; whichever is longer. If issues are encountered during the daily pre-operation inspection, the lift shall be tagged appropriately and removed from service until repaired.
- Operators must use appropriate PPE per the Original Equipment Manufacturers (OEM's) requirements. When required by the manufacturer or regulatory standards, you shall require that all operators of MEWPs use a full body safety harness with appropriate shock-absorbing feature (example self-retracting lanyard secured to the platform's manufacturer's designated anchor point)
- Operators must work while standing on the platform floor, never on the top rail, mid-rails, toe board, or any device or object that would raise the height of the operator or passenger above the platform floor.
- Pre-task plans for MEWP use shall include the pre-inspection of the travel path of the lift for ditches, drop-offs or holes, bumps and obstructions (including and cords crossing the path), debris, untampered earth fills, overhead obstructions, electrical wires, bus ducts, charged utility lines, overhead fire suppression lines, conveyors, underground vaults/structures/duct banks, and other hazards.
- Spotters are required while moving or operating MEWPs and/or other plant and equipment in high-risk environments.
- Barricade(s), with signage, must be used around elevated MEWPs while work is performed in common work areas.
- MEWP operators, spotters and ground personnel shall be properly trained in the safe operations of the MEWP's ground controls.
- You must ensure you have a rescue plan in place should any need arise for Site Personnel to be rescued from elevated equipment.
- Properly identify trapping as a potential hazard (the potential to be caught between an object and the basket or controls of the MEWP). MEWPs used on the Project must have safety features to prevent accidental contact with other plant, equipment or structures. You are responsible for ensuring that all personnel are sufficiently trained in the use of these safety features. It is recommended to use the lowest speed for the boom lift motion or operation and to install and utilize internal handholds.
- Materials should never be suspended from or attached to the lift unless using rated OEM devices in accordance with manufacturer recommendations.

3.6.1.4 Confined Spaces

You agree to the following minimum requirements for Work activities that trigger confined space regulations:

- For activities that meet the local regulatory threshold for "confined space", you shall verify the following conditions. In EMEA, confined space practices should be compliant with all local laws and IOSH standards. For AMER, confined space practices should be compliant with all local laws and OSHA standards. For APJC, working at height practices should be compliant with all local laws and IOSH/OSHA standards in the absence of local guidelines. In addition:
- GC is responsible for authorizing the entry of all confined spaces for all vendors up until the handover of the Project to Amazon.
- The internal atmosphere should be tested, with a calibrated direct-reading instrument, for oxygen content, for flammable gases and vapors, and for potential toxic air contaminants prior to entry and at regular intervals in accordance with local laws and standards referred to above.
- All confined space entry and rescue equipment is on Site, properly maintained, and readily accessible.
- All personnel entering the permit-required confined space have been properly trained and achieved the necessary competency as the local jurisdiction requires and have participated in the task's pre-task planning and task-based risk assessment review process.
- Coordination of adjacent site activities that could impact the permit-required confined space work, for example, hot work.

3.6.1.5 Electrical Safety

You agree to the following minimum requirements for work activities that trigger electrical safety regulations:

- Develop an energized work Permit Program with the minimum requirements including equipment and conductor de-energization occurring under (LOTO) controls.
- You shall comply with the requirements of the Amazon Data Center Construction Lock Out Tag Out Minimum Requirements document. Please note that this document applies to all hazardous energy sources, including electrical energy.
- LOTO must be undertaken only by an Authorized Person (authorized by the GC) with reasonable experience and all necessary qualifications.
- Where a deviation from the GCs LOTO process is required, the GC must create an amended task specific process that ensures the safety of all Site Personnel. All workers involved or impacted by the activity must be trained in the amended process.
- Adhere to Amazon Foreign Object Detection process/mechanism for newly installed/startup of electrical equipment.
- All electrical cords must not present a tripping hazard and must be secured in main walkways when they cannot be strung overhead.
- All temporary electrical installations must comply with all relevant Amazon and LOTO processes.
- Substations, electrical closets, and temporary electrical panels, shall be labeled, closed, and always locked once energized (i.e., once power is switched on in the room).
- Temporary portable electrical switchboards should not obstruct an exit, be accessible for emergency shutdown; and protected from damage from objects, equipment, or people that are near or must pass near them.
- Provisions are made such that first time breaker rack-in is not done manually and will be done using remote breaker rack-in device.

3.6.1.6 Roof Access / Work

You agree to the following minimum requirements:

- Before completion of the roof area and installation of edge protection the roof area shall be designated an Uncommon Work Area, and access shall be restricted to only authorized personnel through the implementation of a strictly enforced roof access permit program that addresses all appropriate safety measures necessary and specific site conditions. The requirements of the roof access Permit Program shall be included in your site-specific safety induction.
- Wherever practical and required, warning lines shall be set up in accordance with regulatory standards for the work taking place, have appropriate signage to warn of fall hazards at roof edges, and shall be

- inspected frequently regardless of any work activities taking place on the roof.
- Rope grabs are not permissible for fall protection use outside of the warning line. Anyone working outside of the roof warning line system shall use a leading edge self-retracting PFAS.
- Roof monitors are not permissible in lieu of fall protection.
- No employees shall be allowed to access or work on the roof alone or unsupervised.

3.6.1.7 Protection of the Public

You agree to the following minimum requirements for work activities that could affect the public:

- Adhere to the local laws.
- Ensure effective measures have been taken to prevent injuries and incident(s) involving the general public.
- Adequate maintenance of traffic (MOT) provisions shall be installed by a qualified person to safely redirect the flow of vehicles and pedestrians and protect them from construction activity when roadways and walkways are encroached by construction activities.
- The construction Site must be secured to prevent unauthorized access.
- Potentially hazardous conditions should be mitigated at the end of each shift and/or at any times the Site is left unattended. For example, equipment should be locked with keys removed, excavations and trenches should be barricaded, fuel storage containers should be locked, access to buildings should be limited if possible, and materials should be secured.

3.6.1.8 Hot Work / Fire protection

You agree to the following minimum requirements for Work activities involving welding or hot work:

- Hot Work permit process that addresses all regulatory requirements.
- Competent personnel to perform the activity.
- Provision of fire protection at the source for hot work.
- Specific site conditions and shall be included in your site-specific safety orientation.
- Temporary membrane construction enclosures and partitions, which are susceptible to burning, shall be protected from open flame source and shall be made of fire-retardant material(s).
- Fire management is to be regularly monitored and maintained with evidence of every inspection and maintenance maintained by the GC.

3.6.1.9 Lifting and Rigging

You agree to the following minimum requirements for Work activities involving lifting and/or rigging:

- A Lift Plan for all lifting and rigging activities.
- Nature, timing, location and characteristics of the rigging and lifting task.
- Personnel training records review.
- Necessary personnel, load weights, equipment, capacity charts and rigging hardware for the task.
- Determining the optimum rigging configuration to properly support and maneuver the load into place.
- In EMEA, anchorage points for lift equipment should be installed in accordance with all local laws and IOSH standards. In AMER, anchorage points for lift equipment should be installed in accordance with all local laws and OSHA standards. In APJC, anchorage points for lift equipment should be installed in accordance with all local laws and IOSH/OSHA standards in the absence of local guidelines.
- Identification of adjacent site activities that could be affected by the lift.
- The plan must be communicated in a manner that ensures the commitment and understanding of all individuals involved or who can be affected by the lift.
- Established communication method(s) between signal person and crane operator.

- Barricading the lift area to ensure unexpected traffic is aware of the lift in progress.
- Damaged rigging that has been tagged out of service shall not be accessible for use or left on the Site longer than 24 hours. For large pieces of rigging equipment that require transportation arrangements, you may grant exceptions to this requirement on a case-by-case basis.
- Develop a written communication plan between signal people and crane operators when multiple cranes are working on site and ensure this is communicated to them.
- Barricading with signage shall be in place to prevent unauthorized personnel within the swing radius of the crane and exposure to overhead suspended loads.
- Ensure safe clearance distance when lifting near or over unprotected equipment or services.
- Steel Erection
- You agree to the following minimum requirements for work involving the erection of steel:
- Confirm, through third-party inspection reports, that site conditions (e.g., soil compaction for crane paths, foundation concrete strength, etc.) meet regulatory standards and local laws to begin steel erection, and you shall provide the steel erector with the required written notifications to commence steel erection work. These written notifications and third-party inspection reports shall be made available to Owner upon request.
- The outer boundary swing radius of the crane used for steel erection shall be barricaded, with signage, and unauthorized personnel must receive written permission from the steel erector prior to accessing this area. The procedures for safely accessing this restricted, uncommon work area shall be reviewed with all Site Personnel in your site-specific safety orientation. You must ensure that the steel erection Subcontractor keeps unauthorized personnel out of this area by frequent monitoring of the area and hazard mitigation controls, and through administration of disciplinary action against those individuals who ignore proper signage and barricading of restricted areas.
- Ensure that the steel erection Subcontractor conducts daily per-use rigging inspections.
- If a perimeter safety cable system is used, it shall be installed by qualified workers and corners must have the appropriate length stiffeners, so that the system can be tightened to specification without deflection of the vertical corner post.
- Implement a permit program for lifts conducted on the site.

3.6.1.10 Ground Penetration, Excavation and Trenching

You agree to the following minimum requirements for work activities involving excavation and trenching work:

- Ensure that all excavation and trenching work, including inspections, performed on the site is done in accordance with regulatory standards and local laws.
- Where excavation takes place inside or adjacent to any Amazon OZ, an excavation permit must be obtained and sufficient notice of the planned works must be given to Amazon Operations teams (for builds where the GC is engaged by a Landlord, via Landlord) in accordance with Amazon Operations excavation requirements. These requirements may vary between regions, so you should determine whether there are any additional requirements and that all necessary permits are obtained and notification submitted to Amazon Operations team (for builds where the GC is engaged by a Landlord, via Landlord) prior to the commencement of the excavation work (for example, in AMER any work using heavy equipment requires a separate “Operational Zone Heavy Machinery Permit” to be submitted to Amazon in addition to an Excavation Permit).
- In addition to any permits required in accordance with the above section, you are required to establish your own ground penetration, excavation and trenching permit program. At a minimum, this permit program must identify the excavation competent person, scope of work details, utility locations if any and means of identification, means of protective systems for employees entering the excavation, confined space considerations, soil management plan, non-destructive excavation within 3m of a live service and surrounding site activities that could impact workers’ safety.
- Excavation Subcontractors should submit new permits for each new area of excavation that was not covered under a previously submitted excavation permit.
- Dust should be kept to a minimum and tracked materials onto roadways shall be frequently cleaned to minimize site impacts on the personnel on site and/or members of the public.
- Ensure excavation area is safely barricaded, and signage is posted, gate access to be removed once

- there is no ongoing risk associated with the excavation.
- Provide delineation of the haul roads required along exit route of muck away trailers.

3.7 Management of Change

- 3.7.1 You must develop a management of change process that helps ensure any change to the Work, Site or Project are risk assessed, and changes incorporated into the project risk register. Management of change process should include proactive communication to Owner of any relevant changes. The process must clearly identify and assess risks associated with changes, and steps to implement controls and corrective and preventative actions to mitigate those risks.

3.8 Documentation Control

- 3.8.1 Your CSMP must be reviewed and updated by you in line with the principle of continuous improvement (as set out in Section 3.30). This means updating the CSMP in accordance with your experience on the project, changes in practices and technology, relevant local laws, and posting any findings identified in lessons learned from significant incidents.
- 3.8.2 When you change or update your H&S procedures, you should update the CSMP to reflect those changes. Substantive changes to your CSMP should be communicated to the Amazon POC for the Project. You are required to share, at Amazon's request, any procedures that may impact on your management of H&S risks.
- 3.8.3 You must provide copies of this CSMP to Amazon, through an electronic medium that Amazon specifies.
- 3.8.4 You must distribute updates of the CSMP with the key leadership and project personnel responsible for the application and enforcement of the CSMP, and the Project H&S Committee.

3.9 Subcontractor Management

- 3.9.1 You shall have a Subcontractor Safety Management Program in place on the Project that defines the qualification, outlines the policies, processes, instruction, and documentation for subcontractors. This will serve to establish the management of safety and understanding for all tiers of Subcontractors involved on the Project, overseen by the GC.

3.10 Project H&S Committee Governance

- 3.10.1 You are encouraged to establish a health and safety committee for the duration of the Project. All local laws must be followed when establishing a Project H&S Committee. If no requirements under local laws exist, the following can be used as a guideline:

The Project H&S Committee make up should be a 2:1 workforce to management ratio from any company or trade.

The Project H&S Committee should be responsible for the following activities, at a minimum:

- Participation in pre-task planning and risk assessment development.
- Consultation on proposed changes to the CSMP.

Consultation on findings of workplace inspections by GC Owner/Owner Representative or subcontractors.

- Providing input to incident investigations, including the review of containment and corrective actions.
- Participation in monthly H&S management meetings.
- Other duties as agreed by contract with companies or labor unions.
- Committees must be structured for participation by all trades and companies at the Site.
- Rules regarding meeting governance shall be agreed to by the Project H&S Committee and should be communicated to all Site Personnel and made visible on Site.

3.11 Project Health & Safety Metric Targets and Reporting

- 3.11.1 The GC must submit the information contained in Appendix A, within Procore (, monthly (at a minimum) by the third working day of every month (for builds where the GC is engaged by a Landlord shall be submitted to Amazon via Landlord). The information to be submitted must include the updated statistics from the GC and all Subcontractor activities during the previous month. This is inclusive of, but not limited to:
1. Incident Reports
 2. Observations including unsafe Acts, Unsafe Conditions, Positive Safety Observations. (and closed within 7 days)
- 3.11.2 The GC is responsible for measuring and reporting all metrics of all work activities they control (directly or indirectly).

3.12 Site-specific H&S Requirements in the Construction Zone(s)

- 3.12.1 The Lifesaving Rules Minimum Requirements (Lifesaving Rules) identified in this section are minimum requirements focused on mitigating the high probability and high severity risks that commonly exist on construction sites.
- 3.12.2 You must ensure that the Lifesaving Rules are adequately incorporated into the CSMP, and that adequate safety systems and processes are in place to ensure that the Lifesaving Rules are complied with. The Lifesaving Rules are minimum requirements and are not intended to capture all safety requirements associated with these activities, but do capture key safety critical aspects. GC, and if the Project is a build where the GC is engaged by a Landlord, also Landlord, are ultimately responsible for ensuring compliance with the local laws.
- 3.12.3 You must ensure that your Discipline policy (stated in section 3.31 of this document) addresses the management of noncompliance(s) with these Lifesaving Rules.
- 3.12.4 The Lifesaving Rules are as follows:
1. **Permit to Work (PTW)** - Adhere to a permit to work process where required including but not limited to hot works, confined space entry, excavation, work on energy systems, work on height. Never do a job task, operate, or interfere with equipment without appropriate PTW.
 2. **Confined Space** - Only enter a confined space if a confined space permit is in place, you are trained and authorized by a competent person to do so, there is a spotter in place, and a rescue plan and equipment are in place and gas test has been carried out.
 3. **Driving** - Operate vehicles safely, respect the speed limit, wear the seatbelt, follow the site journey management plan, and do not use any handheld mobile devices.
 4. **Energy Isolation** - Only work on energy systems if a permit to work on energy systems is in place and you are trained and authorized by a competent person to do so, and a rescue plan and equipment are in place. Confirm all energy sources are isolated, locked out and tagged out before work starts and follow the isolation process in place.
 5. **Hot Work** - Only commence with hot work if a hot work permit is in place and you are trained and authorized by a competent person to do so, a fire watcher deployed, all combustible materials are removed, rescue plan and equipment are in place and the atmosphere is tested and monitored for flammable/combustible gas presence as appropriate.
 6. **Exclusion Zones** - Never walk under a suspended load and keep yourself at a safe distance from plant/machinery radius of movement/catchment area. Ensure clear and segregated access routes to works

areas are always in place. When working on height make sure all items are fixed and/or away from the edges to prevent anything from falling.

7. **Lifting and Rigging** - Always follow a documented lift plan and only perform lifting and rigging operations with certified equipment for which you are trained and certified and a rescue plan is in place.
8. **Safety Controls** - Never bypass, override or disable safety controls/devices.
9. **Work at Height** - Only work at height if a work at height permit is in place and you are trained and authorized by a competent person to do so, and a rescue plan and equipment are in place. Keep yourself protected against a fall by wearing inspected fall protection equipment and tie only on the approved anchor points. Make sure scaffolding and other equipment/plant for working at height are inspected by competent person prior to use.
10. **Excavations and Trenches** - Only enter an excavation or trench if excavation/trench permit is in place and you are trained and authorized by a competent person to do so, and a rescue plan and equipment are in place. Make sure the trench or excavation has been inspected and secured before entering through safe access and egress points. Report any changing conditions to the competent person.

3.13 High Risk Activities (HRA's)

3.13.1 A High-Risk Activity (HRA) is any work task or operation that presents a significant potential for serious injury, fatality, or catastrophic equipment/property damage. The GC is responsible for identifying, classifying, and controlling all HRAs on their projects, which at a minimum should include:

- First time energization of electrical equipment.
- Connection and tie-in of de-energized to energized busduct sections.
- Work at height where exposure to a leading edge is present without the benefit of guardrails. This includes work where restraining lines are used in lieu of guardrails.
- Excavations >12' in depth.
- Confined space entry.
- Crane lifting activities.
- Work with chemicals requiring special protection and controls. The requirements should point to notifying the GC by submitting the Safety Data Sheet (SDS) prior to bringing new chemicals onsite, reviewing the SDS with any trade partners working with or near the chemical.

3.13.2 For all identified HRAs, the GC shall:

- Develop and implement appropriate risk assessment and control measures according to their safety management system and prior to commencing work, ensure all workers and field supervision involved in the task perform a complete field walk of the work area, including reviewing each work step for the day, prior to signing the daily pre-work forms
- Communicate all HRAs with AWS Construction Managers within 2 days of planned work
- Ensure adequate supervision and oversight during the completion of the work
- Verify worker qualification and training
- Establish emergency response procedures and related equipment requirements
- Document all safety planning and controls
- Maintain records of all safety-related activities

3.14 Violence and Weapons

3.14.1 It is a requirement that you maintain a Site free from the following: physical or verbal acts toward another person, the result of which causes that person to reasonably fear for their safety or the safety of others; physical or verbal acts that threatens bodily harm to another person or damage to the property of another, and physical act, whether or not it causes actual body harm to another person or damage to the property of another.

- 3.14.2** You shall develop a policy that holds workers accountable for aggressive behavior, and prohibits all workers from bringing firearms, ammunition, and other weapons to the Site, or associated parking lot.
- 3.14.3** GCs violence and weapons policy shall not supersede any legal right to carry a weapon. Knives and power activated tools (with valid user licenses) may be used where permitted by local laws and required for legitimate construction purposes and captured in the method statement and risk assessment matrix for the task.

3.15 Ready to Work / Fitness for Duty

- 3.15.1** You must develop a process which gives you reasonable assurance that individuals working at the Site are physically fit to safely perform their assigned duties without risk or harm to themselves or others.
- 3.15.2** This policy or process requires an individual to be in a physical, mental and emotional state to enable the individual(s) to perform their assigned duties and must also ensure that all individual(s) are physically able to carry out the tasks that are assigned to them.

3.16 Lone Worker Requirement

- 3.16.1** GCs must ensure that you have effective mechanisms in place for controlling risks arising from lone work/workers conducting work alone. This includes conducting risk assessments for any tasks that involve lone work, implementing control measures to eliminate or mitigate identified risks, as well as compliance with any local laws that relate to lone working.

3.17 Incidents and Incident Reporting

- 3.17.1** You are responsible for investigating incidents, injuries and events (collectively, “Incidents”) that occur on any Sites, including those involving Site Personnel, Visitors or members of public, and reporting to Owner and, for any builds where the GC is engaged by a Landlord, to Amazon, in accordance with this section. For any builds where the GC is engaged by a Landlord, Landlord shall ensure that GC complies with these minimum requirements and shall report all non-compliances to Amazon promptly using Procure.
- 3.17.2** You must ensure that you and any of your subcontractors comply with any incident reporting requirements required to comply with local laws and any that are imposed by Amazon throughout your involvement in the Project. In the case of subcontractors, this includes establishing a formal contractual obligation for subcontractors to comply with incident reporting requirements under local laws. On builds where the GC is engaged by a Landlord, Landlord must ensure that GC complies with these minimum requirements. Landlord must comply with and provide Amazon with all information received under this section, promptly upon receipt in accordance with this section and the Work Letter.
- 3.17.3** The following “Incidents” trigger the reporting and investigation obligations set out in this section:
- Fatal Incident: The death of any person that is the result of a work-related incident or exposure.
 - Lost Time Injury (LTI): A work related injury or illness which results in the injured/ill person being absent from work for one day or more (consecutive or not). The first day of absence counted is the day following the day of the incident irrespective of next working day.
 - Medical Treatment Case (MTC): A work related injury/illness that is not severe enough to be reported as disability, a fatality or lost time incidents but requires treatment by medically qualified personnel (e.g. in a hospital) and which cannot be treated by just administering first aid. A medical treatment case does not necessitate time off work beyond the date of the injury to be classified as such.
 - Restricted Work Injury (RWI): A work related injury/illness which results in persons being able to report for work but who have been advised by a medical professional that they are unable to perform their normal duties because of their restrictions.

- Transfer Related Injury: A work related injury/illness which results in an injured employee to a job other than his or her regular job for all or part of the workday. Number of days that an employee was unable to perform their usual job duties. Resulting in persons being transferred to job that is different from the normal job function or duties than their original assignment, due to their injury
- First Aid Injuries: First aid injuries refer to medical attention that is usually administered immediately after the injury occurs and at the location where it occurred. It often consists of a one-time, short-term treatment and requires little technology or training to administer.
- Near Miss: Near Miss incidents include events that could have resulted in a serious injury or work stoppage and include the following:
 - Any regulatory citations and correspondence related to the site under construction.
 - Property damage events where no people are injured.
 - All emergency events where emergency medical services are dispatched.
 - Security issues – theft, threats (verbal, written, or symbolic), or acts of violence – including the discovery of any racially-charged or discriminatory graffiti or symbol.
 - A substance escaping, spilling, or leaking impacting the environment.
 - Any other near miss or ‘dangerous occurrence’ which does not result in a reportable injury, but which clearly could have done.

3.17.4 For the purpose of reporting H&S metrics set out in Appendix A, Recordable Incidents are classified as any work-related injury resulting in one or more of the following outcomes:

- Fatal Incident
- Medical Treatment Case
- Lost Time Injury Incident
- Restricted Work Injury Case
- Transfer Related Injury Case

3.17.5 You must investigate all Incidents that occur on a Site under GC control, and report to Owner in accordance with the 2-24-2 timeline in this Section.

3.17.6 **Notification and Containment:** Within **2 hours** of Incident occurrence, you must verbally inform the Owner of the event and the containment actions performed.

3.17.7 **Case Management and Initial Report:** Within **24 hours** of the Incident occurring, you must compile a report containing the following information, and submit it to the Owner via Procore. If no system has been determined, the form at Appendix B must be used and submitted via email. At a minimum, within this report you must:

- Confirm all known incident facts at the time of the communication and.
- Determine any corrective and preventative actions put in place to ensure that further incidents do not occur.
- Establish an initial determination of whether there is a residual risk of a similar incident occurring and setting out where else the underlying hazard(s) likely exists.
- Set out any Required Interim Actions (RIA) necessary to ensure there will be no additional harm caused by the underlying hazard(s) suspected to have caused the Incident and timelines for implementation.

3.17.8 Where an Incident results in an injury to a person, you will use your best effort to provide regular updates to Owner on that person’s status (at a minimum, every 24 hours) and any medical treatment that is being provided to the injured person.

3.17.9 **Investigation Report/Case Closure and Lessons Learned:** Within **2 weeks**, you are responsible for

investigating all Incidents and setting out those findings within a written incident investigation. You must conduct your incident investigation and submit the report findings using an industry-recognized method (such as the 5-Why or Taproot method, ICAM or similar) via Procore.

- 3.17.10 You must provide Owner with a detailed investigation report by submitting it into the Owner designated system. The detailed investigation report should include the following as a minimum:
- a) Description, date and time of incident
 - b) the task being performed
 - c) consequence (incident, injury, theft, vandalism, equipment damage)
 - d) severity (first aid, recordable injury, lost time injury)
 - e) days lost
 - f) category of hazard (thermal, mechanical, noise, vibration, etc.)
 - g) Description of incident
 - h) identified contributing factors (Absent or Failed Defences, Individual or Team Actions, Task or Environmental Conditions, and Organizational Factors)
 - i) The final root cause of the event
 - j) corrective actions which, to the satisfaction of Owner, address the identified root cause, the associated contributing factors and either eliminate or reduce the risk of reoccurrence to as low as reasonably practicable. .
- 3.17.11 As soon as reasonably practicable, you must provide information to Owner about any communications you or Site Personnel receive from any AHJ relating to the Incident, advance notice of any upcoming regulatory or emergency service inspections, meetings or filings, and a copy of any proposed submissions or communications, as well as all correspondence submitted to any AHJ.
- 3.17.12 At Owner's request, at any time following the Incident and during the incident investigation you must provide updates on the investigation within Procore in a timely fashion (for builds where the GC is engaged by a Landlord shall be submitted to Amazon via Landlord). This may include your preliminary findings, the medical status of any injured persons and draft investigation reports.
- 3.17.13 All corrective actions must be implemented to sufficiently control the risk which led to the Incident. Any corrective actions relevant to Site Personnel should be communicated to those personnel and any other personnel on Amazon Sites or Projects which the GC is currently operating or managing. Following the identification of corrective actions, you are responsible for establishing a process that ensures corrective actions are tracked and closed out according to established deadlines or escalated to the Owner if not closed within the time identified in the investigation report.
- 3.17.14 The GC must conduct a risk assessment to determine residual risk following implementation of corrective actions and ensure the corrective actions that have sufficiently reduced severity and probability for reoccurrence.
- 3.17.15 Owner or, if a build where the GC is engaged by a Landlord, Amazon, may choose to conduct its own investigation into an Incident. Where Owner or (where applicable) Amazon conducts its own investigation GC will, upon reasonable notice, provide Owner and (where applicable) Amazon with access to any documents, witnesses, and other information related to that incident as requested by Owner and (where applicable) Amazon to facilitate its investigation.

3.18 Emergency Response Plan

- 3.18.1 You must develop a documented Emergency Response Plan (ERP) for any disaster that may occur on the site once identified and include within the Construction Safety Management Plan (CSMP). Your ERP must meet local laws and cover all Project-related risks, including but not limited to the following:
- o Emergency evacuation (communication plan, egress routes, muster points, utility POC(s), emergency

services, etc.);

- Shelter locations (tornado, lightning or other severe weather or geological events).
- Nuclear emergency response plan (if applicable; response methodology: incident command, shelter in place/evacuation, 24/7 communication, and re-entry evaluations).
- Emergency medical services (site-level medical response capabilities and roles, communication plan, emergency vehicle ingress routes, hospital(s) locations, helicopter landing zones).
- Workplace violence (communication plan, escape routes, shelter options).
- Wildfires (if applicable).

3.18.2 At a minimum of 4 weeks prior to mobilization at the Site, you must provide Owner with an ERP covering all aspects of the Project as part of the CSMP.

3.18.3 You must ensure that all Site Personnel on the Site are trained on the ERP and that periodic audits/drills/testing are conducted to gauge efficiency and identify opportunities for improvement, which should then be incorporated into the ERP.

3.18.4 You must ensure ERP training is given to new hires, visitors or members of the public, subcontractors, Site Personnel and that retraining takes place in the event of any ERP changes.

3.18.5 This ERP must be evaluated prior to the start of construction and monthly thereafter. This evaluation shall be completed by the GC, all construction trade representatives, and associated safety committees.

3.18.6 You must ensure the ERP is posted visibly in common areas throughout the Project, along with emergency activation numbers, name of the incident commander, muster points, egress routes, first aid station locations, local emergency phone numbers, project map, and maps of two emergency services such as hospital occupational, clinic, or emergency shelter, and post-incident testing facilities.

3.19 Traffic Management Plan

3.19.1 You must develop a site-specific traffic management plan (TMP). The TMP must document the controls required to mitigate the risks of traffic impacting the worksite as well as the worksite impacting the public. Your TMP must meet local laws and include at minimum the following :

- Date, revision date and site location
- Physical security arrangements
- Speed limits
- Speed limit controls
- Pedestrian/cyclist safety measures
- Employee parking
- Contractor parking
- Visitor parking
- Overflow car parking measures
- Bicycle shelter/storage
- Courier and/or delivery pick-up and drop off points
- Other traffic management controls
- Safety equipment arrangements
- Large vehicle deliveries
- Indoor traffic considerations
- Emergency response arrangements
- Temporary change of design
- Safe walkways for access and egress to all work areas must be established and maintained to ensure delineation between people and vehicles.
- Adequate maintenance of traffic devices shall be installed by a qualified person to safely redirect the flow of vehicles and pedestrians and protect them from construction activity when roadways and walkways are encroached by construction activities. Traffic control devices shall be used in

conformance with the local regulatory requirements.

3.20 Communications

3.20.1 Orientation

- You shall develop a Site-specific safety induction and provide orientation training for all Site Personnel
- The orientation training must be made available to accommodate different languages of workers if needed, including interpreter services.
- You shall maintain records of all Site Personnel that have completed orientation training and provide a unique identification system for each participant that has satisfactorily completed the orientation training.
- Orientation training shall address all components identified in the GC's CSMP, and the orientation training requirements included in this document.

3.20.2 Plan of the Day Meetings

- You shall conduct a daily meeting, in coordination with all actively working subcontractor's management representatives, to review the day's activities and associated hazards (potential and any actual).
- Subcontractor management representatives shall have an opportunity to speak openly about their own activities and hazards and ask questions of other management representatives.

3.20.3 Project Safety Communication

- You are required to implement a policy for ongoing safety communication during the Project. It is essential to keep safety in the forefront by communicating the importance of safety on a regular basis.
- At a minimum, you shall establish regularly scheduled site meetings with subcontractor management, at which site-specific safety must be discussed openly between you and the subcontractor management teams.
- Communication must be reinstated and commenced following any planned work stoppage such as incidents, holidays, weather-induced stoppages etc.
- It is essential that safety-related information discussed at these meetings be disseminated to the employees performing the work, and this should be accomplished through project safety boards, daily pre-task plan meetings, safety huddles, toolbox talks, and/or other such initiatives.

3.20.4 Amazon Correspondence

- You shall review ongoing Amazon-initiated communications within 24 hours of receipt or next business day. This could primarily be issued via email, memos or through any electronic medium that Amazon uses to collaborate with the GC for managing the project, or sharing of lessons learned when immediate dangers to life and health (IDLH) occur. For builds where the GC is engaged by a Landlord, Landlord will provide any Amazon-initiated communications to GC.
- Documentation of completion may be required by Amazon and shall be provided to the Owner or, if a build where the GC is engaged by a Landlord, to the Tenant Representative upon request.

3.21 Personal Protective Equipment

- 3.21.1 You are required to ensure the appropriate personal protective equipment (PPE) is used on the Site, and that equipment meets the local regulatory requirements for PPE design and use. Examples of personal protective equipment that may be required, includes but isn't limited to the following:
- Protective footwear;
 - Safety helmet (Specific to GCs in operating in the Americas region- GCs must adopt Type 1 Class E-

- rated Mountaineering-style Safety Helmet that meets (1) ANSI z89.1 and/or EN 397 Certification, (2) Rated Type 1 and (3) Rated Class E, (4) Equipped with Manufacturers-installed 'chinstrap');
- Anti-impact eye protection;
- Gloves – based on risks (chemical, cut resistant, heat, etc.);
- Hearing protection;
- Reflective clothing (vests);
- Fall protection, prevention, rescue equipment for working at heights;
- Respiratory protection based upon identified respiratory risks for task;
- Sun-protective clothing (AS 4399:2020 or any equivalent approved standard).
- Suitably rated electrical PPE and clothing.

3.22 Training Needs Analysis

- 3.22.1** You are required to ensure that all Site Personnel have received general and specialized training required for the work they are assigned.
- 3.22.2** You must identify the relevant H&S training needs/requirements for the Project. The training must include at a minimum:
- Specific H&S training requirements by job classification before Work commences.
 - Provision of all activity-based H&S training required by local laws or activity based risk assessment for Site Personnel prior to the performance of that type of Work on Site.
 - A means of both documenting completed training (i.e. a training record or register) as well as visually identifying which training has been completed by individuals, (i.e. visual indicators like (i.e., badge, stickers, training cards etc.)

3.23 Critical Equipment Procurement Risk Assessment Process

- 3.23.1** You must have a process to ensure that the appropriate risks have been identified, considered and mitigated when procuring critical equipment or services.
- 3.23.2** This should also include equipment or services that are purchased by your subcontractors and vendors.

3.24 Simultaneous Operations Management Plans (Inclusive of CZ and OZ)

- 3.24.1** Where the Project involves the conduct of construction activity inside of or adjacent to an Amazon OZ, you must prepare a Simultaneous Operations (Sim-Ops) Management Plan and adhere to any processes or reasonable directions established by Amazon Operations teams (for example, Amazon's MCM process) while working in or adjacent to the OZ.
- 3.24.2** Prior to the preparation of a Sim-Ops Management Plan, you must complete a site walk with Amazon Construction and Operational POCs and Landlord (where applicable). The Sim-Ops Management Plan must contain at a minimum:
- Definition of the scope of activities and duration of the work to be conducted.
 - Building plans/construction drawings which clearly establish the demarcation/boundaries between the area where the construction activities will be carried out and the OZ.
 - Project-specific H&S requirements in the CZ (including, but not limited to PPE, permit to work requirements, noise monitoring & dust exposure control and chemical exposure, if any);
 - Project sequencing and logistics coordination mechanisms (including but not limited to any anticipated impacts to live and redundant services which will require energized electrical work to be performed or will require lockout tagout (LOTO) controls to be applied).
 - Details of how hazards that exist in the CZ and those that might impact the OZ will be identified and controlled.
 - Where there has been agreement with Amazon Operations regarding any safety responsibilities, rules or processes, these should be incorporated into the Sim-Ops plan.

- 3.24.3 An ERP that addresses, at a minimum, any anticipated blockage of Emergency Exits, Evacuation paths or Emergency Equipment and provision of adequate signs to prevent any accidental crossover of personnel from the OZ to the CZ or vice versa.
- 3.24.4 The Sim-Ops Management Plan must be reviewed and approval obtained jointly by the Amazon Project POC and any designated Amazon Operations POC, Landlord (where applicable) prior to the start of work.
- 3.24.5 You must ensure that the Sim-Ops Management Plan has been briefed to all Site Personnel that are required to work to the details mentioned in the Sim-Ops Management Plan.

3.25 Planned Inspections

- 3.25.1 You must develop and implement a planned inspection process to monitor the effectiveness of your CSMP and associated safety programs as work is performed on site.
- 3.25.2 These inspections must include area inspections, behavior inspections and equipment inspections as a minimum, and occur on a frequent and regular basis, to identify and correct any unsafe acts, behaviors or conditions related to your safety programs and minimum requirements set out in this document.
- 3.25.3 You must document all inspections and provide this documentation to Owner upon request using Procore.
- 3.25.4 Amazon may review your safety program(s) at any time, without providing prior notification to you. Data collected from these reviews may be evaluated and benchmarked against industry standards and best practices.
- 3.25.5 You are required to conduct any inspections required by local laws (for example, inspection of mobile equipment and/or plant and machinery) and must retain records of those inspections.

3.26 Mobile Technology

- 3.26.1 The use of mobile technology (such as personal cell phones) on a construction site can distract the user from the hazards around them.
- 3.26.2 You shall develop a mobile technology use policy with the objective of prohibiting Site usage, except as necessary to protect safety (e.g., in the event of an emergency) or for the performance of work-related tasks in a safe and secure manner.
- 3.26.3 The policy shall address the following, at a minimum:
- A requirement not to use mobile phones in work zones except in designated mobile phone use areas
 - A “No Walking While Talking/Texting/Browsing” requirement; and
 - Confirmation that mobile technology is not to be used while operating a vehicle or mobile equipment.
- 3.26.4 Your mobile technology policy shall be reviewed in your site-specific orientation and strictly enforced on Site.
- 3.26.5 You shall require each of your subcontractors to provide their lead field/on-site management representative with the appropriate mobile technology equipment needed to meet the reporting and documentation requirements of the GMRs.

3.27 GC and Subcontractor Project Safety Officers

- 3.27.1 You shall always have a designated Project Safety Officer on Site at all times while Work is being

performed, and when high risk activities are conducted.

- 3.27.2 The designated Project Safety Officer shall be appropriately qualified in accordance with any relevant local laws.
- 3.27.3 Each Subcontractor shall also designate a Project Safety Officer when conducting work on site, and that the Project Safety Officer is present and working on site.
- 3.27.4 This Subcontractor Project Safety Officer will typically be an employee of the Subcontractor, who may also function as a superintendent, foreman, or crew leader, on the Project.
- 3.27.5 The Subcontractor Project Safety Officer shall meet the same job qualification requirements as the GC Project Safety Officer.
- 3.27.6 You shall hold the Subcontractor's Project Safety Officer accountable for their responsibility to ensure that the Subcontractor has a robust safety management program in place on the site that is actively managed and enforced by the Subcontractor's leadership team.

3.28 Recordkeeping

- 3.28.1 You shall collect the safety training records of all your employees, contract workers, subcontractors, suppliers and their respective personnel in order to verify that all contractors on site are meeting their obligations to ensure that employees are adequately trained.
- 3.28.2 You shall establish means and methods to ensure that all the Subcontractor's lower tier Subcontractors, vendors, inspectors, or any other personnel invited onto the Site by the Subcontractor, have been identified to you, their visit to the Site has been documented, and that the Subcontractor has collected and verified the safety training records for those individuals.
- 3.28.3 Permit to Work documents issued in accordance with your Permit to Work Program shall be kept on file for the full duration of the Project or a period of one year from issuance; whichever is longer.
- 3.28.4 You shall develop the means and methods needed to collect, analyze, organize, and store all required documentation necessary to comply with applicable local laws, including the additional documentation set out in these GMRs, and documentation shall be made available to Amazon upon request. For builds where the GC is engaged by a Landlord, the below information shall be submitted to Amazon via Landlord.

3.29 Stop Work Authority

- 3.29.1 You should have a documented mechanism which highlights the ability for all Site Personnel to exercise their Stop Work Authority (SWA).
- 3.29.2 SWA provides all site staff with the responsibility to stop work when a perceived unsafe condition or behavior may result in harm to any person, the environment, or property.
- 3.29.3 To stop work, any person raising the concern shall bring the concern to the attention of the person conducting the work. The person raising concern should then notify the manager of the person(s) who were carrying out the work activities, and the Subcontractor/GC project safety officer/representative.
- 3.29.4 All individuals shall be made aware during your project orientation of their SWA ability and their ability to bring their concerns to GC personnel, and that the GC will investigate the concern immediately.

3.30 Continuous Improvement Program

- 3.30.1** You, and all Subcontractors, shall have a continuous improvement program in place that captures best practices and lessons learned to further enhance your, and their overall safety programs.
- 3.30.2** Following the investigation of an Incident in accordance with Section 3.17, any relevant lessons learned shall be documented (including relevant lessons identified by Subcontractors), and disseminated to relevant Site Personnel, Subcontractors and your personnel on other projects being managed or constructed on behalf of Amazon, including via Landlord. Lessons learned should be disseminated within two business days from the incident Root Cause completion.
- 3.30.3** You and your Subcontractors' lessons learned documentation shall be made available to Amazon upon request. For builds where the GC is engaged by a Landlord, the below information shall be submitted to Amazon via Landlord.

3.31 Discipline Policy

- 3.31.1** You shall develop and implement a Discipline Policy on the Project, and the expectations of this policy shall be well communicated to all Site Personnel in your safety orientation training.
- 3.31.2** This policy shall include disciplinary procedures to be utilized for noncompliance/nonconformance with your safety program and/or elements of the CSMP. You shall maintain a detailed disciplinary action report that identifies how your safety program is enforced on the Site.
- 3.31.3** Your disciplinary action report shall be made available to Owner. For builds where the GC is engaged by a Landlord, the below information shall be submitted to Amazon via Landlord

Appendix A: Health and Safety Metric Reporting

Amazon evaluates construction H&S performance based on trailing and leading indicators. GC must capture the metrics described in Appendix A in this specification document monthly for each Project and submit them by the third working day of every new month for the previous month to Amazon. Reporting of these metrics is based on the definition of incident specified and is independent of local jurisdictional requirement reporting that GC must comply with.

Recordable Incident refers to any work-related injury resulting in a Medical Treatment Case, Lost Time Incident, Restricted Work Injury Case or Transfer Case:

Total Recordable Incident Rate (TRIR)		Days Away, Restricted, Transfer Rate (DART)	
$\frac{\# \text{ Recordable Cases} \times 200,000}{\# \text{ Hours Worked}}$		$\frac{\# \text{ Recordable Cases with DART} \times 200,000}{\# \text{ Hours Worked}}$	
Monthly	Cumulative	Monthly	Cumulative
Lost Time Incident Frequency Rate (LTIFR)		Lost Time Incident Rate (LTIR)	
$\frac{\# \text{ Lost Time Cases} \times 1,000,000}{\# \text{ Hours Worked}}$		$\frac{\# \text{ Lost time Cases} \times 200,000}{\# \text{ Hours Worked}}$	
Monthly	Cumulative	Monthly	Cumulative
Total Recordable Incident Rate (TRIR)		Days Away, Restricted, Transfer Rate (DART)	
Severity Rate (SR)		Subcontractor Minimum TRIR Performance Rate Requirement (annual)	
$\frac{\# \text{ Lost Workdays}}{\# \text{ Recordable Incidents}}$			
Monthly	Cumulative		
Safety & Health Meetings		Percent of incident/injury cases without root causes	
$\frac{\# \text{ of Meetings Held}}{\# \text{ Weeks of Active Construction}}$		$\frac{\# \text{ Incidents/injuries without root causes}}{\# \text{ of incidents and injuries}}$	
Target	Actual	Target	Actual
Percent of Lost Time Injury (LTI) events with corrective actions		Percent of near miss cases with corrective actions	
$\frac{\# \text{ LTI Cases with corrective actions}}{\# \text{ Total LTI Cases}}$		$\frac{\# \text{ Near miss Cases with Corrective Actions}}{\# \text{ Total near miss cases}}$	
Target	Actual	Target	Actual
Risk Reduction Rate		Stop Work Activation with Root Cause Analysis (RCA)	
Reserved		$\frac{\# \text{ Stop work authority activations with RCA}}{\# \text{ Stop work authority activations}}$	
Percent Behavioral Safety Observations		Hazardous Observation Action Closure Rate	
$\frac{\# \text{ Positive Safety Interactions}}{\# \text{ Unsafe Acts} + \# \text{ Unsafe Conditions}}$		$\frac{\# \text{ Total Hazardous Observation actions raised}}{\# \text{ Hazardous Observation actions closed}}$	
$\frac{\# \text{ Total Behavioral Observations}}{\# \text{ Total Behavioral Observations}}$			
Target	Actual	Target	Actual

Appendix B: H&S Incident Reporting Form

If you are injured, involved in, or witness to a safety incident (Injury, Illness, Near Miss, or Property Damage) you must fill out this form. Injured people should fill out this form once immediate medical attention is no longer required. This or similar form must be submitted to Owner via Procore

INJURED/INVOLVED/WITNESS PERSON’S DETAILS:			
Statement Type:	Injured Person ☐ Involved Person ☐ Witness ☐		
Last Name:		Given Name(s):	
Employer:	GC/Vendor ☐ Other ☐ ☐	Company Name:	
Email:		Phone Number:	
Supervisor:		Department:	
BASIC INCIDENT DETAILS:			
Incident Type:	Injury ☐ Illness ☐ Property Damage ☐ Near Miss ☐		
Incident Date:		Incident Time:	
Incident Location:	(Site Code, Floor, Room Name/Number, Specific Area)		
Were others involved in the incident?	(Full Name, Organization)		
Did anyone else see/witness what happened?	(Full Name, Organization)		
STATEMENT:			
Describe the events leading up to the incident.			
Describe the incident (please restrict to facts i.e. what happened, what you saw and heard and when).			
Describe what happened after the incident (i.e. actions in response).			
What conditions (if any) do you believe influenced the incident (e.g. weather, time of day, fatigue, equipment malfunction, etc.)?			
How did human action (if any) influence the incident (e.g. actions, body position, etc.)?			
(OPTIONAL) Additional Information:			
(Please add any other relevant details here)			
(OPTIONAL) Drawings/diagrams can be added as attachment. Are there attachments:			Yes ☐ No ☐

AWS - ATL089 Procedures Manual

INJURED PERSON(S) ONLY			
Was first aid offered?	Yes ☐ No ☐	Was outside medical treatment offered?	Yes ☐ No ☐
Was first aid decline?	Yes ☐ No ☐	Was outside medical treatment declined?	Yes ☐ No ☐
Was any medical treatment sought post incident?	Yes ☐ No ☐	If yes, brief describe treatment and provide a medical record.	
What body part		Type of injury	

NEAR MISS AND PROPERTY DAMAGE INCIDENTS ONLY:
What injuries that could have occurred during this Near Miss or Property Damage incident?
Describe all the property damage.

SIGNATORIES:			
INJURED/INVOLVED/WITNESS PERSON'S DECLARATION			
I declare that the above statement is true and accurate.			
Name:		Date:	
Signature:		Time:	
MANAGER RECEIPT OF REPORT			
I received this statement directly from the above signatory.			
Name:		Date:	
Signature:		Time:	